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Selecting Subgame Perfect Nash Equilibria in Stackelberg Games

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Introduction

In this thesis we are interested in describing *Stackelberg games*, namely interactions between two agents which act sequentially in two stages:

- first stage: the *leader* chooses an action x in his action set X ;
- second stage: the *follower* watches what the leader does and selects an action from his set of possible actions, Y .

At this point, each player receives his reward: the leader receives $L(x, y)$, and the follower receives $F(x, y)$. The critical distinction between the two players is that the leader understands how the follower arrives at his decisions. In contrast, the follower can only observe the leader's choice after it has been made.

The games can be represented as $\Gamma = (X, Y, L, F)$, where X and Y denote the action sets of the leader and the follower respectively. Additionally, $L : X \times Y \rightarrow \mathbb{R}$ and $F : X \times Y \rightarrow \mathbb{R}$ are real-valued functions that represent the payoff functions of the leader and the follower respectively.

Please assume that the players aim to maximize their payoff: once seen the decision x of the leader, the follower faces the optimization problem

$$\max_{y \in Y} F(x, y). \tag{P_x}$$

Namely, he reacts to the leader's choice by picking an action in the *follower's optimal reaction set*:

$$M(x) = \operatorname{Arg} \max_{y \in Y} F(x, y) = \{y \in Y \mid F(x, y) = \sup_{z \in Y} F(x, z)\}.$$

Observe that, without any assumptions on the payoff functions and the action sets, $M(x)$ can be an empty set. In the upcoming chapters, we will discover that this scenario does not occur, even without imposing too restrictive conditions.

So we assume that $M(x) \neq \emptyset$, and we define the *follower's best reply correspondence* as the set-valued map $M : X \rightrightarrows Y$ that assigns, to any action x of the leader, the follower's optimal reaction set $M(x)$.

Now it is the leader's turn: if he can foresee the action $y(x) \in M(x)$ chosen by the follower, he has to face the problem:

$$\max_{x \in X} L(x, y(x)).$$

Of course, predicting the follower's optimal responses is nearly always hard for the leader, unless he is aware of some additional information. Based on the additional information provided, we can link the Stackelberg game to various types of mathematical problems, which are examined in Section 2.1.

Let us note that, with this interpretation, Stackelberg games can be considered static games. Essentially, we freeze the situation and examine, for every choice of the leader, the corresponding set of choices available to the follower. Consequently, the follower does not observe the leader's actions before making their move; both players decide simultaneously: the leader chooses his action x , and the follower chooses a function that depends on x .

Let us provide a brief overview of the main topics covered in each chapter.

In the first chapter, we outline some well-known definitions and theorems that we will utilize later on.

In the second chapter, we elaborate on the topic of Stackelberg games. Specifically, we examine the main variants of Stackelberg games, we introduce an improved version of the Nash equilibrium called the subgame perfect Nash equilibrium (SPNE), which will serve as the equilibrium concept for Stackelberg games. We also introduce critical aspects of these games, with an emphasis on the selection problem, which will be the central theme of this thesis. Additionally, we outline our goals and strategy for resolving the selection problem.

In the third chapter, we introduce the concept of cost of change: altering a strategy comes with associated costs. Initially, we examine the scenario involving a single agent, and we adapt the payoff function to incorporate influences from the starting point and a cost function. Subsequently, we extend our analysis to non-cooperative games involving a group of n players. We relax the convexity assumption and we focus our investigation to Stackelberg games.

In the fourth chapter, we introduce the proximal operator of a given proper function f and establish a connection between its fixed points and the minimizers of f . We then proceed to demonstrate that the sequence derived from iteratively applying the proximal operator converges to a fixed point of the operator itself, which is also a minimizer of the function f .

In the fifth chapter, we introduce a specific class of Stackelberg games G to which we apply the *Cost-to-Move procedure*. This procedure defines a sequence of perturbed games and their corresponding subgame perfect Nash equilibria, starting from a given game. We prove that this procedure is well-defined when applied to a game in G . Additionally, we prove that the argmax correspondence used to define $M(x)$, which represents the follower's best reply correspondence, is both single-valued and continuous. Furthermore, by exploiting the SPNEs obtained from the perturbed games, we construct a subgame perfect Nash equilibrium for the initial game.

In the sixth chapter, we explore alternative procedures that lead to an SPNE of the initial game. Firstly, we consider a general result where the perturbed payoffs are not derived from specific selection methods. Later, we utilize Tikhonov regularization, which modifies the payoff by subtracting a penalty term that increases as the size of the player's variable increases. Finally, we define a perturbation based on the altruistic behavior of the players, i.e., the players incorporate a small proportion of the other player's payoff into their own payoffs.

Chapter 1

Background

This chapter aims to recall some well-known definitions and theorems, which will be helpful later.

1.1 Limit inferior and Limit superior

Let (x_n) be a sequence of real numbers. Then, the boundaries that the sequence approaches are the superior and inferior limits. They are defined by

$$\begin{aligned}\liminf_{n \rightarrow \infty} x_n &= \sup_{n \geq 0} \inf_{m \geq n} x_m, \\ \limsup_{n \rightarrow \infty} x_n &= \inf_{n \geq 0} \sup_{m \geq n} x_m.\end{aligned}\tag{1.1}$$

If the standard limit exists, then both the limit inferior and the superior will be equal to it; therefore, each can be considered a generalization of the standard limit. When, on the other hand, $\liminf_{n \rightarrow \infty} x_n$ and $\limsup_{n \rightarrow \infty} x_n$ both exist, we have

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.\tag{1.2}$$

Let us outline some important features of $\liminf$ and $\limsup$. Let (x_n) be a sequence in $\mathbb{R} \cup \{+\infty, -\infty\} = \bar{\mathbb{R}}$, then

$$\limsup_{n \rightarrow \infty} (-x_n) = -\liminf_{n \rightarrow \infty} (x_n),\tag{1.3}$$

and the sequence converges in $\bar{\mathbb{R}}$ if and only if

$$\limsup_{n \rightarrow \infty} (x_n) = \liminf_{n \rightarrow \infty} (x_n).$$

Let (a_n) and (b_n) be two sequences of real numbers; if the right side of the inequalities (1.4) is defined (i.e. it is not $\infty - \infty$ or $-\infty + \infty$), the limit superior shows subadditivity. Analogously, the limit inferior shows superadditivity:

$$\begin{aligned} \limsup_{n \rightarrow \infty} (a_n + b_n) &\leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n, \\ \liminf_{n \rightarrow \infty} (a_n + b_n) &\geq \liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n. \end{aligned} \tag{1.4}$$

1.2 Proper functions

Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a function. We will consider the Euclidean norm in $\mathbb{R}^n$:

$$\begin{aligned} \|\cdot\| : \mathbb{R}^n &\rightarrow \mathbb{R}, \\ \|x\| &= \sqrt{x_1^2 + \cdots + x_n^2}. \end{aligned}$$

Definition 1.2.1. We say that f is a *proper function* if

- $f(x) \neq -\infty$ for every $x \in \mathbb{R}^n$,
- $f \not\equiv +\infty$.

Definition 1.2.2. Let $\varepsilon \in \mathbb{R}$. The *lower level set* of f at height ε is

$$\text{lev}_{\leq \varepsilon} f = \{x \in \mathbb{R}^n \mid f(x) \leq \varepsilon\},$$

and the *upper level set* of f at height ε is

$$\text{lev}_{\geq \varepsilon} f = \{x \in \mathbb{R}^n \mid f(x) \geq \varepsilon\}.$$

Definition 1.2.3. We say that a function f is *closed* if and only if, for every $\varepsilon \in \mathbb{R}$, the lower level set $\text{lev}_{\leq \varepsilon} f$ is closed.

Definition 1.2.4. Let $f : D \subseteq \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a function, and D be a convex set. The function f is *quasiconcave* if $\text{lev}_{\geq \varepsilon}$ is a convex set for every $\varepsilon \in \mathbb{R}$, and it is *quasiconvex* if $\text{lev}_{\leq \varepsilon}$ is a convex set for every $\varepsilon \in \mathbb{R}$.

Definition 1.2.5. The *epigraph* of a function f is the set

$$\text{Epi } f = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}.$$

Definition 1.2.6. The *effective domain* of a proper function f is the set

$$\text{Dom } f = \{x \in \mathbb{R}^n \mid f(x) < +\infty\}.$$

Namely, it is the set of points for which f takes on real values.

Definition 1.2.7. The *codomain* of a function f is the set

$$\text{Co } f = \{y \in \mathbb{R} \cup \{+\infty\} \mid \exists x \in \mathbb{R}^n \text{ such as } f(x) = y\}.$$

Definition 1.2.8. A proper function is *convex* if its epigraph is a non-empty and convex set.

Definition 1.2.9. A convex function f is *strongly convex* if $g(x) = f(x) - \frac{\nu}{2}\|x\|^2$ is convex for some $\nu \geq 0$.

Definition 1.2.10. Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$, where D is an unbounded set. We say that f is *coercive* if

$$\lim_{x \in D, \|x\| \rightarrow +\infty} f(x) = +\infty.$$

Definition 1.2.11. Let (X, d) be a metric space, and $f : X \rightarrow \mathbb{R}$ a function. We say that f is *upper semicontinuous* on X if

$$f(x_0) \geq \limsup_{x \rightarrow x_0} f(x)$$

for every $x_0 \in X$. We say that f is *lower semicontinuous* on X if

$$f(x_0) \leq \liminf_{x \rightarrow x_0} f(x)$$

for every $x_0 \in X$.

Remark 1.2.12. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The following statements are equivalent:

- f is lower semicontinuous,
- $\text{Epi } f$ is a closed set,
- $\text{lev}_{\leq \varepsilon} f$ is a closed set for every $\varepsilon \in \mathbb{R}$.

Theorem 1.2.13. (*Weierstrass generalization*) Let $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a lower semicontinuous function, and one of the following two conditions applies:

- D is a compact set,
- D is a closed set and f is coercive.

Then, f admits a global minimum.

1.3 Correspondences

Definition 1.3.1. A *correspondence* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a map

$$T : \mathbb{R}^n \rightrightarrows \mathbb{R}^m,$$

i.e. $T(x) \subset \mathbb{R}^m$ for every $x \in \mathbb{R}^n$. Moreover, T is nonempty-valued, closed-valued, or convex-valued if, for each $x \in \mathbb{R}^n$, $T(x)$ is, respectively, a nonempty, closed, or convex subset of $\mathbb{R}^m$.

Definition 1.3.2. The *domain* of a correspondence T is the set

$$\text{Dom}(T) = \{x \in \mathbb{R}^n \mid T(x) \neq \emptyset\}.$$

Definition 1.3.3. The *graph* of a correspondence T is the set

$$\text{Graph } T = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^m \mid y \in T(x)\}.$$

Definition 1.3.4. A correspondence T is *upper semicontinuous* at $\bar{x}$ if, for each sequence $\{x^k\} \subset \mathbb{R}^n$ converging to $\bar{x} \in \mathbb{R}^n$ and each open set $\mathcal{Y}^* \subset \mathbb{R}^m$ such that $T(\bar{x}) \subset \mathcal{Y}^*$, there is $k_0 \in \mathbb{N}$ such that, for each $k \geq k_0$, $T(x_k) \subset \mathcal{Y}^*$. Moreover, T is said to be upper semicontinuous on $\mathbb{R}^n$ if it is upper semicontinuous at every point of $\mathbb{R}^n$.

Definition 1.3.5. A correspondence T is *lower semicontinuous* at $\bar{x}$ if for each sequence $\{x^k\} \subset \mathbb{R}^n$ converging to $\bar{x} \in \mathbb{R}^n$ and each open set $\mathcal{Y}^* \subset \mathbb{R}^m$ such that $T(\bar{x}) \cap \mathcal{Y}^* \neq \emptyset$ there is $k_0 \in \mathbb{N}$ such that, for each $k \geq k_0$, $T(x_k) \cap \mathcal{Y}^* \neq \emptyset$. Moreover, T is said to be lower semicontinuous on $\mathbb{R}^n$ if it is lower semicontinuous at every point of $\mathbb{R}^n$.

Remark 1.3.6. If the correspondence is a function (i.e. it is single-valued), both upper and lower semicontinuity proprieties reduce to the standard definition of continuity of a function. Indeed, let $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function. Note that φ is upper semicontinuous at x if for each neighborhood $U(\varphi(x))$ of $\varphi(x)$ there exists a neighborhood $V(x)$ of x such that $\varphi(x') \subset U$ for every $x' \in V$.

Suppose that $\#\varphi(x) = 1$, i.e. for every $x \in \mathbb{R}^n$ $\varphi(x) = \tilde{y} \in \mathbb{R}^m$; we obtain that for every neighborhood $U(\tilde{y})$ of $\tilde{y}$ there exists a neighborhood $V(x)$ of x such that $\varphi(x') \subset U(\tilde{y}) \forall x' \in V(x)$, that is the definition of continuity. Likewise, φ is lower semicontinuous in x if for every $U(\varphi(x))$ there exists $V(x)$ such that $\varphi(x') \cap U(\varphi(x)) \neq \emptyset \forall x' \in V(x)$. If $\#\varphi(x) = 1$ we obtain that for every $U(\tilde{y})$ there exists $V(x)$ such that $\varphi(x') \cap U(\tilde{y}) \neq \emptyset \forall x' \in V(x)$, which means that for every $U(\tilde{y})$ there exists $V(x)$ such that $\varphi(x') \subset U(\tilde{y}) \forall x' \in V(x)$, being $\varphi(x')$ a singleton in $\mathbb{R}^m$.

Remark 1.3.7. Please be aware that the definition of upper semicontinuity for correspondences differs from that of functions (Definition 1.2.11).

Definition 1.3.8. A correspondence T is said to be *closed* on $\mathbb{R}^n$ if for any sequence $(x_n, y_n) \in \mathbb{R}^n \times \mathbb{R}^m$ such that $y_{n_k} \in T(x_{n_k})$ for a subsequence n_k , and $(x_{n_k}, y_{n_k}) \rightarrow (x_0, y_0)$, then $y_0 \in T(x_0)$.

Definition 1.3.9. A correspondence T is said *subcontinuous* on $\mathbb{R}^n$ if, for any sequence (x_n) converging in $\mathbb{R}^n$, every sequence (y_n) such that $y_n \in T(x_n)$ for any $n \in \mathbb{N}$ has a convergent subsequence.

Remark 1.3.10. Note that if $T(x) \subseteq K$, with $K \subseteq \mathbb{R}^m$ compact, then T is trivially subcontinuous.

Definition 1.3.11. Let $T : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ be a correspondence. We say that T is *monotone* if

$$\langle x^* - y^*, x - y \rangle \geq 0 \quad \text{for all } (x, x^*) \in \text{Graph } T, \text{ and for all } (y, y^*) \in \text{Graph } T.$$

Let us state two remarkable theorems regarding correspondence:

Theorem 1.3.12. (*Kakutani Fixed-Point Theorem*) Let $X \subset \mathbb{R}^n$ be a nonempty, convex, and compact set. Let $T : X \rightrightarrows X$ be an upper semicontinuous, nonempty-valued, closed-valued, and convex-valued correspondence. Then, there is $\bar{x} \in X$ such that $\bar{x} \in T(\bar{x})$, i.e T has a fixed-point.

Theorem 1.3.13. (*Closed Graph Theorem*) Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be compact sets. A correspondence T from X to Y is closed if and only if it is upper semicontinuous and closed-valued.

1.4 Subdifferential

We present the concept of *subgradient* of a function at a given point.

Definition 1.4.1. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a closed and proper function, and $x \in \text{Dom } f$.

We say that $v \in \mathbb{R}^n$ is a *subgradient* of f at x if, for all $y \in \text{Dom } f$,

$$f(y) \geq f(x) + \langle v, y - x \rangle.$$

The *Clarke-Rockafellar subdifferential* of f at x is the (possibly empty) set of all subgradients of f at the point x :

Definition 1.4.2. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a closed and proper function, and $x \in \text{Dom } f$.

The *subdifferential* of f at x is the set

$$\partial f(x) = \{v \in \mathbb{R}^n \mid f(y) \geq f(x) + \langle v, y - x \rangle \quad \forall y \in \text{Dom } f\}.$$

Definition 1.4.3. A function f is called *subdifferentiable* at x if at least one subgradient exists at x .

Let us note that 0 belongs to the subdifferential of f at x if and only if x minimizes f , as stated in the following theorem:

Theorem 1.4.4. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a closed and proper function, and $x \in \text{Dom } f$, then

$$0 \in \partial f(x) \iff f(y) \geq f(x) \quad \text{for all } y \in \text{Dom } f.$$

Proof. Let us suppose that $0 \in \partial f(x)$, then, by Definition 1.4.2,

$$f(y) \geq f(x) + \langle 0, y - x \rangle = f(x) \quad \forall y \in \text{Dom} f,$$

so x minimizes f . Now, suppose that $f(y) \geq f(x)$ for all $y \in \text{Dom} f$, then

$$f(y) \geq f(x) = f(x) + \langle 0, y - x \rangle \Rightarrow 0 \in \partial f(x),$$

and the proof is complete. □

Another important property is the following:

Theorem 1.4.5. *Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ and $g : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be closed, proper and convex functions and let $\bar{x} \in \text{int}(\text{Dom} f \cap \text{Dom} g)$. Then, for all $x \in \mathbb{R}^n$*

$$\partial(f + g)(x) = \partial f(x) + \partial g(x).$$

Additionally, the subdifferential concept extends the derivative concept, as explained in the following result:

Theorem 1.4.6. *Let f be a closed and proper convex function. If f is Gâteaux differentiable at x , then*

$$\partial f(x) = \{\nabla f(x)\}.$$

Theorem 1.4.7. *Let f be a closed, convex, and proper function. Then ∂f is a monotone operator.*

Chapter 2

Stackelberg games

2.1 Models

This paragraph aims to describe the main types of Stackelberg problems.

2.1.1 Classical Stackelberg problem

This problem occurs when the leader can fully foresee what the follower will do. Namely, we suppose that the follower's optimal response to any choice of the leader exists and is unique: $M(x) \neq \emptyset$ and $\#M(x) = \{1\}$, for every $x \in X$.

In this case, assuming $M(x) = \{m(x)\}$, the leader faces the so-called *classical Stackelberg problem*:

$$\max_{x \in X} L(x, m(x)), \quad (\text{SP})$$

where $m(x)$ is the solution to (P_x) . The classical Stackelberg game is named after von Stackelberg, who first investigated it in [22].

Remark 2.1.1. Let us note that the solution of (SP) may not necessarily exist. In order to guarantee its existence, we make the following assumptions:

- X and Y are compact sets,
- $F : X \times Y \rightarrow \mathbb{R}$ and $L : X \times Y \rightarrow \mathbb{R}$ are upper semicontinuous functions,
- $m : X \rightarrow Y$ is continuous.

Indeed, the upper semicontinuity of F and the compactness of Y ensure the well-posedness of $m(x)$. Moreover, since L is upper semicontinuous and m is continuous, we obtain that:

$$\limsup_{(x,y) \rightarrow (x_0,y_0)} L(x,y) \leq L(x_0,y_0), \quad \lim_{x \rightarrow x_0} m(x) = m(x_0),$$

so

$$\limsup_{x \rightarrow x_0} L(x, m(x)) \leq L(x_0, m(x_0)).$$

Thus, $L(x, m(x))$ is upper semicontinuous; this, together with the compactness of X , ensures that (SP) is well-posed.

2.1.2 Weak Stackelberg problem

In this scenario, the leader faces an extreme situation where he believes that the follower will deliberately choose, within his optimal reaction set, the response that will result in the most severe penalty for the leader. So, assuming $M(x) \neq \emptyset$, the leader faces the so-called *weak Stackelberg problem*:

$$\max_{x \in X} \min_{y \in M(x)} L(x, y), \quad (\text{WSP})$$

where $M(x) = \text{Arg max}_{y \in Y} F(x, y)$.

Let us note that, although this is a very simple case, the existence of a solution is not guaranteed.

Remark 2.1.2. Consider $X = Y = [-1, 1]$, and

$$L(x, y) = -x + y, \quad F(x, y) = xy.$$

Then we have

$$M(x) = \text{Arg max}_{y \in [-1, 1]} xy = \begin{cases} -1 & \text{if } x \in [-1, 0) \\ [-1, 1] & \text{if } x = 0 \\ 1 & \text{if } x \in (0, 1], \end{cases}$$

and

$$\min_{y \in M(x)} -x + y = \begin{cases} -x - 1 & \text{if } x \in [-1, 0] \\ -x + 1 & \text{if } x \in (0, 1]. \end{cases}$$

As we can see in Figure 2.1,

$$\operatorname{Argmax}_{x \in X} \min_{y \in M(x)} L(x, y) = \emptyset,$$

and the problem has no solution.

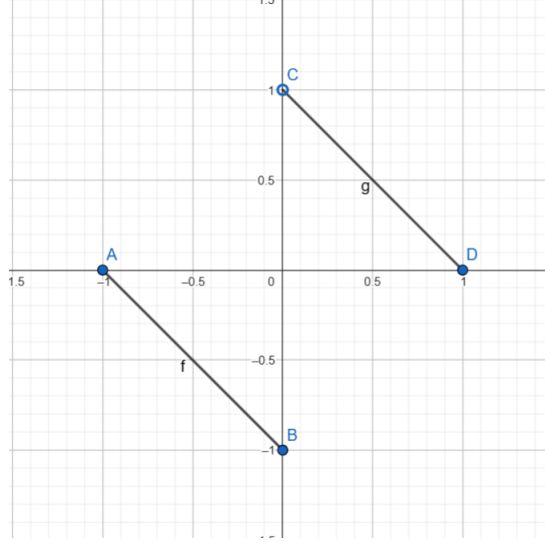


Figure 2.1: Weak Stackelberg solutions may not exist

2.1.3 Strong Stackelberg problem

In contrast to the previous case, in this scenario, the leader holds the belief that the follower will choose the best possible response for the leader, within the follower's optimal reaction set. Assuming $M(x) \neq \emptyset$, the leader faces the so-called *strong Stackelberg problem*:

$$\max_{x \in X} \max_{y \in M(x)} L(x, y), \quad (\text{SSP})$$

where $M(x) = \operatorname{Argmax}_{y \in Y} F(x, y)$.

In this case, under mild assumptions, the existence of a solution is guaranteed. To prove this, we will need a modified version of the Berge Theorem:

Theorem 2.1.3. (*Berge Maximum Theorem*) Let $F : X \times Y \rightarrow \mathbb{R}$ be a continuous function. Define the value function $m : X \rightarrow \mathbb{R}$ by

$$m(x) = \max_{y \in Y} F(x, y),$$

and the correspondence $\varphi : X \rightrightarrows Y$ of maximizers by

$$\varphi(x) = \{y \in Y \mid F(x, y) = m(x)\} = \operatorname{Arg} \max_{y \in Y} F(x, y).$$

Then:

- the value function m is continuous,
- the “argmax” correspondence φ has nonempty compact values,
- the “argmax” correspondence φ is upper semicontinuous.

In our model, F need not be continuous. Nevertheless, we overcome this lack of continuity assumption by utilizing the following property: for any $(x, y) \in X \times Y$ and for any sequence $(x_k)_k \subset X$ converging to x there exists a sequence $(\tilde{y}_k)_k \subset Y$ converging to y such that

$$\liminf_{k \rightarrow \infty} F(x_k, \tilde{y}_k) \geq F(x, y). \quad (F2)$$

Later, we will assume this property to be valid and call it (F2).

Theorem 2.1.4. *Let $F : X \times Y \rightarrow \mathbb{R}$ be an upper semicontinuous function such that (F2) holds. Define the value function $m : X \rightarrow \mathbb{R}$ by*

$$m(x) = \max_{y \in Y} F(x, y),$$

and the correspondence $\varphi : X \rightrightarrows Y$ of maximizers by

$$\varphi(x) = \{y \in Y \mid F(x, y) = m(x)\} = \operatorname{Arg} \max_{y \in Y} F(x, y).$$

Then, the “argmax” correspondence φ is continuous.

Proof. Thanks to Theorem 1.3.13, if we show that φ is closed, we obtain the upper semicontinuity of the correspondence. Let $(x_n, y_n) \in \operatorname{Graph}(\varphi)$ such that $(x_n, y_n) \rightarrow (\bar{x}, \bar{y})$, namely

$$\begin{cases} x_n \rightarrow \bar{x}, \\ y_n \rightarrow \bar{y}, \\ y_n \in \varphi(x_n). \end{cases}$$

By definition, φ is closed if $(\bar{x}, \bar{y}) \in \text{Graph}(\varphi)$, equivalently if $\bar{y} \in \varphi(\bar{x})$. We observe that

$$\bar{y} \in \varphi(\bar{x}) \iff F(\bar{x}, \bar{y}) = m(\bar{x}) = \max_{y \in Y} F(\bar{x}, y) \iff F(\bar{x}, \bar{y}) \geq F(\bar{x}, y) \quad \forall y \in Y,$$

and

$$F(\bar{x}, \bar{y}) \geq \limsup_n F(x_n, y_n) = \limsup_n \max_{y \in Y} F(x_n, y) = \limsup_n m(x_n) \geq \liminf_n m(x_n),$$

where the first inequality holds thanks to the upper semicontinuity of F , the first equality applies because $y_n \in \varphi(x_n)$, and the second equality follows from the definition of m .

Let $y \in Y$. Thanks to (F2), for every $\bar{x} \in X$ and for every $\{x_n\} \subseteq X$, $x_n \rightarrow \bar{x}$, there exists $(\tilde{y}_n)_n \subseteq Y$ such that

$$\tilde{y}_n \rightarrow y \quad \text{and} \quad \liminf_{n \rightarrow \infty} F(x_n, \tilde{y}_n) \geq F(\bar{x}, y). \quad (2.1)$$

In particular, since

$$m(x_n) \geq F(x_n, y) \quad \forall y \in Y$$

we have that

$$m(x_n) \geq F(x_n, \tilde{y}_n),$$

and

$$\liminf_n m(x_n) \geq \liminf_n F(x_n, \tilde{y}_n) \geq F(\bar{x}, y),$$

where the second inequality follows from (2.1). So $F(\bar{x}, \bar{y}) \geq F(\bar{x}, y)$ for every $y \in Y$, so $\bar{y} \in \varphi(\bar{x})$, and φ is closed. This implies that φ is upper semicontinuous.

Moreover, F is strictly concave, so $\#\varphi(x) = 1$, and in this case being upper semicontinuous means being continuous, as shown in Remark 1.3.6. We can conclude that φ is continuous. $\square$

Theorem 2.1.5. *If X is a compact and nonempty set, Y is a closed and nonempty set, M is a closed and subcontinuous correspondence on X , and $f : X \times Y \rightarrow \mathbb{R}$ is an upper semicontinuous function, then the problem*

$$\begin{cases} \max_{(x,y) \in X \times Y} f(x, y) \\ y \in M(x) \end{cases} \quad (P)$$

has a solution.

Proof. Let (x_n, y_n) be a maximizing sequence for the problem (P) , i.e.

$$\begin{cases} (x_n, y_n) \in X \times Y, \\ y_n \in M(x_n) \text{ for any } n \in \mathbb{N}, \\ f(x_n, y_n) \geq \sup_{(x,y) \in X \times Y, y \in M(x)} f(x, y) - \varepsilon_n, \end{cases}$$

where (ε_n) tends to zero and $\varepsilon_n > 0$ for any $n \in \mathbb{N}$.

Since X is compact (x_n) admits a converging subsequence (x_{n_k}) ; suppose that this sequence converge to $x_0 \in X$. From the subcontinuity of M , every sequence (y_n) such that $y_n \in M(x_n)$ for any $n \in \mathbb{N}$ has a convergent subsequence, so there exists a subsequence (y_{n_k}) of (y_n) wich converges; suppose that this sequece converges to y_0 . For the closedness of M , $y_0 \in M(x_0)$. Moreover, since $X \times Y$ is closed, we have that $(x_0, y_0) \in X \times Y$, and thanks to the upper semicontinuity of f it results that

$$f(x_0, y_0) \geq \limsup_{k \rightarrow \infty} f(x_{n_k}, y_{n_k}) = \sup_{x \in X, y \in M(x)} f(x, y).$$

□

From the previous theorem, we obtain the following one:

Theorem 2.1.6. *Let X and Y be compact and non-empty sets, and $F : X \times Y \rightarrow \mathbb{R}$ be an upper semicontinuous function such that for any $(x, y) \in X \times Y$ and for any sequence $(x_k)_k \subseteq X$ converging to x there exists a sequence $(\tilde{y}_k)_k \subseteq Y$ converging to y such that*

$$\liminf_{k \rightarrow \infty} F(x_k, \tilde{y}_k) \geq F(x, y). \quad (\text{F2})$$

Then, the problem (SSP) has a solution.

Proof. It follows from Theorem 2.1.5 that there exists $(\bar{x}, \bar{y})$ such that $F(\bar{x}, \bar{y}) \geq F(x, y)$ for every $(x, y) \in \text{Graph}(M)$. Indeed, Theorem 2.1.5 applies since $M(x) \subseteq Y$ is closed for Theorem 2.1.4, and since Y is a compact set and $M(x) \subseteq Y$, M is subcontinuous thanks to Remark 1.3.10. Now we will prove that $(\bar{x}, \bar{y})$ also corresponds to the maximum of (SSP). Suppose that there exists $x' \in X$ such that

$$\max_{y \in M(\bar{x})} F(\bar{x}, y) < \max_{y \in M(x')} F(x', y).$$

Since $M(x)$ is compact for all $x \in X$ and F is upper semicontinuous, there exists a point $y' \in M(x')$ where $F(x', \cdot)$ reaches his maximum. Thus,

$$F(\bar{x}, \bar{y}) = \max_{y \in M(\bar{x})} F(\bar{x}, y) < F(x', y'),$$

which leads to a contradiction since $(x', y') \in \text{Graph}(M)$. $\square$

Note that the continuity of

$$\omega(x) = \max_{y \in M(x)} L(x, y)$$

is not guaranteed even for nice functions, as shown in the following example:

Example 2.1.7. Let $X = Y = [0, 1]$, and

$$L(x, y) = x + y, \quad F(x, y) = y \left(x - \frac{1}{2} \right).$$

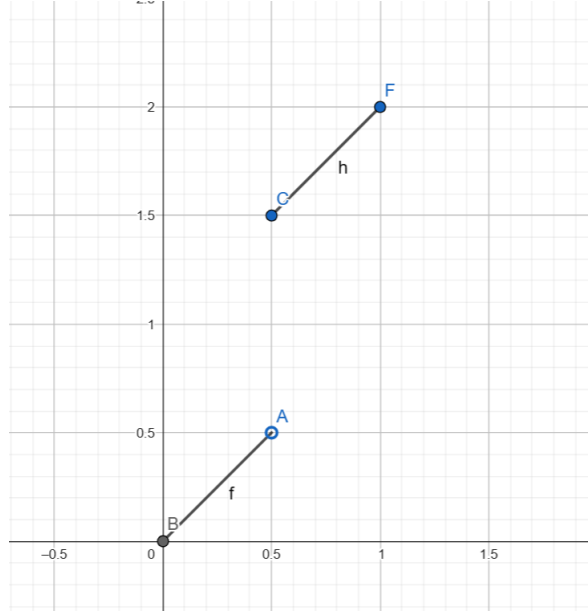
For these functions, we have

$$M(x) = \text{Arg max}_{y \in [0,1]} y \left(x - \frac{1}{2} \right) = \begin{cases} \{0\} & \text{if } x \in [0, \frac{1}{2}) \\ [0, 1] & \text{if } x = \frac{1}{2} \\ \{1\} & \text{if } x \in (\frac{1}{2}, 1]. \end{cases}$$

So,

$$\omega(x) = \max_{y \in M(x)} (x + y) = \begin{cases} x & \text{if } x \in [0, \frac{1}{2}) \\ \frac{3}{2} & \text{if } x = \frac{1}{2} \\ x + 1 & \text{if } x \in (\frac{1}{2}, 1], \end{cases}$$

which is not continuous at $x = \frac{1}{2}$, as shown in Figure 2.2.

Figure 2.2: $\omega(x)$ can be discontinuous.

2.1.4 Intermediate Stackelberg problem

As a last case, we have an intermediate situation: the leader has some information on the follower's way of choosing, so he can attribute a probability distribution on $M(x)$ for any $x \in X$. To simplify, we can suppose that $M(x)$ has a finite number of elements $k(x)$ for any $x \in X$, i.e.

$$M(x) = \{y_1(x), \dots, y_{k(x)}(x)\}.$$

In this scenario, the leader faces the so-called *intermediate Stackelberg problem*:

$$\max_{x \in X} \sum_{i=1}^{k(x)} p_i(x) L(x, y_i(x)), \quad (\text{ISP})$$

where $p_i(x)$ is the probability that the leader attributes to every follower's choice $y_i(x)$ in $M(x)$. In a more general case, suppose $M(x)$ to be not discrete for at least one $x \in X$ and measurable with respect to the Lebesgue measure, with non-zero measure. The leader attributes a probability measure $\nu_x(y)$ on $M(x)$ for any $x \in X$, and the problem becomes:

$$\max_{x \in X} \int_{M(x)} L(x, y) d\nu_x(y), \quad (\text{ISP}_2)$$

where $\nu_x(\cdot)$ is well defined for every $x \in X$.

Note that, depending on the probability distribution, the intermediate problem could become the pessimistic or optimistic one.

2.2 Solution concept

Let us recall that a *strategic game* is a mathematical model that describes interactive situations among several players. If we denote the set of players by $N = \{1, \dots, n\}$, the formal definition of a strategic game is:

Definition 2.2.1. An n -player strategic game with set of players N is a pair $G = (A, u)$, where

- A is the *set of strategies*: for each $i \in N$, A_i is the nonempty set of strategy of player i , and $A = \prod_{i=1}^n A_i$ is the set of strategies profiles,
- u is the *payoff function*: for each $i \in N$, $u_i : A \rightarrow \mathbb{R}$ is the payoff function of player i , and $u = \prod_{i=1}^n u_i$.

In strategic games, the crucial solution concept is Nash equilibrium. This refers to a strategy profile where no player benefits from altering their strategy.

Definition 2.2.2. Let $G = (A, u)$ be a strategic game. A *Nash equilibrium* of G is a strategy profile $a^* \in A$ such that, for each $i \in N$ and each $\hat{a}_i \in A_i$,

$$u_i(a^*) \geq u_i(a_{-i}^*, \hat{a}_i),$$

where $(a_{-i}, \hat{a}_i)$ denote the profile $(a_1, \dots, a_{i-1}, \hat{a}_i, a_{i+1}, \dots, a_n)$.

The concept of equilibrium that we will consider is an advancement of Nash equilibrium, known as *subgame perfect Nash equilibrium* (henceforth *SPNE*). This solution concept was introduced by Selten in [21], where the author assumed that the players will act according to the *principle of sequential rationality*, so he declared that “the equilibria to select are those whereby the players behave optimally from any point of the game onwards”.

Definition 2.2.3. A strategy profile $(\bar{x}, \bar{\varphi}) \in X \times Y^X$ is an *SPNE* of Γ if the following conditions are satisfied:

- $\bar{\varphi}(x) \in M(x) \quad \forall x \in X,$
- $\bar{x} \in \text{Arg max}_{x \in X} L(x, \bar{\varphi}(x)),$

where $Y^X = \{\varphi : X \rightarrow Y\}$ is the set of all function from X to Y .

It is important to note that the first part of this definition implies that the follower's objective is to maximize his payoff function for every choice made by the leader. Similarly, the second part states that the leader also aims to maximize his own payoff function and formulates a strategy based on his prediction of $\bar{\varphi}(x)$.

Therefore an SPNE is a strategy profile composed of one action of the leader and one strategy of the follower.

2.3 Crucial issues

In this section, we will list all the crucial issues associated with Stackelberg games:

- **Existence:** Do solutions to the problem exist without making excessively restrictive assumptions about the action set and payoff function?

Note that we have already examined some examples and results concerning the existence of strong and weak Stackelberg problems in Section 2.1.2 and 2.1.3,

- **“Variational stability”:** Suppose to modify the action sets and/or payoff functions of the game Γ creating a sequence of perturbed games (Γ_n) . If we consider the sequence of solutions $(x_n, \varphi_n(x_n))$ to (Γ_n) , will it converge to a solution to the original problem?

We might have this situation, for example, if the problem is affected by uncertainty. In this condition, we can not achieve exact equilibria, but we aim to predict the solution using perturbed problems,

- **Well-posedness:** Is there a unique solution to the problem?

- **Approximation:** There may be situations where we do not have solutions, and in such cases, we need to find a way to estimate them,
- **Approximation via numerical methods:** Our goal is to restate the problem by transforming it into a better-behaved equivalent one and to find a solution to the numerical issues that come up due to the non-single-valuedness of the best reply correspondence of the follower,
- **Selection:** When faced with multiple solutions, how can we determine a procedure that determines the best one to choose? What are the reasons that make the players follow this selection process?

As we already mentioned in the introduction, we will focus on the problem of selections of SPNEs, which comes up when the follower's best reply correspondence M is not single-valued. When this happens, we can find infinitely many SPNEs, as shown in the following example:

Example 2.3.1. Consider $X = Y = [-1, 1]$, $L(x, y) = -x - y$, and $F(x, y) = xy$.

$$M(x) = \operatorname{Arg\,max}_{y \in [-1, 1]} xy = \begin{cases} \{-1\} & \text{if } x \in [-1, 0) \\ [-1, 1] & \text{if } x = 0 \\ \{1\} & \text{if } x \in (0, 1]. \end{cases}$$

If we denote by φ^σ the function on $[-1, 1]$ defined by

$$\varphi^\sigma = \begin{cases} -1 & \text{if } x \in [-1, 0) \\ \sigma & \text{if } x = 0 \\ 1 & \text{if } x \in (0, 1] \end{cases}$$

we obviously have that $\varphi^\sigma(x) \in M(x)$ for any $x \in [-1, 1]$, and

$$\operatorname{Arg\,max}_{x \in [-1, 1]} L(x, \varphi^\sigma(x)) = \{-1\}.$$

So, for any $\sigma \in [-1, 1]$, the strategy profile $(-1, \varphi^\sigma)$ is an SPNE: the game has infinitely many SPNEs. Therefore, predicting the follower's strategy is hard for the leader.

Our goal is to define constructive methods that enable the leader to select an SPNE with the following features:

- to relieve the leader of predicting the follower's best reply correspondence M ,
- to overcome the difficulties arising from the non-single-valuedness of M ,
- to provide reasons why players should behave in a certain way.

The idea is to consider a generalized sequence of games Γ_n obtained by modifying the payoff functions of the players in the game Γ . So, we consider

$$\Gamma_n = (L_n, F_n) \quad \text{for any } n \in \mathbb{N},$$

i.e. Γ_n is a two-player Stackelberg game with action set X and Y , and payoff functions L_n and F_n . For every $n \in \mathbb{N}$, we assume that Γ_n has an SPNE denoted as $(\bar{x}_n, \bar{\varphi}_n)$. Now, the question is whether the sequence $(\bar{x}_n, \bar{\varphi}_n)_n$ converges to an SPNE of the original game Γ .

When the follower's optimal reaction is unique, i.e. M is single-valued, $(\bar{x}_n, \bar{\varphi}_n)_n$ converges to an SPNE of Γ , as shown in the following result:

Theorem 2.3.2. *Assume that*

- Y is compact,
- M is single-valued,
- $(L_n)_n$ and $(F_n)_n$ continuously converge to L and F respectively, i.e.

$$\lim_{n \rightarrow \infty} L_n(x_n, y_n) = L(x, y), \quad \lim_{n \rightarrow \infty} F_n(x_n, y_n) = F(x, y)$$

for any sequence $(x_n, y_n)_n \subseteq X \times Y$ converging to (x, y) .

Let $(\bar{x}_n, \bar{\varphi}_n)$ be an SPNE of Γ_n for any $n \in \mathbb{N}$ and $(\bar{x}, \bar{\varphi})$ be the limit of $(\bar{x}_n, \bar{\varphi}_n)$, i.e.

$$\bar{x} = \lim_{n \rightarrow \infty} \bar{x}_n, \quad \bar{\varphi}(x) = \lim_{n \rightarrow \infty} \bar{\varphi}_n(x)$$

for any $x \in X$. Then:

- the sequence $(\bar{\varphi}_n)_n$ continuously converges to $\bar{\varphi}$,
- the strategy profile $(\bar{x}, \bar{\varphi})$ is an SPNE of Γ .

Proof. First of all, let us note that the following inequalities trivially hold:

- (F) For any $(x, y) \in X \times Y$, for any sequence x_n converging to x in X ,
there exists a sequence $y_n \in Y$ such that $\lim_{n \rightarrow \infty} F_n(x_n, y_n) \geq F(x, y)$.
- (L) For any $x \in X$, there exists a sequence x_n converging to x in X such that,
for every $y \in Y$, and for every $y_n \rightarrow y$, $\lim_{n \rightarrow \infty} L_n(x_n, y_n) \geq L(x, y)$.

Let us prove that $(\bar{\varphi}_n)_n$ continuously converges to $\bar{\varphi}$, i.e. that $\lim_{n \rightarrow \infty} \bar{\varphi}_n(x_n) = \bar{\varphi}(x)$ for every $x_n \rightarrow x$. Let $x_n \rightarrow x$ be a converging sequence, and consider $\bar{\varphi}_n(x_n) \subseteq Y$. Thanks to the compactness of Y , there exists a subsequence of $\bar{\varphi}_n(x_n)$, which we will denote again by $\bar{\varphi}_n(x_n)$, that converges to $\bar{y} \in Y$. From the continuous convergence, it holds that

$$\lim_{n \rightarrow \infty} F_n(x_n, \bar{\varphi}_n(x_n)) = F(x, \bar{y}).$$

Moreover, thanks to (F) it holds that, for every $y \in Y$, there exists $y_n \in Y$ such that

$$\lim_{n \rightarrow \infty} F_n(x_n, y_n) \geq F(x, y),$$

and, from the definition of SPNE,

$$F_n(x_n, y_n) \leq F_n(x_n, \bar{\varphi}_n(x_n)),$$

so

$$F(x, y) \leq \lim_{n \rightarrow \infty} F_n(x_n, \bar{\varphi}_n(x_n)) = F(x, \bar{y})$$

for every $y \in Y$, and this implies that $\bar{y} \in M(x)$. But M is single-valued and we denote the unique solution by $\bar{\varphi}$, therefore $\bar{y} = \bar{\varphi}(x)$. We obtain the same result for any convergent subsequence of the initial sequence. Then, the entire sequence $\bar{\varphi}_n(x_n)$ converges to $\bar{\varphi}(x)$ when $n \rightarrow +\infty$. In order to prove the second item, we have to prove that

$$\bar{x} \in \operatorname{Argmax}_{x \in X} L(x, \bar{\varphi}(x)).$$

By the assumptions $\bar{x}_n \rightarrow \bar{x}$, so

$$\bar{\varphi}_n(\bar{x}_n) \rightarrow \bar{\varphi}(\bar{x})$$

from the continuous convergence of $(\bar{\varphi}_n)_n$, and

$$L_n(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n)) \rightarrow L(\bar{x}, \bar{\varphi}(\bar{x}))$$

because of the continuous convergence of $(L_n)_n$.

By using (L) we have that for every $x \in X$ there exists a sequence $z_n \rightarrow x$ such that for every $y \in Y$, and for every $y_n \rightarrow y$,

$$\lim_{n \rightarrow \infty} L_n(z_n, y_n) \geq L(x, y).$$

Let us consider $y_n = \bar{\varphi}_n(z_n) \rightarrow y = \bar{\varphi}(x)$. From the definition of SPNE,

$$L_n(z_n, \bar{\varphi}_n(z_n)) \leq L_n(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n)),$$

thus

$$L(x, \bar{\varphi}(x)) = \lim_{n \rightarrow \infty} L_n(z_n, \bar{\varphi}_n(z_n)) \leq \lim_{n \rightarrow \infty} L_n(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n)) = L(\bar{x}, \bar{\varphi}(\bar{x}))$$

for every $x \in X$, and the proof is complete. $\square$

On the other hand, when M is not single-valued, the limit of the sequence $(\bar{x}_n, \bar{\varphi}_n)_n$ is not necessarily an SPNE of Γ , as shown in Example 7.0.2. However, it holds the following theorem:

Theorem 2.3.3. *Assume that the sequences (L_n) and (F_n) continuously converge to L and F , respectively. Let $(\bar{x}_n, \bar{\varphi}_n)$ be an SPNE of Γ_n for any $n \in \mathbb{N}$ and let $\bar{x} = \lim_{n \rightarrow \infty} \bar{x}_n$. If the sequence $(\bar{\varphi}_n)_n$ continuously converges to $\bar{\varphi}$, then $(\bar{x}, \bar{\varphi})$ is an SPNE of Γ .*

Remark 2.3.4. The continuous convergence can be replaced with the following weaker conditions:

($\bar{F}1$) For any $(x, y) \in X \times Y$, for any sequence (x_n, y_n) converging to (x, y) in $X \times Y$ we have

$$\limsup_{n \rightarrow \infty} F_n(x_n, y_n) \leq F(x, y),$$

($\bar{F}2$) For any $(x, y) \in X \times Y$, for any sequence x_n converging to x in X , there exists a sequence y_n in Y such that

$$\liminf_{n \rightarrow \infty} F_n(x_n, y_n) \geq F(x, y),$$

($\bar{L}1$) For any $(x, y) \in X \times Y$, for any sequence (x_n, y_n) converging to (x, y) in $X \times Y$ we have

$$\limsup_{n \rightarrow \infty} L_n(x_n, y_n) \leq L(x, y),$$

($\bar{L}2$) For any $(x, y) \in X \times Y$, there exists a sequence x_n converging to x in X , such that, for any sequence y_n converging to y in Y , we have

$$\liminf_{n \rightarrow \infty} L_n(x_n, y_n) \geq L(x, y).$$

Let us give the proof of Theorem 2.3.3:

Proof. First of all, we prove that $\bar{\varphi}(x) \in M(x)$ for every $x \in X$. Let $x \in X$ and x_n converging to x in X when $n \rightarrow \infty$. For the continuous convergence of $(\bar{\varphi}_n)_n$, we have that $\bar{\varphi}_n(x_n)$ converges to $\bar{\varphi}(x)$ when $n \rightarrow \infty$. From ($\bar{F}1$),

$$\limsup_{n \rightarrow \infty} F_n(x_n, \bar{\varphi}_n(x_n)) \leq F(x, \bar{\varphi}(x)).$$

From ($\bar{F}2$), for any $y \in Y$ there exists a sequence y_n such that

$$\liminf_{n \rightarrow \infty} F_n(x_n, y_n) \geq F(x, y).$$

Furthermore, for the definition of SPNE,

$$F_n(x_n, \bar{\varphi}_n(x_n)) \geq F_n(x_n, y_n) \text{ for any } y_n \in Y.$$

So we get $F(x, \bar{\varphi}(x)) \geq F(x, y)$ for any $y \in Y$, hence $\bar{\varphi}(x) \in \text{Arg max}_{y \in Y} F(x, y) = M(x)$, and the first condition of SPNEs is proved. The second condition of SPNEs holds by applying Theorem 3.4.2, whose hypotheses are trivially satisfied thanks to the continuous convergence. $\square$

Nevertheless, the continuous convergence of $(\bar{\varphi}_n)_n$ could not be satisfied even under nice data of Γ and for a nice type of perturbation, as shown in Example 7.0.2.

Thus, in this thesis, we want to generate a sequence of perturbed games such that their SPNEs converge to an SPNE of Γ , without assuming nor the single-valuedness of M neither the continuous convergence of $(\bar{\varphi}_n)$. We will employ the concept of *cost of change* to adjust the payoffs in order to achieve this goal.

Chapter 3

Cost of change

This chapter aims to provide a different perspective on how players make their decisions. The fundamental concept is that altering a strategy comes with associated costs. Thus, prior to taking any action, each player assesses whether the advantages of making a move outweigh the costs involved. In particular, when analyzing a game theory model with a single agent, three controversial features arise:

- without any hesitation, players can move to the best choice,
- players take into account only their goals and do not consider any cost of change,
- player's actions do not depend on the past or the status quo.

In a more realistic model, players move step-wise, incurring a cost of change, and start from some initial point of departure. If the model involves multiple players, we must consider another crucial aspect: how do players interact with each other? How do they anticipate and respond to actions taken by others?

In social sciences, two hypotheses aim to answer this question: one suggests that agents always behave as if they are in an equilibrium state, even if they are not. The other hypothesis assumes that each player acts as if they have foreseen everything, being perfectly competent and rational. Obviously, these are two extreme models, and we seek a more realistic approach. Therefore, we replace complete optimization with incremental improvements and a global view with a local perception.

3.1 Single agent model

To begin with, let us consider the case of one agent only. Let $X \subseteq \mathbb{R}^n$ be a non-empty set.

Definition 3.1.1. We call *net benefit* the function $b(\hat{x} \mid x) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty\}$, which represents the improvement that the player obtains if he moves from x to $\hat{x}$:

$$(\hat{x}, x) \rightarrow b(\hat{x} \mid x) \in \mathbb{R} \cup \{-\infty\}.$$

In particular, assuming $x \in X$, $b(\hat{x} \mid x) = -\infty$ if $\hat{x} \notin X$.

By adopting this approach, a player's choices will be influenced by their point of departure: they may seek to maximize $b(\hat{x} \mid x)$ subject to $\hat{x} \in X$. Classical models let

$$b(\hat{x} \mid x) = \beta(\hat{x}) - \beta(x), \tag{3.1}$$

for some benefit function $\beta : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty\}$ that does not incorporate adjustment costs. More realistic models, as proximal point methods (see Chapter 5), suppose

$$b(\hat{x} \mid x) = \beta(\hat{x}) - \beta(x) - c(\hat{x} \mid x), \tag{3.2}$$

for some cost function $c : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^+ \cup \{+\infty\}$, where $\mathbb{R}^+$ contains every $x \in \mathbb{R}$ such that $x \geq 0$.

Remark 3.1.2. Please note that the variable c can only take positive values, indicating that players do not appreciate changes. This is in line with our model.

The function c must respect two other features:

- $c(\cdot \mid \cdot)$ must vanish on the diagonal, i.e.:

$$c(x \mid x) = 0 \text{ for any } x \in X,$$

- $c(\cdot \mid \cdot)$ must respect the triangle inequality. i.e.:

$$c(\hat{x} \mid x) + c(\bar{x} \mid \hat{x}) \geq c(\bar{x} \mid x).$$

The second condition implies that the optimal method for transitioning between two points is to move directly to the final point. Deviating from this path to reach an intermediate point would lead to increased costs. In addition, the cost of going from one point to another can be different from the cost of moving oppositely, i.e.

$c(\hat{x} \mid x) \neq c(x \mid \hat{x})$ may happens.

Definition 3.1.3. A move from x to $\hat{x}$ is called *strictly improving* if $b(\hat{x} \mid x) > 0$.

If the player does not move, he has no improvement: $b(x \mid x) \leq 0$ for all $x \in X$; in many cases $b(x \mid x) = 0$.

Definition 3.1.4. A point $x \in X$ is called *stationary* for the map $(\hat{x}, x) \rightarrow b(\hat{x} \mid x)$ if

$$x \in \operatorname{Arg\,max}_{\hat{x} \in X} b(\hat{x} \mid x).$$

Namely, if we move from a stationary point to any other point, we lose benefit: given x a stationary point, $b(\cdot \mid x) \leq 0$ on X .

Given the above, a question arises: do stationary states exist?

First of all, let us recall the *Ky Fan Inequality* and a lemma that will be needed in the proof of the inequality:

Lemma 3.1.5. *Let X be an arbitrary set in a Hausdorff topological vector space E . For every $x \in X$ suppose that there exists a closed set $F(x)$ in E such that $F(x)$ is compact for at least one $x \in X$. If the convex hull of every finite subset $\{x_1, x_2, \dots, x_n\}$ of X (i.e. the smallest convex set that contains $\{x_1, x_2, \dots, x_n\}$) is contained in the corresponding union $\bigcup_{i=1}^n F(x_i)$, then $\bigcap_{x \in X} F(x) \neq \emptyset$.*

Theorem 3.1.6 (Ky Fan Inequality). *Let X be a compact convex set in $\mathbb{R}^n$, and let f be a real-valued function defined on $X \times X$ such that:*

- *for each fixed $x \in X$, $f(x, y)$ is a lower semicontinuous function of y on X ,*
- *for each fixed $y \in X$, $f(x, y)$ is a quasi-concave function of x on X .*

Then the following inequality holds:

$$\min_{y \in X} \sup_{x \in X} f(x, y) \leq \sup_{x \in X} f(x, x).$$

Proof. Observe that $\sup_{x \in X} f(x, y)$ is a lower semicontinuous function of y on X since $f(x, y)$ is, and therefore its minimum $\min_{y \in Y} \sup_{x \in X} f(x, y)$ on the compact set X exists.

Let $\mu = \sup_{x \in X} f(x, x)$. We may assume that $\mu < +\infty$, otherwise is trivial. We want to prove that there is a point $y_0 \in X$ such that $f(x, y_0) \leq \mu$ for all $x \in X$. For each $x \in X$, let $F(x) = \{y \in X \mid f(x, y) \leq \mu\}$, which is a closed subset of X by the lower semicontinuity of f . The existence of a point $y_0 \in X$ satisfying $f(x, y_0) \leq \mu$ for all $x \in X$ is equivalent to the relation $\bigcap_{x \in X} F(x) \neq \emptyset$. According to Lemma 3.1.5, it is sufficient to show that the convex hull of any finite subset $\{x_1, x_2, \dots, x_n\}$ of X is contained in the corresponding union $\bigcup_{i=1}^n F(x_i)$.

Suppose that this is false. Then for some choice of $\{x_1, x_2, \dots, x_n\} \subset X$ and $\alpha_j \geq 0$ for every j such as $1 \leq j \leq n$ with $\sum_{j=1}^n \alpha_j = 1$, we would have $\sum_{j=1}^n \alpha_j x_j \notin \bigcup_{i=1}^n F(x_i)$, i.e. $f(x_i, \sum_{j=1}^n \alpha_j x_j) > \mu$ for $1 \leq i \leq n$. Then, by the quasi-convexity of f it follows that $f(\sum_{j=1}^n \alpha_j x_j, \sum_{j=1}^n \alpha_j x_j) > \mu$, which is absurd. Hence, $\bigcap_{x \in X} F(x) \neq \emptyset$ and the theorem is proved. $\square$

Theorem 3.1.7. Suppose $X \subseteq \mathbb{R}^n$ is a non-empty, compact and convex set. Also suppose that:

- for each fixed $\hat{x} \in X$, $b(\hat{x} \mid x)$ is a lower semicontinuous function of $x \in X$,
- for each fixed $x \in X$, $b(\hat{x} \mid x)$ is a quasi-concave function of $\hat{x} \in X$,
- $b(x \mid x) \leq 0$ for every $x \in X$.

Then there exists at least one stationary state.

Proof. From the Ky Fan Inequality 3.1.6, we have that

$$\min_{y \in X} \sup_{x \in X} b(x \mid y) \leq \sup_{x \in X} b(x \mid x).$$

Equivalently, there exists $\hat{y} \in X$ such that

$$\sup_{x \in X} b(x \mid \hat{y}) \leq \sup_{x \in X} b(x \mid x) \leq 0,$$

so $b(x \mid \hat{y}) \leq 0$ for every $x \in X$, and this prove that $\hat{y}$ is stationary. $\square$

Now, another question arises: how can players reach and maintain a stationary state? To answer this question, we introduce a correspondence $A : X \rightrightarrows X$, defined as

$$A(x) = \{\hat{x} \in X \mid b(\hat{x} \mid x) \geq 0\}. \quad (3.3)$$

Note that $A(x)$ includes every point $\hat{x}$ that enables the player to improve their situation by moving from x to $\hat{x}$. Given a starting point x_0 , let us define the iterative process

$$x^{k+1} \in A(x^k), \quad k = 0, 1, \dots \quad (3.4)$$

We can state the following theorem:

Theorem 3.1.8. *Let $X \subseteq \mathbb{R}^n$ be a closed subset. Suppose that (3.4) generates a summable sequence $k \mapsto b(x^{k+1} \mid x^k) \geq 0$, i.e.*

$$\sum_{k \in \mathbb{N}} b(x^{k+1} \mid x^k) < +\infty.$$

Further suppose that for each non-stationary point $x \in X$ there exists a neighborhood $\mathcal{N}$ of x and a positive number δ such that

$$b(\hat{\chi} \mid \chi) \geq \delta \text{ for all } \hat{\chi} \in A(\chi) \text{ when } \chi \in X \cap \mathcal{N}. \quad (3.5)$$

Then, either the sequence (x^k) is finite with a stationary last point or every cluster point of the infinite sequence must be stationary.

Proof. Let K be an infinite subsequence of $\mathbb{N}$, and $x = \lim_{k \in K} x^k$ be a cluster point. Suppose that x is not stationary. With no loss of generality, take $x^k \in \mathcal{N}$ for all $k \in K$. Then, the contradiction follows:

$$+\infty > \sum_{k=0}^{\infty} b(x^{k+1} \mid x^k) \geq \sum_{k \in K} b(x^{k+1} \mid x^k) = +\infty.$$

The first inequality follows from the summability of the sequence, the second is trivial since we are adding fewer terms, and the last equality holds thanks to (3.5). $\square$

Remark 3.1.9. Is important to note that if the process (3.4) reaches a stationary point x , and if we suppose that

$$\begin{cases} \hat{x} \in A(x) \\ \hat{x} \neq x \end{cases} \implies b(\hat{x} | x) > 0, \quad (3.6)$$

then

$$A(x) = \{x\} \text{ and } b(x | x) = 0.$$

Indeed, if x is stationary we have that $b(\hat{x} | x) \leq 0$ for every $\hat{x} \in X$. This, together with (3.6), implies that

$$\nexists \hat{x} \neq x \text{ such that } \hat{x} \in A(x).$$

Moreover, if $\hat{x} \in A(x)$ is such that $\hat{x} = x$, we have that $b(x | x) \geq 0$; this together with the assumption that x is stationary, implies that $b(x | x) = 0$.

3.2 Non-cooperative games

Let N denote the set of players with $n = \#N \geq 2$, and $X_i \in \mathbb{R}^n$ for every $i \in N$.

Definition 3.2.1. A *strategy profile* $x = (x_i)$ is a map that, for each $i \in N$, associates a strategy $x_i \in X_i \subseteq \mathbb{R}^n$. So, $x \in X = \prod_{i \in I} X_i$. We denote by x_{-i} the profile of strategies played by players in $N \setminus \{i\}$.

Given a strategy profile x , the net benefit of each player i upon deviating from x_i to $\hat{x}_i$ is

$$b_i(\hat{x}_i | x) \in \mathbb{R}, \text{ with } x = (x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n).$$

In this way, each agent assumes that other players will not change their strategy: when a player changes strategy, he acts as if the new profiles of strategies is $(\hat{x}_i, x_{-i})$.

Definition 3.2.2. A strategy profile $x = (x_i) \in X$ is called *multi-players stationary* if

$$x_i \in \operatorname{Argmax}_{\hat{x}_i \in X_i} b_i(\hat{x}_i | x) \quad \forall i \in N. \quad (3.7)$$

If we define b as in (3.1), i.e.

$$b_i(\hat{x}_i \mid x) = \beta_i(\hat{x}_i, x_{-i}) - \beta_i(x_i, x_{-i}),$$

then this definition of stationarity coincides with the concept of Nash equilibrium, as explained in the following theorem:

Theorem 3.2.3. *For every $i \in N$ let $\bar{x} \in X$ fixed, $\beta_i : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty\}$, and suppose that player i aims to maximize*

$$b_i(\hat{x}_i \mid \bar{x}) = \beta_i(\hat{x}_i, \bar{x}_{-i}) - \beta_i(\bar{x}_i, \bar{x}_{-i})$$

with respect to $\hat{x}_i$. Then, a state $x \in X$ is multi-players stationary if and only if it is a Nash equilibrium of the game $G = (\beta, X)$ (where $\beta = \prod_{i=1}^N \beta_i$ and $X = \prod_{i=1}^N X_i$), meaning

$$x_i \in \operatorname{Argmax}_{\hat{x}_i \in X_i} \beta_i(\hat{x}_i, \bar{x}_{-i}) \quad \forall i \in N.$$

Proof. The proof is trivial since $\beta_i(\bar{x}_i, \bar{x}_{-i})$ does not depend on $\hat{x}_i$, so maximizing $b_i(\cdot \mid \bar{x})$ is the same as maximizing $\beta_i(\cdot, \bar{x}_{-i})$. $\square$

Taking into account the cost function, we define b as in (3.2):

$$b_i(\hat{x}_i \mid x) = \beta_i(\hat{x}_i, x_{-i}) - \beta_i(x) - c_i(\hat{x}_i \mid x). \quad (3.8)$$

Note that c_i depends on the entire profile x , so we are taking into account the interactions between players. Moreover, $c(x_i \mid x_i) = 0$, meaning that the cost of change disappears if we reach the equilibrium. Nevertheless, the cost of change influences the player's behavior in any other state.

Proposition 3.2.4. *For each $i \in N$, suppose X_i is a nonempty, compact and convex subset of $\mathbb{R}^n$, and $X = \prod_{i=1}^N X_i$. Further, let $F_i : X_i \times X \Rightarrow \mathbb{R}$ and $f_i : X \rightarrow \mathbb{R}$ be functions such that*

$$F_i(\hat{x}_i \mid x) > 0 \implies f_i(\hat{x}_i, x_{-i}) > f_i(x), \quad (3.9)$$

with f_i concave in $\hat{x}_i \in X_i$ and jointly continuous in $x \in X$. Then, at least one Nash equilibrium exists in the game $G = (f = \prod_{i=1}^N f_i, X)$. Each such equilibrium x is

stationary for

$$F(\hat{x} \mid x) = \sum_{i \in I} [f_i(\hat{x}_i, x_{-i}) - f_i(x)]. \quad (3.10)$$

When moreover, $F_i(x_i \mid x) = 0$ for all $i \in N$, the equilibrium is multi-player stationary.

Proof. Observe that $F(\hat{x} \mid x)$ is concave in $\hat{x}$ (as a sum of concave functions), continuous in x (as a sum of continuous functions), and $F(x \mid x) = 0$ for all $x \in X$ (see the definition of F (3.10)). Then by Theorem 3.1.7, there exists a stationary point $x \in X$, i.e. a point such that $F(\hat{x} \mid x) \leq 0$ for all $\hat{x} \in X$.

Consequently, $f_i(\hat{x}_i, x_{-i}) \leq f_i(x)$ for each $\hat{x}_i \in X_i$ and $i \in N$. Indeed, fix a point $\hat{x} \in X$, then from (3.10) we have that

$$F(\hat{x} \mid x) \leq 0 \implies \sum_{i \in I} [f_i(\hat{x}_i, x_{-i}) - f_i(x)] \leq 0.$$

Suppose that exists $\tilde{i} \in N$ such that

$$f_{\tilde{i}}(\hat{x}_{\tilde{i}}, x_{-\tilde{i}}) - f_{\tilde{i}}(x) > 0,$$

and consider

$$\bar{x} = (x_1, \dots, \hat{x}_{\tilde{i}}, \dots, x_n) \in X.$$

We have that

$$F(\bar{x} \mid x) = \sum_{i \in I} [f_i(\bar{x}_i, x_{-i}) - f_i(x)] = f_{\tilde{i}}(\hat{x}_{\tilde{i}}, x_{-\tilde{i}}) - f_{\tilde{i}}(x) > 0$$

but this is absurd since $F(\hat{x} \mid x) \leq 0$ for every $\hat{x} \in X$, and $\bar{x} \in X$. Given this, the fact that this inequality also holds for every $\hat{x} \in X$ is trivial. So, $x = (x_i)$ is a Nash equilibrium for G as well as a stationary point of F . Moreover, provided $F_i(x_i \mid x) = 0$ for every $i \in N$, condition (3.7) is also satisfied, since for every other point $\hat{x}_i \in X_i$ and for every $i \in N$ we have $f_i(\hat{x}_i, x_{-i}) \leq f_i(x)$, that implies $F(\hat{x} \mid x) \leq 0$ from (3.9). $\square$

Remark 3.2.5. Let us suppose that $c_i(\hat{x}_i \mid x) \geq 0$ is convex with respect to $\hat{x}_i$ and continuous with respect to x . We can define F_i as in (3.8):

$$F_i(\hat{x}_i \mid x) = \beta_i(\hat{x}_i, x_{-i}) - \beta_i(x) - c_i(\hat{x}_i \mid x).$$

If the conditions required for f_i apply to β_i , then the previous theorem remains in force.

The theorems discussed earlier directly lead to the following result:

Theorem 3.2.6. *Suppose $X \subseteq \mathbb{R}^n$ is a non-empty, compact and convex set. We define*

$$b(\hat{x} \mid x) = \sum_{i \in I} b_i(\hat{x}_i \mid x)$$

with the cost of change already considered in $b_i(\hat{x}_i \mid x)$, and suppose that:

- *for each fixed $\hat{x} \in X$, $b(\hat{x} \mid x)$ is a quasi-concave function of $x \in X$,*
- *for each fixed $x \in X$, $b(\hat{x} \mid x)$ is a lower semicontinuous function of $\hat{x} \in X$,*
- *$b(x \mid x) \leq 0$ for every $x \in X$.*

Then there exists a generalized Nash equilibrium.

3.3 Non convex case

In this section, we want to remove the assumption of convexity. We will replace the set A with a stage-dependent correspondences $A^k : X \rightrightarrows X$, to have

$$x^{k+1} \in A^k(x^k), \quad k = 0, 1, \dots \quad (3.11)$$

We would also replace the functions β and c with stage-dependent versions:

$$\begin{aligned} \beta^k &: X \rightarrow \mathbb{R}, \\ c^k &: X \times X \rightarrow \mathbb{R}^+ \cup \{+\infty\}, \end{aligned}$$

and let

$$A^k(x) = \left\{ \hat{x} \in X \mid \beta^{k+1}(\hat{x}) - \beta^k(x) \geq c^k(\hat{x} \mid x) \right\} = \left\{ \hat{x} \in X \mid b^k(\hat{x} \mid x) \geq 0 \right\}, \quad (3.12)$$

where we set

$$b^k(\hat{x} \mid x) = \beta^{k+1}(\hat{x}) - \beta^k(x) - c^k(\hat{x} \mid x).$$

Definition 3.3.1. We say that a limiting correspondence $A : X \rightrightarrows X$ *closes* the sequence (A^k) if

$$(\hat{x}^k, x^k) \rightarrow (\hat{x}, x) \text{ with } \hat{x}^k \in A^k(x^k) \text{ implies } \hat{x} \in A(x). \quad (3.13)$$

The sequence (A^k) is called *asymptotically regular* if $x^{k+1} \in A^k(x^k)$ for every $k \in \mathbb{N}$ implies $\|x^{k+1} - x^k\| \rightarrow 0$.

Let us give some definitions in order to define the *graphical upper limit*:

Definition 3.3.2 (Kuratowski limits). Let $(K_n)_{n \in \mathbb{N}}$ be a sequence of subsets of $\mathbb{R}^n$. We say that the subset

$$\limsup_{n \rightarrow \infty} K_n = \{x \in \mathbb{R}^n \mid \liminf_{n \rightarrow \infty} d(x, K_n) = 0\}$$

is the *Kuratowski upper limit* of the sequence K_n , and that the subset

$$\liminf_{n \rightarrow \infty} K_n = \{x \in \mathbb{R}^n \mid \lim_{n \rightarrow \infty} d(x, K_n) = 0\}$$

is its *Kuratowski lower limit*. A subset K is said to be the *Kuratowski limit* of the sequence K_n if

$$K = \limsup_{n \rightarrow \infty} K_n = \liminf_{n \rightarrow \infty} K_n = \lim_{n \rightarrow \infty} K_n.$$

Definition 3.3.3. Let us consider metric spaces X and Y and a sequence of correspondences $F_n : X \rightrightarrows Y$. The correspondences $\lim_{n \rightarrow \infty}^u F_n$ and $\lim_{n \rightarrow \infty}^l F_n$ from X to Y defined by

$$\begin{aligned} \text{Graph}(\lim_{n \rightarrow \infty}^u F_n) &= \limsup_{n \rightarrow \infty} \text{Graph}(F_n), \\ \text{Graph}(\lim_{n \rightarrow \infty}^l F_n) &= \liminf_{n \rightarrow \infty} \text{Graph}(F_n) \end{aligned}$$

are called the *graphical upper and lower limits* of the correspondence F_n respectively. When the graphical lower and upper limits coincide, it is referred to as the *graphical limit*.

The definitions provided above lead to the following result:

Remark 3.3.4. Condition (3.13) requires that A contains or equals the *graphical upper limit* of (A^k) .

Theorem 3.3.5. *Let $X \subseteq \mathbb{R}^n$ be a non-empty and closed set, and the sequence A^k closed by the correspondence*

$$x \in X \Rightarrow A(x) = \{\hat{x} \in X \mid \beta(\hat{x}) - \beta(x) \geq c(\hat{x} \mid x)\}.$$

Suppose that for any sequence $x^{k+1} \in A^k(x^k)$, $k = 0, 1, \dots$, $\limsup_k \beta^k(x^k) \in \mathbb{R}$ and

$$c^k(x^{k+1} \mid x^k) \geq \delta \|x^{k+1} - x^k\|$$

for some positive δ . Then, (A^k) is asymptotically regular, and any sequence (x^k) generated by (3.11) and (3.12) converges to a stationary point.

Proof. Consider a sequence (x^k) such that $x^{k+1} \in A^k(x^k)$. From the hypothesis

$$\limsup_k \beta^{k+1}(x^{k+1}) - \beta^0(x^0) \in \mathbb{R}.$$

Note that

$$\begin{aligned} \beta^{k+1}(x^{k+1}) - \beta^0(x^0) &= \beta^{k+1}(x^{k+1}) - \beta^k(x^k) + \beta^k(x^k) + \dots - \beta^0(x^0) \\ &= \sum_{n=0}^k (\beta^{n+1}(x^{n+1}) - \beta^n(x^n)) \end{aligned}$$

and $x^{n+1} \in A^n(x^n)$, so that

$$\beta^{n+1}(x^{n+1}) - \beta^n(x^n) \geq c^n(x^{n+1} \mid x^n).$$

Thus we have that

$$\begin{aligned} +\infty &> \limsup_k \beta^{k+1}(x^{k+1}) - \beta^0(x^0) \\ &= \limsup_k (\beta^{k+1}(x^{k+1}) - \beta^0(x^0)) \\ &= \limsup_k \sum_{n=0}^k (\beta^{n+1}(x^{n+1}) - \beta^n(x^n)) \\ &\geq \sum_{n=0}^{\infty} c^n(x^{n+1} \mid x^n) \geq \delta \sum_{n=0}^{\infty} \|x^{n+1} - x^n\|. \end{aligned} \tag{3.14}$$

Implying that

$$\|x^{n+1} - x^n\| \rightarrow 0$$

so (A^k) is asymptotically regular. Moreover, x^k is a Cauchy sequence, and since $\mathbb{R}^n$ is complete, that implies that $x^k \rightarrow x \in X$. Hence, by the closeness, $x \in A(x)$, and stationary obtains. □

The following theorem states the existence of a stationary point:

Theorem 3.3.6. *Let $X \subseteq \mathbb{R}^n$, $A : X \rightrightarrows X$ be a correspondence with non-empty values and $\beta : X \rightarrow \mathbb{R}$ a function bounded above and upper semicontinuous. If, for each $x \in X$ and $\hat{x} \in A(x)$,*

$$\beta(\hat{x}) - \beta(x) \geq \|\hat{x} - x\|, \quad (3.15)$$

then A has a fixed point x at which $A(x)$ reduces to a singleton.

To demonstrate this theorem, we need to recall some results:

Definition 3.3.7. A binary relation $\succsim$ on $S \times S$ defines a *preorder* on a set S if:

- it is reflexive, i.e. $s \succsim s$ for all $s \in S$,
- it is transitive, i.e. $s \succsim t$ and $t \succsim u$ implies $s \succsim u$, for all $s, t, u \in S$.

If, moreover, $\succsim$ is antisymmetric, i.e. $s \succsim u$ and $u \succsim s$ implies $s = u$, we say that $\succsim$ defines a *partial order*.

Definition 3.3.8. Given a set S and a preorder $\succsim$, we say that S is *totally ordered* if, selecting any two points from the set S , they are comparable through $\succsim$. We say that S is *inductively ordered* if every totally ordered subset T of the set of the parts of S admits a maximal element.

Let us recall the following classical result:

Zorn's Lemma 3.3.9. *Let $(S, \succsim)$ be a preorder, and suppose that S is inductively ordered. Then S admits a maximal element, i.e. there exists $\hat{s} \in S$ such that $\hat{s} \succsim s$ for each $s \in S$.*

Proof of Theorem (3.3.6). The binary relation

$$\hat{x} \succsim x \iff \beta(\hat{x}) - \beta(x) \geq \|\hat{x} - x\|$$

defines a partial order on X , indeed it is:

- reflexive:

$$\beta(x) - \beta(x) \geq \|x - x\| \iff 0 \geq 0$$

which is always true,

- transitive: if

$$\beta(\hat{x}) - \beta(x) \geq \|\hat{x} - x\|, \quad \text{and} \quad \beta(x) - \beta(y) \geq \|x - y\|$$

we have that

$$\beta(\hat{x}) - \beta(y) = \beta(\hat{x}) - \beta(x) + \beta(x) - \beta(y) \geq \|\hat{x} - x\| + \|x - y\| \geq \|\hat{x} - y\|,$$

where the last inequality holds thanks to the triangle inequality,

- antisymmetric: by adding up

$$\beta(\hat{x}) - \beta(x) \geq \|\hat{x} - x\|, \quad \text{and} \quad \beta(x) - \beta(\hat{x}) \geq \|x - \hat{x}\|$$

we have

$$0 \geq 2\|x - \hat{x}\| \Rightarrow \|x - \hat{x}\| \leq 0 \Rightarrow x = \hat{x}.$$

Let us consider totally ordered subsets of X . The subsets are ordered with respect to set inclusion, and their elements are ordered with respect to the relation $\succsim$. Since every chain in this collection has an upper bound, by Zorn's lemma, there exists a totally ordered subset $\mathcal{X} \subseteq X$, which is maximal under set inclusion.

Let $\mathcal{X} = \{x_n \mid n \in \mathcal{N}\}$ for some index set $\mathcal{N}$, such as $x_{\hat{n}} \succsim x_n \iff \hat{n} \succsim n$. This means that the set $\{n \in \mathcal{N}\}$ inherit the ordering of the $\{x_n\}$. Then we have:

$$\hat{n} \succsim n \Rightarrow x_{\hat{n}} \succsim x_n \Rightarrow \beta(x_{\hat{n}}) - \beta(x_n) \geq \|x_{\hat{n}} - x_n\| \geq 0 \Rightarrow \beta(x_{\hat{n}}) \geq \beta(x_n).$$

So, the function $n \mapsto \beta(x_n) \in \mathbb{R}$ is increasing. Being bounded from above, $\lim_{n \rightarrow \infty} \beta(x_n) = r \in \mathbb{R}$ is well defined. Consequently, given any $\varepsilon > 0$, there exists $n(\varepsilon) \in \mathcal{N}$ such that

$$n \succsim n(\varepsilon) \implies r - \varepsilon \leq \beta(x_n) \leq r$$

Therefore, if $\hat{n} \succsim n \succsim n(\varepsilon)$, then

$$\varepsilon \geq \beta(x_{\hat{n}}) - \beta(x_n) \geq \|x_{\hat{n}} - x_n\|$$

where the second inequality follows from the definition of $\succsim$, and the first one holds because:

$$\begin{cases} \hat{n} \succsim n(\varepsilon) \implies \beta(x_{\hat{n}}) \leq r \\ n \succsim n(\varepsilon) \implies -\beta(x_n) \leq \varepsilon - r \end{cases} \implies \beta(x_{\hat{n}}) - \beta(x_n) \leq r + \varepsilon - r = \varepsilon.$$

This proves that $(x_n)_{n \in N}$ is a Cauchy sequence; hence, thanks to the completeness of $\mathbb{R}$, $\lim_{n \rightarrow \infty} x_n = x$. From the upper semicontinuity, we get that

$$\begin{aligned} \beta(x) - \beta(x_n) &\geq \limsup_{\hat{n} \rightarrow \infty} \beta(x_{\hat{n}}) - \beta(x_n) \\ &= \limsup_{\hat{n} \rightarrow \infty} (\beta(x_{\hat{n}}) - \beta(x_n)) \\ &\geq \limsup_{\hat{n} \rightarrow \infty} \|x_{\hat{n}} - x_n\| \\ &= \|x - x_n\|, \end{aligned} \tag{3.16}$$

and thereby, $x \succsim x_n$. Since $\mathcal{X}$ is maximal under inclusion, $x \in \mathcal{X}$.

Suppose that there exists $\hat{x} \in A(x)$, $\hat{x} \neq x$; then it follows from (3.15) that $\hat{x} \succsim x$. So, again since $\mathcal{X}$ is maximal under inclusion, $\hat{x} \in \mathcal{X}$. Thereby, $x \succsim \hat{x}$, being $x = \lim_{n \rightarrow \infty} x_n$, and $x_{\hat{n}} \succsim x_n \iff \hat{n} \succsim n$. So we can conclude that $\hat{x} = x$ thanks to the antisymmetric property, and therefore $\{x\} = A(x)$. $\square$

If we consider the model defined in (3.3) and (3.2), from Theorem 3.3.6 we have the following result:

Corollary 3.3.10. *Let $X \subseteq \mathbb{R}^n$, and $\beta : X \rightarrow \mathbb{R}$ bounded above and upper semicontinuous. For each $x \in X$, suppose that the set*

$$A(x) = \{\hat{x} \in X \mid b(\hat{x} \mid x) \geq 0\} = \{\hat{x} \in X \mid \beta(\hat{x}) - \beta(x) \geq c(\hat{x} \mid x)\}$$

is non-empty, with $c(\cdot \mid x)$ upper semicontinuous and $c(\cdot \mid x) \geq \|\cdot - x\|$ for every $x \in X$. Then, there exists a stationary point x at which

$$\{x\} = A(x).$$

3.4 Stackelberg games

In Section 3.2, we discussed games in which players make simultaneous moves. Now, we will focus on analyzing Stackelberg's games involving two players: the leader and the follower. Players alternate moves and have, respectively, payoff functions $L(x, y)$ and $F(x, y)$. Let us recall what the players do: the follower chooses

$$y \in M(x) = \operatorname{Arg} \max_{\hat{y} \in Y} F(x, \hat{y}),$$

and the leader chooses

$$x \in \operatorname{Arg} \max_{\hat{x} \in X} L(\hat{x}, M(\hat{x})).$$

In our model, we assume that they are evaluating some conditional payoff functions:

$$\begin{aligned} (x, y) &\mapsto L^k(x \mid x^k, y) \leq L(x, y), \\ (x, y) &\mapsto F^k(y \mid x, y^k) \leq F(x, y). \end{aligned} \tag{3.17}$$

The functions L^k and F^k approximate their real payoffs, but they also respect the following features:

- players who choose to change their previous decision will have to pay a cost,
- β_i^k worth less than β_i .

Suppose also that

$$(x^k, y^k) \rightarrow (x, y) \implies \limsup_k L^k(\hat{x} \mid x^k, y^k) \geq \liminf_k L(\hat{x}, y^k) \forall \hat{x} \in X, \tag{3.18}$$

and

$$\begin{cases} y^k \rightarrow y \\ (\hat{x}^k, \hat{y}^k) \rightarrow (\hat{x}, \hat{y}) \end{cases} \implies \limsup_k F^k(\hat{y}^k \mid \hat{x}^k, y^k) \geq \liminf_k F(\hat{x}^k, \hat{y}^k). \tag{3.19}$$

Conditions (3.18) and (3.19) mean that, asymptotically, the cost of making changes does not impact the player's payoff.

Remark 3.4.1. For example, we can consider

$$\begin{aligned} L^k(x | x^k, y^k) &= L(x, y^k) - c_x^k(x | x^k), \\ F^k(y | x^k, y^k) &= F(x^k, y) - c_y^k(y | y^k), \end{aligned}$$

with $c_x^k \geq 0$, $c_y^k \geq 0$, and (3.17) holds. Moreover, (3.18) and (3.19) holds when $c_x^k \rightarrow 0^+$ and $c_y^k \rightarrow 0^+$ for $k \rightarrow \infty$.

So, at stage k of the game, the leader tries to predict the response of the follower $r^k : X \rightarrow Y$, expecting that:

$$F^k(r^k(x) | x, y) \geq \sup_{\hat{y} \in Y} F^k(\hat{y} | x, y) - \varepsilon^k, \text{ with } \varepsilon^k \rightarrow 0^+. \quad (3.20)$$

It is important to note that we are assuming that r^k is single-valued and that $r^k(x)$ does not necessarily lie in $M(x)$. So the leader chooses

$$x^{k+1} \in A_x^k(x^k) = \text{Arg max}_{\hat{x} \in X} L^k(\hat{x} | x^k, r^k(\hat{x})). \quad (3.21)$$

After observing x^{k+1} , the follower chooses

$$y^{k+1} \in A_y^k(x^{k+1}, y^k) = \text{Arg max}_{\hat{y} \in Y} F^k(\hat{y} | x^{k+1}, y^k). \quad (3.22)$$

Note that our goal is to achieve a Nash equilibrium in a single stage, rather than an equilibrium in the stage game as a whole.

According to the following result, the Stackelberg game we described will converge. In particular, we also provide proof of the existence of an equilibrium in games of this nature.

Theorem 3.4.2. *Let hypotheses (3.17)-(3.20) be strengthened. Suppose that both functions $(x, y) \in X \times Y \mapsto L(x, y)$ or $(x, y) \in X \times Y \mapsto F(x, y)$ are upper semicontinuous, and that the leader's objective $L(x, y)$ is lower semicontinuous w.r.t. $y \in Y$. Suppose in addition that for any point $\chi = (\chi_1, \chi_{-1}) \in X \times Y$ and sequence $\chi_1^k \in X$, such that $\chi_1^k \rightarrow \chi_1$, there exists a sequence $\{\chi_{-1}^k\} \subseteq Y$, with $\chi_{-1}^k \rightarrow \chi_{-1}$ such that $\chi^k = (\chi_1^k, \chi_{-1}^k)$ yields*

$$\liminf_{k \rightarrow \infty} F(\chi^k) \geq F(\chi). \quad (3.23)$$

If r^k converges continuously to some $r : X \rightarrow Y$, meaning that

$$x^k \rightarrow x \implies r^k(x^k) \rightarrow r(x), \quad (3.24)$$

then for each limit $x = \lim_{k \rightarrow \infty} x^k \in X$ of the leader's play (3.21) it holds that

$$x \in \text{Arg max } L(\cdot, r(\cdot)) \text{ and } y = r(x) \in \text{Arg max } F(x, \cdot) = M(x). \quad (3.25)$$

Furthermore, if $\|x_1^{k+1} - x_1^k\| \rightarrow 0$, then each cluster point x of (x^k) satisfies (3.25).

Proof. By definition of the game, the leader chooses $x^{k+1} \in X$ at stage k .

Suppose $x = \lim_{k \rightarrow \infty} x^k$ and define $y = \lim_{k \rightarrow \infty} r^k(x^{k+1})$. Then, by the continuous convergence of r^k ,

$$y = \lim_{k \rightarrow \infty} r^k(x^{k+1}) = \lim_{k \rightarrow \infty} r^k(x^k) = r(x)$$

is well-defined. Moreover, for any $\hat{x} \in X$ it holds that

$$\begin{aligned} L(x, y) &\geq \limsup_{k \rightarrow \infty} L(x^{k+1}, r^k(x^{k+1})) \\ &\geq \limsup_{k \rightarrow \infty} L^k(x^{k+1} \mid x^k, r^k(x^{k+1})) \\ &\geq \limsup_{k \rightarrow \infty} L^k(\hat{x} \mid x^k, r^k(\hat{x})) \\ &\geq \liminf_{k \rightarrow \infty} L(\hat{x}, r^k(\hat{x})) \\ &\geq L(\hat{x}, r(\hat{x})). \end{aligned}$$

The first inequality derives from the upper semicontinuity of L , the second from (3.17), the third from the definition of A_x^k (3.21), and the second-to-last from (3.18). Finally, the last follows from the lower semicontinuity of $L(\hat{x}, \cdot)$ in Y and (3.24). Thus, $x \in \text{Arg max } L(\cdot, r(\cdot))$.

Further, for the same sequence $x^k \rightarrow x$ and any $\chi_{-1} \in Y$ there exists a sequence $\chi_{-1}^k \rightarrow \chi_{-1}$ such that (3.23) holds with $(x^k, \chi_{-1}^k) \rightarrow (x, \chi_{-1})$. So,

$$\begin{aligned}
F(x, y) &\geq \limsup_{k \rightarrow \infty} F\left(x^{k+1}, r^k\left(x^{k+1}\right)\right) \\
&\geq \limsup_{k \rightarrow \infty} F^k\left(r^k\left(x^{k+1}\right) \mid x^{k+1}, y^k\right) \\
&\geq \limsup_{k \rightarrow \infty} \left[F^k\left(\chi_{-1}^{k+1} \mid x^{k+1}, y^k\right) - \varepsilon^k\right] \\
&\geq \liminf_{k \rightarrow \infty} F\left(x^{k+1}, \chi_{-1}^{k+1}\right) \\
&\geq F\left(x, \chi_{-1}\right).
\end{aligned}$$

The first inequality derives from the upper semicontinuity of F , the second from (3.17), the third one from (3.20), with $\varepsilon^k \rightarrow 0$, and the second-to-last from (3.19). Finally, the last follows from (3.23). Thus, $y \in \text{Arg max } F(x, \cdot)$, and the proof is complete. $\square$

There are still some questions about this model that have not been answered yet:

- how do we select the correct L^k and F^k ?
- what learning scheme should we use to determine our choice of r^k ?
- when will these functions converge continuously?

One possible solution for a game with a continuous action space is the proximal point procedure, which can be found in Chapter 5.

Chapter 4

Proximal algorithms

4.1 Definition and interpretations

Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a closed and proper convex function.

Definition 4.1.1. The *proximal operator* $\text{prox}_f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of f is given by

$$\text{prox}_f(v) = \text{Arg min}_x \left(f(x) + \frac{1}{2} \|x - v\|^2 \right).$$

Likewise we can consider the scaled function λf and define the *proximal operator* with parameter $\lambda > 0$:

$$\text{prox}_{\lambda f}(v) = \text{Arg min}_x \left(f(x) + \frac{1}{2\lambda} \|x - v\|^2 \right).$$

It is important to note that the function $g_v(x) = f(x) + \frac{1}{2} \|x - v\|^2$, where $v \in \mathbb{R}^n$ is fixed, admits a unique minimizer. To do this, let us recall the following results:

Theorem 4.1.2. Let f be a proper, convex, lower-semicontinuous function defined on $\mathbb{R}^n$. Then there exists $\hat{x}_0, x_1 \in X$ such that $f(\hat{x}_0)$ is finite and

$$f(x) \geq f(\hat{x}_0) + \langle x - \hat{x}_0, x_1 \rangle, \quad x \in X.$$

Corollary 4.1.3. Let f be a proper, convex, lower-semicontinuous function defined on $\mathbb{R}^n$. If f is strongly convex, then f is coercive.

Proof. Consider the convex function $g(x) = f(x) - \frac{\nu}{2}\|x\|^2$, which is proper and lower semicontinuous. Apply the preceding theorem and then substitute the definition of g to get

$$f(x) \geq f(\hat{x}_0) + \frac{\nu}{2} [\|x\|^2 - \|\hat{x}_0\|^2] + \langle x - \hat{x}_0, x_1 \rangle.$$

By the Cauchy-Schwartz inequality, $\langle x, x_1 \rangle \geq -\|x\|\|x_1\|$, and so

$$f(x) \geq f(\hat{x}_0) + \frac{\nu}{2} [\|x\|^2 - \|\hat{x}_0\|^2] - \|x\|\|x_1\|,$$

which tends to infinity as $\|x\| \rightarrow \infty$. □

Remark 4.1.4. Note that g_v is strongly convex as a sum of a convex and a strongly convex function. Additionally, it is a proper function because f is proper. Furthermore, the closure of g_v follows from the closure of f , which implies that g_v is lower-semicontinuous, as stated in Remark 1.2.12. As a result, Corollary 4.1.3 holds, so g_v is coercive. Consequently, we can deduce that Theorem 1.2.13 holds, leading to the conclusion that the function admits a global minimum. Moreover, it's worth noting that g_v is strictly convex, as it is the sum of a convex and a strictly convex function. This ensures that the global minimum is unique.

The following remark will be essential in Chapter 5.

Remark 4.1.5. If we have a concave function, we can define the proximal operator as follows:

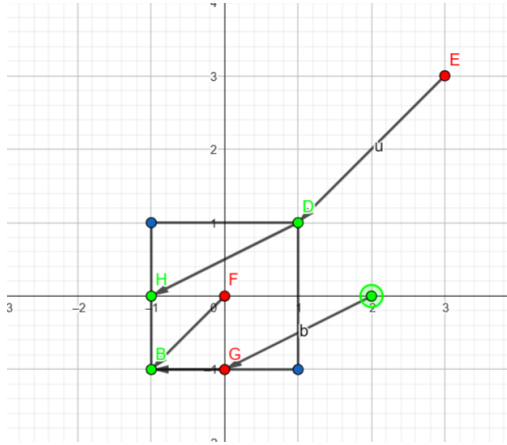
$$\text{prox}_f(v) = \text{Arg max}_x \left(f(x) - \frac{1}{2}\|x - v\|^2 \right),$$

and the previous observations remain valid.

Proximal algorithms are used to solve convex optimization problems by evaluating the proximal operator of the objective function. They have many advantages:

- they work without restrictive conditions: even when dealing with non-smooth and extended real-valued inputs,
- in some instances, functions that are challenging to handle have elementary proximal operators. Consequently, employing proximal algorithms can lead to faster and more efficient solutions compared to traditional methods,

- they are both conceptually and mathematically simple, which makes them easy to implement.



(a) Evaluating proximal operator

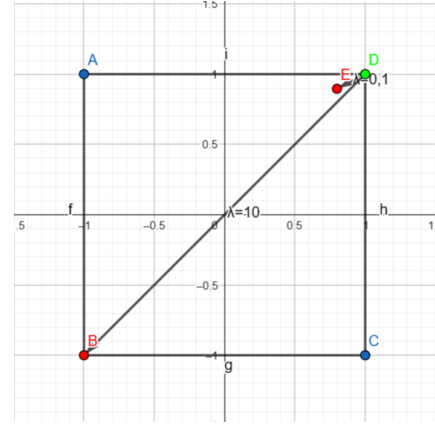
(b) Proximal Operator depends on λ

Figure (4.1(a)) illustrates how the proximal operator works. Our goal is to minimize the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$f(x, y) = 2x + y,$$

over the domain $[-1, 1] \times [-1, 1]$.

From Definition 4.1.1, we observe that $\text{prox}_f(v)$ represents a point that strikes a compromise between minimizing f and being close to v . Let us mention that if v lies within the domain, $\text{prox}_f(v)$ remains within the domain. However, if v is outside the domain, $\text{prox}_f(v)$ will move to the boundary of the domain. In both scenarios, $\text{prox}_f(v)$ tends to approach the minimum of f , which is $(-1, -1)$.

As illustrated in Figure (4.1(b)), the parameter λ governs the speed at which the point moves towards the minimum: a smaller value of λ leads to smaller movements, and vice versa. Therefore, λ acts as a weight, determining the trade-off between minimizing the objective function and being close to v .

Proximal operators can be interpreted as a generalized projection: let us consider

the *indicator function*

$$I_C(x) = \begin{cases} 0 & x \in C \\ +\infty & x \notin C. \end{cases}$$

The proximal operator of the indicator function is

$$\text{prox}_{I_C}(v) = \underset{x \in C}{\text{Arg min}} \|x - v\|,$$

which coincides with the Euclidean projection onto C .

4.2 Fixed points

Let us introduce the definition of a fixed point and analyze the link between the fixed points of the proximal operator of a function and the minimizers of these functions.

Definition 4.2.1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a function. We say that $x \in \mathbb{R}^n$ is a *fixed point* of f if $f(x) = x$. We define the set of the fixed points of f as

$$\text{Fix } f = \{x \in \mathbb{R}^n \mid f(x) = x\}.$$

Theorem 4.2.2. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ be a closed, proper, and convex function. Then:

$$\text{prox}_{\lambda f}(v) = v \iff v \text{ minimizes } \lambda f,$$

i.e. the fixed points of the proximal operator of λf coincides with the minimizers of λf .

Proof. Since $\lambda > 0$, the minimizers of f coincide with the minimizers of λf so, without loss of generality, we can consider $\lambda = 1$. Let us suppose that f is subdifferentiable on its domain, though the result is true in general. If x^* minimizes f , then

$$f(x) \geq f(x^*) \text{ for any } x \in \text{Dom } f,$$

so

$$f(x) + \frac{1}{2}\|x - x^*\|^2 \geq f(x^*) = f(x^*) + \frac{1}{2}\|x^* - x^*\|^2.$$

The first inequality applies because $\frac{1}{2}\|x - x^*\|^2 \geq 0$, while the second equality because $\frac{1}{2}\|x^* - x^*\|^2 = 0$. So x^* minimizes also $f(x) + \frac{1}{2}\|x - x^*\|^2$, and it follows that $x^* = \text{prox}_f(x^*)$.

To show the converse, we use the subdifferential characterization of the minimum of a convex function (Theorem 1.4.4): $\tilde{x}$ minimizes

$$f(x) + \frac{1}{2}\|x - x^*\|^2$$

if and only if

$$\begin{aligned} 0 &\in \partial(f(\tilde{x}) + \frac{1}{2}\|\tilde{x} - x^*\|^2) \\ &= \partial f(\tilde{x}) + \partial(\frac{1}{2}\|\tilde{x} - x^*\|^2) \\ &= \partial f(\tilde{x}) + \nabla(\frac{1}{2}\|\tilde{x} - x^*\|^2) \\ &= \partial f(\tilde{x}) + (\tilde{x} - x^*). \end{aligned}$$

The first equality follows from Theorem 1.4.5, taking

$$\begin{aligned} G(x) &= f(x), \\ F(x) &= \frac{1}{2}\|x - x^*\|^2, \\ \bar{x} &\in \text{Dom } f, \end{aligned}$$

and the second one follows from Theorem 1.4.6 because $\frac{1}{2}\|x - x^*\|^2$ is Gateaux differentiable. Taking $\tilde{x} = x^*$, it follows that $0 \in \partial f(x^*)$, so x^* minimizes f . $\square$

Therefore, we can minimize f by finding the fixed points of its proximal operator. If prox_f were a *contraction* (i.e. Lipschitz continuous with constant less than 1), it would possess a unique fixed point, thanks to the well-known

Banach-Caccioppoli Fixed Point Theorem 4.2.3. *Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a contraction on $\mathbb{R}^n$. Then T admits a unique fixed point $x^* \in \mathbb{R}^n$.*

Unfortunately, prox_f need not be a contraction, but it does enjoy another property, called *firm nonexpansiveness*:

Theorem 4.2.4. $\|\text{prox}_f(x) - \text{prox}_f(y)\|^2 \leq (x - y)^T(\text{prox}_f(x) - \text{prox}_f(y))$ for every $x \in \mathbb{R}^n, y \in \mathbb{R}^n$.

Proof. Set

$$u_1 = \text{prox}_f(x),$$

$$u_2 = \text{prox}_f(y).$$

From the definition of prox_f we have that

$$u_1 = \underset{x}{\text{Arg min}} \left(f(x) + \frac{1}{2} \|x - v\|^2 \right)$$

so, thanks to Theorems 1.4.4, 1.4.5 and 1.4.6

$$0 \in \partial f(u_1) + u_1 - x \Rightarrow x - u_1 \in \partial f(u_1).$$

Similarly,

$$y - u_2 \in \partial f(u_2).$$

Taking into account that ∂f is monotone (Theorem 1.4.7), we obtain that

$$\langle x - u_1 - (y - u_2), u_1 - u_2 \rangle \geq 0 \quad \Rightarrow \quad \langle x - y, u_1 - u_2 \rangle \geq \|u_1 - u_2\|_2^2.$$

□

Firmly nonexpansive operators are a special case of *nonexpansive operators* (i.e. Lipschitz continuous with constant 1). If we consider a nonexpansive operator, it is not guaranteed that the sequence obtained by iteratively applying it will converge to a fixed point in general, as shown in the following example:

Example 4.2.5. Let us consider the rotation by θ

$$R_\theta = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

Let us consider the sequence $\{x_k\}_{k \in \mathbb{R}} \in \mathbb{R}^2$ defined by

$$x_0 = (\bar{x}, \bar{y}),$$

$$x_{k+1} = R_\theta(x_k).$$

This sequence trivially does not converge, but R_θ is a nonexpansive operator:

$$\begin{aligned}
\left\| R_\theta \begin{bmatrix} x \\ y \end{bmatrix} - R_\theta \begin{bmatrix} z \\ t \end{bmatrix} \right\|^2 &= \left\| \begin{bmatrix} x \cos(\theta) - y \sin(\theta) \\ x \sin(\theta) + y \cos(\theta) \end{bmatrix} - \begin{bmatrix} z \cos(\theta) - t \sin(\theta) \\ z \sin(\theta) + t \cos(\theta) \end{bmatrix} \right\|^2 \\
&= \left\| \begin{bmatrix} \cos(\theta)(x - z) - \sin(\theta)(y - t) \\ \sin(\theta)(x - z) + \cos(\theta)(y - t) \end{bmatrix} \right\|^2 \\
&= \cos(\theta)^2(x - z)^2 + \sin(\theta)^2(y - t)^2 \\
&\quad + \sin(\theta)^2(x - z)^2 + \cos(\theta)^2(y - t)^2 \\
&= (x - z)^2 + (y - t)^2 \\
&= \left\| \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} z \\ t \end{bmatrix} \right\|^2.
\end{aligned}$$

Hence, it becomes necessary to find an alternative method to ensure that the sequence obtained by iteratively applying the proximal operator will converge to a fixed point. To achieve this, we will use the firm nonexpansiveness of the proximal operator.

Given a nonexpansive operator N and a number $\alpha \in (0, 1)$, we can define the operator

$$T = (1 - \alpha)I + \alpha N. \quad (4.1)$$

Remark 4.2.6. The operator T shares the same fixed points as the operator N :

$$\begin{aligned}
T(x) = x &\iff (1 - \alpha)x + \alpha N(x) = x \\
&\iff x - \alpha x + \alpha N(x) = x \\
&\iff \alpha(N(x) - x) = 0 \\
&\iff N(x) = x.
\end{aligned} \quad (4.2)$$

Operators of the form (4.1) are called α -averaged operators.

Theorem 4.2.7. *T is a firmly nonexpansive operator if and only if it is $\frac{1}{2}$ -averaged.*

To demonstrate this theorem, we need to restate the definition of firmly nonexpansive operators:

Definition 4.2.8. T is firmly nonexpansive if and only if

$$\|T(x) - T(y)\|^2 + \|(I - T)(x) - (I - T)(y)\|^2 \leq \|x - y\|^2. \quad (4.3)$$

Proof.

$$\begin{aligned} \|(I - T)(x) - (I - T)(y)\|^2 &= \|x - T(x) - y + T(y)\|^2 \\ &= \|x - y\|^2 + \|T(x) - T(y)\|^2 - 2(x - y)^T(T(x) - T(y)). \end{aligned}$$

If we write the second term of (4.3) in this way we obtain

$$\begin{aligned} \|T(x) - T(y)\|^2 + \|x - y\|^2 + \|T(x) - T(y)\|^2 - 2(x - y)^T(T(x) - T(y)) &\leq \|x - y\|^2 \\ \Rightarrow \|T(x) - T(y)\|^2 &\leq (x - y)^T(T(x) - T(y)), \end{aligned}$$

which is the classical definition of a firmly nonexpansive operator. $\square$

Now, let us restate Theorem 4.2.7:

Theorem 4.2.9. T is $\frac{1}{2}$ -averaged if and only if

$$\|T(x) - T(y)\|^2 \leq \|x - y\|^2 - \|(I - T)(x) - (I - T)(y)\|^2.$$

In order to demonstrate this theorem, let us recall the following lemma (see Corollary 2.14 from [3]):

Lemma 4.2.10. Given $\alpha \in \mathbb{R}$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}^n$, the following equation holds:

$$\|\alpha x + (1 - \alpha)y\|^2 + \alpha(1 - \alpha)\|x - y\|^2 = \alpha\|x\|^2 + (1 - \alpha)\|y\|^2.$$

Proof of Theorem 4.2.9. We define $T = \frac{1}{2}I + \frac{1}{2}R \Rightarrow R = 2T - I$. It is sufficient to prove that T is firmly nonexpansive if and only if R is nonexpansive. We have that

$$\|R(x) - R(y)\|^2 = \|2T(x) - x - 2T(y) + y\|^2 = \|2(T(x) - T(y)) - (x - y)\|^2.$$

Using Lemma 4.2.10, we obtain

$$\begin{aligned}
& \|2(T(x) - T(y)) - (x - y)\|^2 \\
&= 2\|T(x) - T(y)\|^2 + (1 - 2)\|x - y\|^2 + 2\|T(x) - T(y) - (x - y)\|^2 \\
&\Rightarrow \|x - y\|^2 - \|R(x) - R(y)\|^2 \\
&= 2(-\|T(x) - T(y)\|^2 + \|x - y\|^2 - \|T(x) - T(y) - (x - y)\|^2) \\
&\Rightarrow \frac{1}{2}(\|x - y\|^2 - \|R(x) - R(y)\|^2) \\
&= \|x - y\|^2 - \|T(x) - T(y)\|^2 - \|(T - I)(x) - (T - I)(y)\|^2.
\end{aligned}$$

R is nonexpansive if and only if the second term is greater than zero, namely if T is firmly nonexpansive. $\square$

Let us state the *Krasnoselskii-Mann Theorem*. We will use this result to demonstrate that if we define

$$x^{k+1} = \text{prox}_{\lambda f}(x^k),$$

this sequence converges to a fixed point of prox_f , that is a minimizer of f .

To demonstrate this theorem we will need the following two results, respectively Corollary 4.18 and Lemma 5.5 from [3]:

Lemma 4.2.11. *Let D be a nonempty closed and convex subset of $\mathbb{R}^n$, let $T : D \rightarrow \mathbb{R}^n$ be nonexpansive, let $(x_n)_{n \in \mathbb{N}}$ be a sequence in D , and let x be a point in $\mathbb{R}^n$. Suppose that $x_n \rightarrow x$ and that $(x_n - T(x_n)) \rightarrow 0$. Then $x \in \text{Fix}(T)$.*

Lemma 4.2.12. *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$ and let C be a nonempty subset of $\mathbb{R}^n$. Suppose that $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C , namely, for all $x \in C$, for all $n \in \mathbb{N}$*

$$\|x_{n+1} - x\| \leq \|x_n - x\|.$$

Then $(x_n)_{n \in \mathbb{N}}$ converges to a point in C .

Theorem 4.2.13. (*Krasnoselskii-Mann Theorem*) *Let D be a nonempty, closed and convex subset of $\mathbb{R}^n$, let $N : D \rightarrow D$ be a nonexpansive operator that has at least one fixed point, let $(\lambda_n)_{n \in \mathbb{N}}$ be a sequence in $[0, 1]$ such that*

$$\sum_{n \in \mathbb{N}} \lambda_n (1 - \lambda_n) = +\infty,$$

and let $x_0 \in D$. Set, $\forall n \in \mathbb{N}$

$$x_{n+1} = x_n + \lambda_n(N(x_n) - x_n). \quad (4.4)$$

Then the following hold:

- $(N(x_n) - x_n)_{n \in \mathbb{N}}$ converges to 0,
- $(x_n)_{n \in \mathbb{N}}$ converges to a fixed point of N .

Proof. Since $x_0 \in D$ and D is convex, (4.4) produces a well-defined sequence in D .

Let $n \in \mathbb{N}$ and $y \in \text{Fix}(N)$; from the definition of x_{n+1} , we have that:

$$\|x_{n+1} - y\|^2 = \|x_n + \lambda_n N(x_n) - \lambda_n x_n - y\|^2.$$

By writing $y = (1 - \lambda_n)y + \lambda_n y$ we obtain

$$\begin{aligned} \|x_n + \lambda_n N(x_n) - \lambda_n x_n - y\|^2 &= \|(1 - \lambda_n)x_n + \lambda_n N(x_n) - (1 - \lambda_n)y - \lambda_n y\|^2 \\ &= \|(1 - \lambda_n)(x_n - y) + \lambda_n(N(x_n) - y)\|^2. \end{aligned}$$

From Lemma 4.2.10 it follows that:

$$\begin{aligned} &\|(1 - \lambda_n)(x_n - y) + \lambda_n(N(x_n) - y)\|^2 \\ &= (1 - \lambda_n) \|x_n - y\|^2 + \lambda_n \|N(x_n) - y\|^2 - \lambda_n(1 - \lambda_n) \|N(x_n) - x_n\|^2. \end{aligned}$$

Since $y \in \text{Fix}(N)$ and N is nonexpansive,

$$\begin{aligned} &(1 - \lambda_n) \|x_n - y\|^2 + \lambda_n \|N(x_n) - N(y)\|^2 - \lambda_n(1 - \lambda_n) \|N(x_n) - x_n\|^2 \\ &\leq \|x_n - y\|^2 - \lambda_n(1 - \lambda_n) \|N(x_n) - x_n\|^2, \end{aligned}$$

therefore

$$\|x_{n+1} - y\|^2 \leq \|x_n - y\|^2 - \lambda_n(1 - \lambda_n) \|N(x_n) - x_n\|^2. \quad (4.5)$$

We derive from (4.5) that

$$\sum_{n \in \mathbb{N}} \lambda_n(1 - \lambda_n) \|N(x_n) - x_n\|^2 \leq \|x_0 - y\|^2,$$

and since

$$\sum_{n \in \mathbb{N}} \lambda_n(1 - \lambda_n) = +\infty$$

we have that

$$\liminf_{n \rightarrow \infty} \|N(x_n) - x_n\| = 0. \quad (4.6)$$

However, for every $n \in \mathbb{N}$

$$\|N(x_{n+1}) - x_{n+1}\| = \|N(x_{n+1}) - x_n - \lambda_n N(x_n) + \lambda_n x_n\|.$$

By adding and removing $(1 - \lambda_n)N(x_n)$ we have

$$\begin{aligned} \|N(x_{n+1}) - x_n - \lambda_n N(x_n) + \lambda_n x_n\| &= \|N(x_{n+1}) + (1 - \lambda_n)(N(x_n) - x_n) - N(x_n)\| \\ &\leq \|x_{n+1} - x_n\| + (1 - \lambda_n) \|N(x_n) - x_n\|, \end{aligned} \quad (4.7)$$

where the last inequality holds because of the nonexpansiveness of N and the triangle inequality. If we write

$$\lambda_n \|N(x_n) - x_n\| = \|x_n + \lambda_n(N(x_n) - x_n) - x_n\| = \|x_{n+1} - x_n\|$$

in (4.7), we can finally conclude that

$$\|N(x_{n+1}) - x_{n+1}\| \leq \|N(x_n) - x_n\|.$$

Consequently, $\|N(x_n) - x_n\|$ converges, and, taking into account (4.6), we must have $(N(x_n) - x_n) \rightarrow 0$. Then it follows from Corollary 4.2.11 that $x \in \text{Fix}(N)$. In view of Theorem 4.2.12 (where $C = \text{Fix}(T)$, and (x_n) is Fejér monotone thanks to (4.5)), the proof is complete. $\square$

From Theorem 4.2.4 we have that prox_f is firmly nonexpansive, so, from Theorem 4.2.7 there exists N nonexpansive such that $\text{prox}_f = \frac{1}{2}I + \frac{1}{2}N$. Define

$$x_{n+1} = x_n + \frac{1}{2}(N(x_n) - x_n) = \frac{1}{2}x_n + \frac{1}{2}N(x_n) = \text{prox}_f(x_n).$$

Thanks to Theorem 4.2.13, we have that x_n converges to a fixed point of N , which is also a fixed point of prox_f .

This suggests the simplest proximal method:

$$x^{k+1} = \text{prox}_{\lambda f}(x^k).$$

If f has a minimum, then x^k converges to the set of minimizers of f , and $f(x^k)$ converges to its optimal value.

Chapter 5

A learning approach via Cost-to-Move

This chapter aims to define a method to approach SPNEs in scenarios where the follower's best reply is not uniquely determined. The idea is to consider perturbed games denoted as $(\Gamma_n)_n$, in which the follower's best reply correspondence becomes single-valued. To generate this sequence, we introduce modifications to the payoff functions of both players. Specifically, we subtract a term from each player's payoff function, representing the physical and behavioral costs they incur when deviating from their current actions.

5.1 Γ -convergence

Let X be a topological space (in our case, $X = \mathbb{R}^n$), and $F_h : X \rightarrow \bar{\mathbb{R}}$ a sequence of functions. Let us introduce the Γ -limits of a sequence of functions (F_h) .

Definition 5.1.1. The Γ -lower limit and the Γ -upper limit of the sequence (F_h) are the functions from X to $\bar{\mathbb{R}}$ defined by

$$\begin{aligned}(\Gamma - \liminf_{h \rightarrow \infty} F_h)(x) &= \sup_{U \in N(x)} \liminf_{h \rightarrow \infty} \inf_{y \in U} F_h(y), \\ (\Gamma - \limsup_{h \rightarrow \infty} F_h)(x) &= \sup_{U \in N(x)} \limsup_{h \rightarrow \infty} \inf_{y \in U} F_h(y),\end{aligned}$$

where $N(x)$ is the set of all open neighbourhoods of x in X (i.e. the set of all the open subset V of X such that $x \in V$). If there exists a function $F : X \rightarrow \bar{\mathbb{R}}$ such that

$$\Gamma - \liminf_{h \rightarrow \infty} F_h = \Gamma - \limsup_{h \rightarrow \infty} F_h = F,$$

then we write

$$F = \Gamma - \lim_{h \rightarrow \infty} F_h,$$

and we say that the sequence (F_h) Γ -converges to F in X .

Remark 5.1.2. It is clear that

$$\Gamma - \liminf_{h \rightarrow \infty} F_h \leq \Gamma - \limsup_{h \rightarrow \infty} F_h,$$

hence (F_h) Γ -converges to F if and only if

$$\Gamma - \liminf_{h \rightarrow \infty} F_h \geq F \geq \Gamma - \limsup_{h \rightarrow \infty} F_h$$

for every $x \in X$.

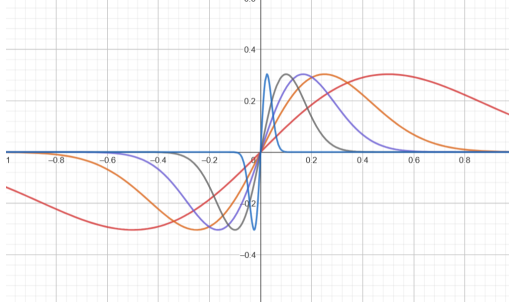
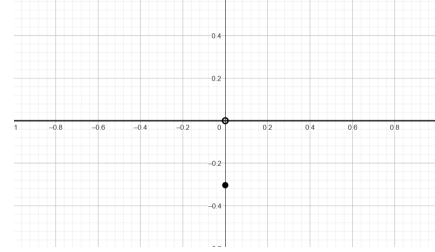
Remark 5.1.3. If $B(x)$ is a base for the neighbourhood system of x in X , then

$$\begin{aligned} (\Gamma - \liminf_{h \rightarrow \infty} F_h)(x) &= \sup_{U \in B(x)} \liminf_{h \rightarrow \infty} \inf_{y \in U} F_h(y), \\ (\Gamma - \limsup_{h \rightarrow \infty} F_h)(x) &= \sup_{U \in B(x)} \limsup_{h \rightarrow \infty} \inf_{y \in U} F_h(y). \end{aligned}$$

This means that if two sequences coincide on an open subset U of X , then their Γ -lower limits and their Γ -upper limits coincide on U .

Example 5.1.4. Let us take $X = \mathbb{R}$ and $F_h(x) = hxe^{-2h^2x^2}$. Then (F_h) Γ -converges to the function

$$F(x) = \begin{cases} -\frac{1}{2}e^{-\frac{1}{2}} & \text{if } x = 0 \\ 0 & \text{if } x \neq 0. \end{cases}$$

(a) $F_h(x)$, as h varies.(b) $F(x)$.

Remark 5.1.5. If the functions (F_h) are independent of x , i.e., for every $h \in \mathbb{N}$ there exists a constant $a_h \in \bar{\mathbb{R}}$ such that $F_h(x) = a_h$ for every $x \in X$, then

$$(\Gamma - \liminf_{h \rightarrow \infty} F_h)(x) = \liminf_{h \rightarrow \infty} a_h \quad \text{and} \quad (\Gamma - \limsup_{h \rightarrow \infty} F_h)(x) = \limsup_{h \rightarrow \infty} a_h$$

for every $x \in X$. If the functions $F_h(x)$ are independent on h , i.e., there exists $F : X \rightarrow \bar{\mathbb{R}}$ such that $F_h(x) = F(x)$ for every $x \in X$ and for every $h \in \mathbb{N}$, then

$$\Gamma - \liminf_{h \rightarrow \infty} F_h = \Gamma - \limsup_{h \rightarrow \infty} F_h = \sup_{U \in N(x)} \inf_{y \in U} F(y)$$

for every $x \in X$.

Remark 5.1.6. The open sets of $\bar{\mathbb{R}}$ consist of the open sets of $\mathbb{R}$ along with intervals such as $[-\infty, a)$, $(a, +\infty]$, and $[-\infty, +\infty]$.

Definition 5.1.7. The sequence (F_h) is *continuously convergent* in X to a function $F : X \rightarrow \bar{\mathbb{R}}$ if for every $x \in X$ and for every neighbourhood V of $F(x)$ in $\bar{\mathbb{R}}$ there exists $k \in \mathbb{N}$ and $U \in N(x)$ such that $F_h(y) \in V$ for every $h \geq k$ and for every $y \in U$.

Remark 5.1.8. From the definitions it follows that (F_h) is continuously convergent to F if and only if (F_h) and $(-F_h)$ Γ -converge to F and $-F$ respectively.

Proposition 5.1.9. Let $\Sigma : \bar{\mathbb{R}} \rightarrow \bar{\mathbb{R}}$ be a continuous increasing function. Then

$$\Gamma - \liminf_{h \rightarrow \infty} (\Sigma \circ F_h) = \Sigma \circ (\Gamma - \liminf_{h \rightarrow \infty} F_h),$$

$$\Gamma - \limsup_{h \rightarrow \infty} (\Sigma \circ F_h) = \Sigma \circ (\Gamma - \limsup_{h \rightarrow \infty} F_h).$$

In particular, if (F_h) Γ -converges to F , then $(\Sigma \circ F_h)$ Γ -converges to $\Sigma \circ F$.

Proof. Since Σ is continuous and increasing, we have

$$\Sigma(\inf A) = \inf \Sigma(A), \quad \Sigma(\sup A) = \sup \Sigma(A)$$

for every subset A of $\bar{\mathbb{R}}$. Since

$$(\Gamma - \liminf_{h \rightarrow \infty} F_h)(x) = \sup_{U \in N(x)} \sup_{k \in \mathbb{N}} \inf_{h \geq k} \inf_{y \in U} F_h(y),$$

the first inequality follows easily. The proof for the second one is analogous. $\square$

Lemma 5.1.10. *Given another sequence of functions (G_h) from X to $\bar{\mathbb{R}}$, it holds that:*

$$\begin{aligned} \Gamma - \liminf_{h \rightarrow \infty} (F_h + G_h) &\geq \Gamma - \liminf_{h \rightarrow \infty} F_h + \Gamma - \liminf_{h \rightarrow \infty} G_h, \\ \Gamma - \limsup_{h \rightarrow \infty} (F_h + G_h) &\geq \Gamma - \limsup_{h \rightarrow \infty} F_h + \Gamma - \limsup_{h \rightarrow \infty} G_h, \end{aligned}$$

provided that the sums occurring in it are well defined on X .

Proposition 5.1.11. *Let $G : X \rightarrow \mathbb{R}$ be a continuous function. Then*

$$\begin{aligned} \Gamma - \liminf_{h \rightarrow \infty} (F_h + G) &= \Gamma - \liminf_{h \rightarrow \infty} F_h + G, \\ \Gamma - \limsup_{h \rightarrow \infty} (F_h + G) &= \Gamma - \limsup_{h \rightarrow \infty} F_h + G. \end{aligned}$$

Proof. Since G is continuous, the constant sequence $G_h = G$ is continuously convergent to G . We prove only the second equality since the proof of the first one is analogous. By Remark 5.1.8, (G_h) Γ -converges to G , so thanks to Lemma 5.1.10 we have

$$\Gamma - \limsup_{h \rightarrow \infty} (F_h + G_h) \geq \Gamma - \limsup_{h \rightarrow \infty} F_h + G.$$

On the other hand $(-G_h)$ Γ -converges to $-G$, so we have

$$\Gamma - \limsup_{h \rightarrow \infty} F_h = \Gamma - \limsup_{h \rightarrow \infty} (F_h + G_h - G_h) \geq \Gamma - \limsup_{h \rightarrow \infty} (F_h + G_h) - G.$$

Hence

$$\Gamma - \limsup_{h \rightarrow \infty} (F_h + G_h) \leq \Gamma - \limsup_{h \rightarrow \infty} F_h + G$$

and equality follows. $\square$

Let us state another definition of Γ -convergence, which holds for metric spaces:

Definition 5.1.12. Let X be a metric space and $F_j : X \rightarrow \bar{\mathbb{R}}$ be a sequence of functions. We say that F_j Γ -converges to $F : X \rightarrow \bar{\mathbb{R}}$ if for all $x \in X$ we have

(i) for every sequence (x_h) converging to x in X , the following inequality holds:

$$F(x) \leq \liminf_{h \rightarrow \infty} F_h(x_h),$$

(ii) there exists a sequence (x_h) converging to x in X such that the following inequality holds

$$F(x) \geq \limsup_{h \rightarrow \infty} F_h(x_h).$$

The function F is called the Γ -limit of (F_j) .

Now we prove that the two Definitions 5.1.1 and 5.1.12 are equivalent. Let us remember the *first axiom of countability*:

Axiom 5.1.13. Let X be a space. We say that X satisfies the first axiom of countability if, for every $x \in X$, there exists a countable base for the neighborhood system of x in X .

Remark 5.1.14. If X is a metric space, it satisfies Axiom 5.1.13. For instance, we can construct a base using open balls with radius $\frac{1}{n}$, where n belongs to the set of natural numbers $\mathbb{N}$.

Proposition 5.1.15. Let X be a space satisfying the first axiom of countability. We define

$$F' = \Gamma - \liminf_{h \rightarrow \infty} F_h, \quad F'' = \Gamma - \limsup_{h \rightarrow \infty} F_h.$$

Then F' and F'' are characterised by the following properties:

(i) for every $x \in X$ and for every sequence (x_h) converging to x in X , the following inequality holds:

$$F'(x) \leq \liminf_{h \rightarrow \infty} F_h(x_h),$$

(ii) for every $x \in X$ there exists a sequence (x_h) converging to x in X such that the following inequality holds

$$F'(x) \geq \liminf_{h \rightarrow \infty} F_h(x_h),$$

(iii) for every $x \in X$ and for every sequence (x_h) converging to x in X , the following inequality holds:

$$F''(x) \leq \limsup_{h \rightarrow \infty} F_h(x_h),$$

(iv) for every $x \in X$ there exists a sequence (x_h) converging to x in X such that the following inequality holds

$$F'(x) \geq \limsup_{h \rightarrow \infty} F_h(x_h).$$

So, (F_h) Γ -converges to F if and only if Definition 5.1.12 holds.

Proof. Let (x_h) be a sequence converging to x in X , and let $U \in N(x)$. Then there exists $k \in \mathbb{N}$ such that $x_h \in U$ for every $h \geq k$, so $\inf_{y \in U} F_h(y) \leq F_h(x_h)$ for every $h \geq k$. Moreover,

$$\liminf_{h \rightarrow \infty} \inf_{y \in U} F_h(y) \leq \liminf_{h \rightarrow \infty} F_h(x_h),$$

$$\limsup_{h \rightarrow \infty} \inf_{y \in U} F_h(y) \leq \limsup_{h \rightarrow \infty} F_h(x_h)$$

for every $U \in N(x)$, so

$$F'(x) \leq \liminf_{h \rightarrow \infty} F_h(x_h), \quad F''(x) \leq \limsup_{h \rightarrow \infty} F_h(x_h),$$

and (i), (iii) are proved. Now, let $x \in X$ such that $F'(x) < +\infty$, and let (U_k) be a countable base for the neighborhood system of x in X such that $U_{k+1} \subseteq U_k$ for every $k \in \mathbb{N}$, and let (s_k) be a sequence converging to $F'(x)$ in $\bar{\mathbb{R}}$ such that $s_k > F'(x)$ for every $k \in \mathbb{N}$. From the definition of $F'(x)$ we get:

$$s_k > \liminf_{h \rightarrow \infty} \inf_{y \in U_k} F_h(y)$$

for every $k \in \mathbb{N}$, so there exists a strictly increasing sequence of natural numbers (h_k) such that

$$s_k > \inf_{y \in U_k} F_{h_k}(y)$$

for every $k \in \mathbb{N}$. Therefore, for every $k \in \mathbb{N}$ there exists $y_k \in U_k$ such that $s_k > F_{h_k}(y_k)$. Now we define the sequence (x_h) as follows: if $h = h_k$ for some $k \in \mathbb{N}$, then $x_h = y_k$, and if $h \neq h_k$ for every $k \in \mathbb{N}$, then $x_h = x$. Observe that $x_h \in U_k$ for every $h \geq h_k$, so (x_h) converges to x in X . Moreover

$$F'(x) = \lim_{k \rightarrow \infty} s_k \geq \liminf_{k \rightarrow \infty} F_{h_k}(y_k) \geq \liminf_{h \rightarrow \infty} F_h(x_h),$$

where the last inequality holds since $x_{h_k} = y_k$, so (iii) is proved. To show (iv), let $x \in X$ be a point such that $F''(x) < +\infty$. Let (U_k) be defined as stated earlier and let (t_k) be a decreasing sequence converging to $F''(x)$ in $\bar{\mathbb{R}}$ such that $t_k > F''(x)$ for every $k \in \mathbb{N}$. From the definition of $F''(x)$ we get

$$t_k > \limsup_{h \rightarrow \infty} \inf_{y \in U_k} F_h(y)$$

for every $k \in \mathbb{N}$, so there exist a strictly increasing sequence of natural numbers h_k such that

$$t_k > \inf_{y \in U_k} F_h(y)$$

for every $h \geq h_k$. This implies that for every $h \geq h_k$ exists a $y_k^h \in U_k$ such that $t_k > F_h(y_k^h)$. Let us define the sequence (x_h) as follows: if $h < h_1$ then $x_h = x$, and if $h_k < h < h_{k+1}$, then $x_h = y_k^h$. Since $x_h \in U_k$ for every $h \geq h_k$, we get

$$F''(x) = \lim_{k \rightarrow \infty} t_k \geq \limsup_{h \rightarrow \infty} F_h(x_h),$$

and (iv) is proved.

The statement that (F_h) Γ -converges to F if and only if Definition 5.1.12 holds indeed follows from (i), (ii), (iii), and (iv). $\square$

5.2 Constructive procedure

Let Γ be a Stackelberg game. In our analysis, we assume that the action sets, denoted by X and Y , remain fixed throughout, so we can identify Γ with the pair (L, F) . First, we introduce a specific class of games to which we will apply the proposed Procedure S_n :

Definition 5.2.1. We say that a Stackelberg game $\Gamma = (L, F)$ belongs to the family G if the following assumptions are satisfied:

(A) X and Y are non-empty compact subsets of $\mathbb{R}^n$, and Y is convex,

(L) – $L : X \times Y \rightarrow \mathbb{R}$ is upper semicontinuous on $X \times Y$,
– $L(x, \cdot)$ is lower semicontinuous on Y , for every $x \in X$,

(F1) $F : X \times Y \rightarrow \mathbb{R}$ is upper semicontinuous on $X \times Y$,

(F2) for any $(x, y) \in X \times Y$ and for any sequence $(x_k)_k \subset X$ converging to x there exists a sequence $(\tilde{y}_k)_k \subset Y$ converging to y such that

$$\liminf_{k \rightarrow \infty} F(x_k, \tilde{y}_k) \geq F(x, y),$$

(F3) $F(x, \cdot)$ is concave on Y , for any $x \in X$.

Remark 5.2.2. Requiring (F1)-(F3) is weaker than requiring the continuity of F , as shown in the following example:

Example 5.2.3. Suppose that $X = [1, 2]$ and Y is the closed ball in $\mathbb{R}^2$ centered in $(1, 0)$ with radius 1. Define F on $X \times Y$ as follows:

$$F(x, (y_1, y_2)) = \begin{cases} -\frac{y_2^2}{2y_1}x & \text{if } (y_1, y_2) \neq (0, 0) \\ 0 & \text{if } (y_1, y_2) = (0, 0). \end{cases}$$

Then F satisfies (F1)-(F3) but is not lower semicontinuous at $(0, 0)$, for any $x \in [1, 2]$.

Firstly, we show that F satisfies (F1)-(F3):

(F1) We must show the upper semicontinuity at $(x, (0, 0))$, as F is continuous for any other point in $X \times Y$. Let $x \in X$ and let $(x_k, (\tilde{y}_{1,k}, \tilde{y}_{2,k}))_k \subseteq X \times Y$ be a sequence converging to $(x, (0, 0))$. Since $F(x, (y_1, y_2)) \leq 0$ for every $(x, (y_1, y_2)) \in X \times Y$, and $F(x, (0, 0))=0$, then

$$F(x, (0, 0)) \leq \limsup_{k \rightarrow \infty} F(x, (y_{1,k}, y_{2,k})).$$

Therefore (F1) holds.

(F2) As in the previous case, the only critical point is $(x, (0, 0))$. Let $x \in X$ and let $(x_k)_k \subseteq X$ be a sequence converging to x . Define $(\tilde{y}_{1,k}, \tilde{y}_{2,k}) = (\frac{1}{k}, 0) \in Y$ for any $k \in \mathbb{N}$. Since $F(x_k, (\tilde{y}_{1,k}, \tilde{y}_{2,k})) = 0$ for any $k \in \mathbb{N}$ and even $F(x, (0, 0)) = 0$, then

$$\liminf_{k \rightarrow \infty} F(x_k, (\tilde{y}_{1,k}, \tilde{y}_{2,k})) = F(x, (0, 0)) \Rightarrow \liminf_{k \rightarrow \infty} F(x_k, (\tilde{y}_{1,k}, \tilde{y}_{2,k})) \geq F(x, (0, 0)).$$

Therefore (F2) holds.

(F3) Let $x \in X$. First we prove the concavity of $F(x, (\cdot, \cdot))$ on $Y \setminus \{(0, 0)\}$: we consider the twice-continuously differentiable function $g : (0, +\infty) \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(y_1, y_2) = -\frac{y_2^2}{2y_1}x.$$

The Hessian matrix of g at (y_1, y_2) is

$$Hg(y_1, y_2) = \begin{bmatrix} -\frac{y_2^2}{y_1^3}x & \frac{y_2}{y_1^2}x \\ \frac{y_2}{y_1^2}x & -\frac{1}{y_1}x \end{bmatrix}.$$

Since

$$\begin{cases} -\frac{y_2^2}{y_1^3}x < 0 \\ \frac{y_2^2}{y_1^3}x^2 - \frac{y_2^2}{y_1^3}x^2 = 0, \end{cases}$$

for any $(y_1, y_2) \in (0, \infty) \times \mathbb{R}$, $Hg(y_1, y_2)$ is negative semi-definite, then g is concave on $(0, \infty) \times \mathbb{R}$. Therefore, $F(x, (\cdot, \cdot))$ is concave on $Y \setminus \{(0, 0)\}$, being equal to g . The concavity on the whole of Y is fulfilled if

$$F(x, t(0, 0) + (1-t)(y_1, y_2)) \geq tF(x, (0, 0)) + (1-t)F(x, (y_1, y_2))$$

for every $t \in [0, 1]$ and $(y_1, y_2) \in Y$. Calculating terms of the inequality we find out that they are both equal to

$$\frac{-(1-t)y_2^2}{2y_1}x,$$

so the inequality holds, and (F3) is satisfied.

Now, let us show that the function $F(x, (\cdot, \cdot))$ is not lower semicontinuous at $(0, 0)$:

let $(\bar{y}_{1,k}, \bar{y}_{2,k}) = (\frac{1}{k}, \frac{1}{\sqrt{k}}) \in Y$ for any $k \in \mathbb{N}$. Note that

$$\begin{cases} F(x, (\bar{y}_{1,k}, \bar{y}_{2,k})) = \frac{-(\frac{1}{\sqrt{k}})^2}{2(\frac{1}{k})}x = \frac{-x}{2} \in [-\frac{1}{2}, -1] \\ F(x, (0, 0)) = 0, \end{cases}$$

so

$$\liminf_{k \rightarrow \infty} F(x, (\bar{y}_{1,k}, \bar{y}_{2,k})) < F(x, (0, 0)).$$

We can finally conclude that the function is not continuous on $X \times Y$.

Remark 5.2.4. Assumptions (F1) and (F2) imply the Γ -convergence: let $x \in X$ and $(x_j)_j \subseteq X$ be a sequence converging to x , and consider the following real-valued functions defined on Y by

$$W_j(y) = F(x_j, y) \text{ for every } j \in \mathbb{N}, \quad W(y) = F(x, y).$$

Then the sequence $(-W_j)_j$ Γ -converges to $-W$. Indeed, let us verify Definition 5.1.12:

- (I) For every $y_j \rightarrow y$, $-W(y) \leq \liminf_j (-W_j(y_j)) = -\limsup_j W_j(y_j)$, so $W(y) \geq \limsup_j W_j(y_j)$, that is true thanks to (F1).
- (S) There exists $\tilde{y}_j \rightarrow y$ such that $-W(y) \geq \limsup_j (-W_j(\tilde{y}_j)) = -\liminf_j W_j(\tilde{y}_j)$, so $W(y) \leq \liminf_j W_j(\tilde{y}_j)$, true thanks to (F2).

Remark 5.2.5. Let us assume that $(U, V) \in G$ and $(\hat{U}, \hat{V}) \in G$. By using Propositions 5.1.9 and 5.1.11 we obtain that:

- $(hU, hV) \in G$ for any $h, k \geq 0$,
- If $\hat{V}$ is continuous, then $(U + \hat{U}, V + \hat{V}) \in G$,
- if φ and ϑ are real-valued and continuous functions defined on $\mathbb{R}$ with ϑ increasing and concave, then $(\varphi \circ U, \vartheta \circ V) \in G$.

Now we illustrate the procedure that allows us to construct a sequence of perturbed games $(\Gamma_n)_n$ and a sequence of strategy profiles $(\bar{x}_n, \varphi_n)_n$. We will refer to this procedure as the *Cost-to-Move Procedure* S_n :

Procedure 5.2.6. Fix an initial point $(\bar{x}_0, \bar{y}_0) \in X \times Y$ and define, for any $n \in \mathbb{N}$

$$\begin{cases} \Gamma_n = (L_n, F_n) \\ \{\varphi_n(x)\} = \text{Arg max}_{y \in Y} F_n(x, y), \text{ for any } x \in X \\ \bar{x}_n \in \text{Arg max } L_n(x, \varphi_n(x)) \end{cases} \quad (S_n)$$

where, for any $(x, y) \in X \times Y$

$$\begin{aligned} F_n(x, y) &= F(x, y) - \frac{1}{2\gamma_{n-1}} \|y - \varphi_{n-1}(x)\|^2, \\ L_n(x, y) &= L(x, y) - \frac{1}{2\beta_{n-1}} \|x - \bar{x}_{n-1}(y)\|^2, \end{aligned}$$

with

$$(\gamma_n)_{n \in \mathbb{N} \cup \{0\}} \subseteq (0, \infty) \text{ and } \lim_{n \rightarrow \infty} \gamma_n = +\infty,$$

$$(\beta_n)_{n \in \mathbb{N} \cup \{0\}} \subseteq (0, \infty) \text{ and } \lim_{n \rightarrow \infty} \beta_n = +\infty,$$

$$\varphi_0(x) = \bar{y}_0 \text{ for any } x \in X.$$

Theorem 5.2.7. Assume that $\Gamma \in G$. Then, the Procedure S_n is well-defined, φ_n is single-valued and it is a continuous function on X , for any $n \in \mathbb{N}$.

Proof. Firstly, let us show that, for any $n \in \mathbb{N}$, $\Gamma_n \in G$

(A) trivially holds since $\Gamma \in G$,

(L) and (F1) are verified because we add a continuous function,

(F2) holds because

$$\begin{aligned} \liminf_{k \rightarrow \infty} F_n(x_k, \tilde{y}_k) &= \liminf_{k \rightarrow \infty} \left\{ F(x_k, \tilde{y}_k) - \frac{1}{2\gamma_{n-1}} \|\tilde{y}_k - \varphi_{n-1}(x_k)\|^2 \right\} \\ &\geq \liminf_{k \rightarrow \infty} F(x_k, \tilde{y}_k) + \liminf_{k \rightarrow \infty} \left\{ -\frac{1}{2\gamma_{n-1}} \|\tilde{y}_k - \varphi_{n-1}(x_k)\|^2 \right\} \\ &\geq F(x, y) - \lim_{k \rightarrow \infty} \frac{1}{2\gamma_{n-1}} \|\tilde{y}_k - \varphi_{n-1}(x_k)\|^2, \end{aligned}$$

since $\Gamma \in G$ and the function $(x, y) \rightarrow \frac{1}{2\gamma_{n-1}} \|y - x\|^2$ is continuous. Thus the ordinary limit exists and coincides with the inferior one. By using the continuity of φ we can conclude:

$$\liminf_{k \rightarrow \infty} F_n(x_k, \tilde{y}_k) \geq F(x, y) - \frac{1}{2\gamma_{n-1}} \|y - \varphi_{n-1}(x)\|^2 = F_n(x, y),$$

(F3) is verified because $\eta(x_k, \cdot) = \frac{1}{2\gamma_{n-1}} \|\cdot - \varphi_{n-1}(x_k)\|^2$ is strictly convex for any $x_k \in X$, and F is concave on Y for any $x \in X$ thanks to the hypotheses.

So, the assumptions of Theorem 2.1.4 holds, and we obtain that φ_n is single-valued and continuous. The Procedure S_n is well defined when $F_n(x, \cdot)$ has a unique maximizer on Y , for any $x \in X$ and for any $n \in \mathbb{N}$, and when $L_n(\cdot, \varphi_n(\cdot))$ admits a maximizer on X , for any $n \in \mathbb{N}$. Furthermore, F_n is strictly concave for any $x \in X$, so it has a unique maximizer on Y (whose existence is guaranteed thanks to the generalization of Weierstrass Theorem 1.2.13): $\varphi_n(x)$ is therefore well defined and the follower's best reply correspondence in Γ_n is single-valued.

Moreover, L is upper semicontinuous in $X \times Y$, which is a compact set, so the Weierstrass Theorem ensures that it has at least a maximizer. We can conclude that the Procedure S_n is well-defined. $\square$

5.3 Interpretation of the procedure

By analyzing Procedure S_n we note that, at every step n of the procedure, the follower chooses a strategy that compromises between maximizing his payoff $F(x, \cdot)$ and being near to his previous strategy φ_{n-1} . The latter purpose is caused by the so-called *anchoring effect*: each agent's action depends on the preceding one. In particular, we can think of actions as routines, and the Cost-to-Move represents the difficulty of quitting a routine or, more generally, reacting quickly. So the agents tend not to move too far from what they were doing till now.

To formalize this anchoring effect we subtract a quadratic Cost-to-Move from the payoff function.

The coefficient γ_{n-1} is like a trade-off parameter between maximizing the payoff and minimizing the changes.

Since this behavior repeats at every step of the algorithm, we will not move very far from the starting point: the limit of $\varphi_n(x)$ will be near y_0 .

An analogous observation holds also for the leader, which will not move far from x_0 .

5.4 Convergence of the procedure

In this paragraph, we will prove that the Procedure S_n produces a sequence $\varphi_n(x)$ that converges to a function in the follower's best reply correspondence.

Before showing the result, we state the following lemmas:

Lemma 5.4.1. *Let G be a real-valued function defined on $X \times Y$ and $\bar{G}$ be the extended real valued function defined on $X \times \mathbb{R}^n$ by*

$$\bar{G}(x, y) = \begin{cases} G(x, y) & \text{if } y \in Y \\ -\infty & \text{if } y \notin Y. \end{cases}$$

Let $x \in X$. If the function $G(x, \cdot)$ is upper semicontinuous and concave on Y , then

- *the function $\bar{G}(x, \cdot)$ is upper semicontinuous and concave on $\mathbb{R}^n$,*
- *$\text{Arg max}_{y \in Y} G(x, y) = \text{Arg max}_{y \in \mathbb{R}^n} \bar{G}(x, y)$,*
- *$\text{Arg max}_{y \in Y} G(x, y) - \frac{1}{2\lambda} \|y - v\|^2 = \text{Arg max}_{y \in \mathbb{R}^n} \bar{G}(x, y) - \frac{1}{2\lambda} \|y - v\|^2$ for any $\lambda > 0$ and for any $v \in \mathbb{R}^n$,*
- *$\varphi^*(x) \in \text{Arg max}_{y \in Y} G(x, y) \iff \{\varphi^*(x)\} = \text{prox}_{\lambda, G(x, \cdot)}(\varphi^*(x))$,*

where $\text{prox}_{\lambda, G(x, \cdot)}(v) = \text{Arg max}_{y \in Y} G(x, y) - \frac{1}{2\lambda} \|y - v\|^2$, for any $v \in Y$.

The proof of this lemma is quite trivial: claims 1, 2, 3 are immediate, and claim 4 follows directly from Theorem 4.2.2.

Lemma 5.4.2. *(proximal-point algorithm) Let F be a proper lower semicontinuous convex function such that $\text{Arg min } F \neq \emptyset$, and let $(\gamma_n)_{n \in \mathbb{N}}$ be a sequence in $(0, +\infty)$ such that $\sum_{n \in \mathbb{N}} \gamma_n = +\infty$ and let $x_0 \in X \times Y$. Set*

$$x_{n+1} = \text{prox}_{\gamma_n f} x_n \quad \forall n \in \mathbb{N}.$$

Then the following hold:

- *$(x_n)_{n \in \mathbb{N}}$ is a minimizing sequence of F ,*

- $(x_n)_{n \in \mathbb{N}}$ converges to a point in $\text{Arg min } F$.

Theorem 5.4.3. *Assume that (A), (F1) and (F3) hold. Then, the sequence $(\varphi_n)_n$ generated by the Procedure S_n pointwisely converges to a function $\varphi \in Y^X$, and $\varphi(x) \in M(x)$ for any $x \in X$, where*

$$M(x) = \text{Arg max}_{y \in Y} F(x, y).$$

Proof. Let $x \in X$ and consider the function $\bar{F}$, defined on $X \times \mathbb{R}^m$ by

$$\bar{F}(x, y) = \begin{cases} F(x, y) & \text{if } y \in Y \\ -\infty & \text{if } y \notin Y. \end{cases} \quad (5.1)$$

By assumption (F1), (F3) and Lemma 5.4.1, $-\bar{F}(x, \cdot)$ is proper lower semicontinuous and convex. Moreover, in light of Lemma 5.4.1, the compactness of Y and (F1) we have that

$$\text{Arg min}_{y \in \mathbb{R}^n} -\bar{F}(x, y) = \text{Arg max}_{y \in \mathbb{R}^n} \bar{F}(x, y) = \text{Arg max}_{y \in Y} F(x, y) \neq \emptyset,$$

thanks to Theorem 1.2.13.

Since $\lim_{n \rightarrow \infty} \gamma_n = +\infty$, the function $-\bar{F}(x, \cdot)$ satisfies the hypothesis for the convergence of Theorem 5.4.2. So, if we define

$$\{z_n\} = \text{Arg min}_{y \in Y} \left(-\bar{F}(x, y) + \frac{1}{2\gamma_{n-1}} \|y - z_{n-1}\|^2 \right) \text{ for any } n \in \mathbb{N},$$

where $z_0 = \bar{y}_0$, Theorem 5.4.2 assures that z_n converges to a point in $\text{Arg min}_{y \in Y} -\bar{F}(x, y)$.

So, equivalently,

$$\{z_n\} = \text{Arg max}_{y \in Y} \left(\bar{F}(x, y) - \frac{1}{2\gamma_{n-1}} \|y - z_{n-1}\|^2 \right) \text{ for any } n \in \mathbb{N}$$

converges to a point in $\text{Arg max}_{y \in Y} \bar{F}(x, y)$.

Since the unique maximizers of $\bar{F}(x, \cdot) - \frac{1}{2\gamma_{n-1}} \|\cdot - z_{n-1}\|^2$ coincides with the unique maximizer of $\tilde{F}(x, \cdot) = F(x, \cdot) - \frac{1}{2\gamma_{n-1}} \|\cdot - z_{n-1}\|^2$ in light of Lemma 5.4.1, then $z_n = \varphi_n(x)$ for any $n \in \mathbb{N}$, thanks to the definition of φ_n . Furthermore, since the set of maximizers of $\bar{F}(x, \cdot)$ over $\mathbb{R}^n$ coincides with the set of maximizers of $\bar{F}(x, \cdot)$ over Y in light of Lemma 5.4.1, the sequence $(\varphi_n(x))_n$ converges to a maximizer of $F(x, \cdot)$ over Y . Hence, the function φ that associates to each $x \in X$ the point $\varphi(x) = \lim_{n \rightarrow \infty} \varphi_n(x) \in Y$ is well defined and $\varphi(x) \in M(x)$ for any $x \in X$. $\square$

5.5 Existence results for SPNEs

In this section, we will prove the existence of the SPNE generated with the Procedure S_n .

Theorem 5.5.1. *Let $\Gamma = (L, F) \in G$ and suppose the sequence of actions profiles $(\bar{x}_n, \varphi_n(\bar{x}_n))_n \subseteq X \times Y$ generated by the Procedure S_n converges to $(\bar{x}, \bar{y}) \in X \times Y$. Then, the strategy profile $(\bar{x}, \bar{\varphi}) \in X \times Y^X$, where*

$$\bar{\varphi}(x) = \begin{cases} \bar{y} & \text{if } x = \bar{x} \\ \lim_{n \rightarrow \infty} \varphi_n(x) & \text{if } x \neq \bar{x}, \end{cases} \quad (5.2)$$

is a subgame perfect Nash equilibrium of Γ .

Before proving this theorem, it is worth noting that if the sequence $(\bar{x}_n, \varphi_n(\bar{x}_n))_n$ does not converge, we can select a subsequence $(\bar{x}_{n_k}, \varphi_{n_k}(\bar{x}_{n_k}))_k$ and, via the compactness of both X and Y , we can assert that the subsequence $(\bar{x}_{n_k}, \varphi_{n_k}(\bar{x}_{n_k}))_k$ is convergent.

By replacing the sequence $(\bar{x}_n, \varphi_n(\bar{x}_n))_n$ with the convergent subsequence $(\bar{x}_{n_k}, \varphi_{n_k}(\bar{x}_{n_k}))_k$, the assertion of Theorem 5.5.1 still holds. Thus, provided that $\Gamma \in G$, we can be certain of the existence of an SPNE, regardless of the lower semicontinuity of the follower's best reply correspondence.

Proof. To prove that $\bar{\varphi}$ is an SPNE, we have to prove that:

$$(S1) \quad \bar{\varphi}(x) \in M(x) \quad \forall x \in X,$$

$$(S2) \quad \bar{x} \in \text{Arg max}_{x \in X} L(x, \bar{\varphi}(x)).$$

Let us prove (S1) first. Let $x \in X$ and $\varphi(x) = \lim_{n \rightarrow \infty} \varphi_n(x)$, as defined in Theorem 5.4.3. If $x \neq \bar{x}$, Theorem 5.4.3 ensures that

$$\bar{\varphi}(x) = \varphi(x) \in M(x).$$

If $x = \bar{x}$, pick $y \in Y$. By assumption (F2), there exists a sequence $(\tilde{y}_n)_n$ converging to y such that

$$\liminf_{n \rightarrow \infty} F(\bar{x}_n, \tilde{y}_n) \geq F(\bar{x}, y). \quad (5.3)$$

By (F1) we have:

$$\begin{aligned}
F(\bar{x}, \bar{y}) &\geq \limsup_{n \rightarrow \infty} F(\bar{x}_n, \varphi_n(\bar{x}_n)) \\
&= \limsup_{n \rightarrow \infty} [F(\bar{x}_n, \varphi_n(\bar{x}_n)) - \frac{1}{2\gamma_{n-1}} \|\varphi_n(\bar{x}_n) - \varphi_{n-1}(\bar{x}_n)\|^2] \\
&= \limsup_{n \rightarrow \infty} F_n(\bar{x}_n, \varphi_n(\bar{x}_n)),
\end{aligned} \tag{5.4}$$

where the first equality holds since the second addend in the superior limit converges to 0, being $(\gamma_n)_n$ a divergent sequence of positive real numbers, and Y a compact set; the second equality comes from the definition of F_n in the Procedure S_n . By the definition of $\varphi_n(\bar{x}_n)$, namely

$$\{\varphi_n(x)\} = \text{Arg max}_{y \in Y} F_n(x, y), \text{ for any } x \in X,$$

we get

$$\begin{aligned}
\limsup_{n \rightarrow \infty} F_n(\bar{x}_n, \varphi_n(\bar{x}_n)) &\geq \limsup_{n \rightarrow \infty} F_n(\bar{x}_n, \tilde{y}_n) \\
&= \limsup_{n \rightarrow \infty} [F(\bar{x}_n, \tilde{y}_n) - \frac{1}{2\lambda_{n-1}} \|\tilde{y}_n - \varphi_{n-1}(\bar{x}_n)\|^2].
\end{aligned} \tag{5.5}$$

Recalling the properties of $(\gamma_n)_n$ and the compactness of Y , by (5.3)-(5.5) we have

$$\begin{aligned}
F(\bar{x}, \bar{y}) &\geq \limsup_{n \rightarrow \infty} [F(\bar{x}_n, \tilde{y}_n) - \frac{1}{2\lambda_{n-1}} \|\tilde{y}_n - \varphi_{n-1}(\bar{x}_n)\|^2] \\
&= \limsup_{n \rightarrow \infty} F(\bar{x}_n, \tilde{y}_n) \\
&\geq \liminf_{n \rightarrow \infty} F(\bar{x}_n, \tilde{y}_n) \\
&\geq F(\bar{x}, \bar{y}).
\end{aligned}$$

Hence, $\bar{y} = \bar{\varphi}(\bar{x}) \in M(\bar{x})$ and (S1) is satisfied.

In order to prove condition (S2), we have to show that $L(\bar{x}, \bar{y}) = L(\bar{x}, \bar{\varphi}(\bar{x})) \geq L(x, \bar{\varphi}(x))$ for any $x \in X$.

So, let $x \in X \setminus \{\bar{x}\}$. In light of (L), we get

$$\begin{aligned}
L(\bar{x}, \bar{y}) &\geq \limsup_{n \rightarrow \infty} L(\bar{x}_n, \varphi_n(\bar{x}_n)) \\
&= \limsup_{n \rightarrow \infty} \left[L(\bar{x}_n, \varphi_n(\bar{x}_n)) - \frac{1}{2\beta_{n-1}} \|\bar{x}_n - \bar{x}_{n-1}\|^2 \right] \\
&\geq \limsup_{n \rightarrow \infty} \left[L(x, \varphi_n(x)) - \frac{1}{2\beta_{n-1}} \|x - \bar{x}_{n-1}\|^2 \right] \\
&\geq \liminf_{n \rightarrow \infty} \left[L(x, \varphi_n(x)) - \frac{1}{2\beta_{n-1}} \|x - \bar{x}_{n-1}\|^2 \right] \\
&= \liminf_{n \rightarrow \infty} L(x, \varphi_n(x)) \geq L(x, \varphi(x)),
\end{aligned}$$

where the first (respectively second) equality holds since the second term in the superior limit (inferior limit respectively) converges to 0, being $(\beta_n)_n$ a divergent sequence of positive real numbers and X a compact set; the second inequality comes from the definition of x_n in the Procedure S_n as the maximizer of L_n , and the last inequality follows by (L).

As $x \in X \setminus \{\bar{x}\}$, then $L(x, \varphi(x)) = L(x, \bar{\varphi}(x))$ and, therefore,

$$L(\bar{x}, \bar{y}) \geq L(x, \bar{\varphi}(x)).$$

Hence (S2) holds, and the proof is complete. $\square$

Let us note that the strategy $\bar{\varphi}$ defined in (5.2) differs from the pointwise limit of the sequence $(\varphi_n)_n$ at most at one point. Indeed, the two limits

$$\lim_{n \rightarrow \infty} \varphi_n(\bar{x}_n), \quad \lim_{n \rightarrow \infty} \varphi_n(\lim_{n \rightarrow \infty} \bar{x}_n)$$

need not to be equal, so $(\bar{x}, \varphi)$ could be not an SPNE of Γ , as shown Example 7.0.1 and 7.0.2 in the Appendix.

Remark 5.5.2. The SPNE selected according to Theorem 5.5.1 depends heavily on the starting point of the Procedure S_n since neither the leader nor the follower wants to detach itself from (x_0, y_0) , as seen in Section 5.3. This dependence is shown in Example 7.0.3 in the Appendix.

Let us note that we can modify the Procedure S_n by leaving unchanged the leader's payoff function (i.e $L_n = L$, for any $n \in \mathbb{N}$); in this case, an existence result analogous to Theorem 5.5.1 holds.

We can also modify the starting point. For the Procedure S_n we took inspiration from the proximal point method, which requires a fixed starting point. However, we can also choose a continuous function $\varphi_0 \in Y^X$ as starting point, and each result would still be valid. Moreover, we have some further properties if we change the starting point in the following way:

Theorem 5.5.3. *Let $\Gamma \in G$ and let the follower's initial point $\varphi_0 \in Y^X$ be a continuous function. Assume that $\varphi_0(x) \in M(x)$ for any $x \in X$. Then $\varphi_n = \varphi_0$ for any $n \in \mathbb{N}$. Moreover, φ_0 is the strategy chosen by the follower in the SPNE selected according to Theorem 5.5.1.*

Namely, if the starting point is already in the follower's best reply set, the follower's strategy will not change.

Proof. We prove the first part of the result by induction. Firstly, note that the function F satisfies the assumption of Lemma 5.4.1 as $\Gamma \in G$.

Let $n = 1$. Since $\varphi_0(x) \in M(x)$ implies that $\{\varphi_0(x)\} = \text{Arg max}_{y \in Y} F(x, y)$ for any $x \in X$, in light Lemma 5.4.1 and the definition of φ_1 we have:

$$\{\varphi_0(x)\} = \text{Arg max}_{y \in Y} F(x, y) - \frac{1}{2\gamma_0} \|y - \varphi_0(x)\|^2 = \{\varphi_1(x)\}$$

for any $x \in X$. So, the base case is satisfied.

Let $n > 1$ and suppose that $\varphi_n = \varphi_0$. Then $\varphi_n(x) \in M(x)$ for any $x \in X$ and, by Lemma 5.4.1 and definition of φ_{n+1} , we get

$$\{\varphi_n(x)\} = \text{Arg max}_{y \in Y} F(x, y) - \frac{1}{2\gamma_n} \|y - \varphi_n(x)\|^2 = \{\varphi_{n+1}(x)\}$$

for any $x \in X$. So, the inductive step is provided. Hence, $\varphi_n = \varphi_0$ for any $n \in \mathbb{N}$ and the first part of the proof is complete. Since $\varphi_n = \varphi_0$ for any $n \in \mathbb{N}$ and φ_0 is continuous, then for any sequence $(x_n)_n \subseteq X$ converging to $x \in X$, the sequence $\varphi_n(x_n)$ converges to $\varphi_0(x)$. So, from Theorem 5.5.1:

$$(x_n, \varphi_n(x_n)) \rightarrow (x, \varphi_0(x)) \quad \forall x_n \rightarrow x,$$

and

$$\bar{\varphi}(x) = \begin{cases} \bar{y} = \varphi_0(x) & \text{if } x = \bar{x} \\ \lim_{n \rightarrow \infty} \varphi_n(x) = \lim_{n \rightarrow \infty} \varphi_0(x) = \varphi_0(x) & \text{if } x \neq \bar{x} \end{cases} = \varphi_0(x).$$

Therefore, $\varphi_0(x)$ is the follower's strategy in the SPNE selected according to Theorem 5.5.1. □

Theorem 5.5.4. *Let $\Gamma \in G$ and let the follower's initial point $\varphi_0 \in Y^X$ be a continuous function. Assume that there exists $\nu \in \mathbb{N}$ such that $\varphi_\nu = \varphi_{\nu-1}$. Then $\varphi_\nu(x) \in M(x)$ for any $x \in X$ and $\varphi_n = \varphi_\nu$ for any $n > \nu$. Moreover, φ_ν is the strategy chosen by the follower in the SPNE selected according to Theorem 5.5.1.*

Proof. By the definition of φ_ν and since $\varphi_\nu = \varphi_{\nu-1}$, we have

$$\begin{aligned} \{\varphi_\nu(x)\} &= \text{Arg max}_{y \in Y} F(x, y) - \frac{1}{2\gamma_{\nu-1}} \|y - \varphi_{\nu-1}(x)\|^2 = \{\varphi_{n+1}(x)\} \\ &= \text{Arg max}_{y \in Y} F(x, y) - \frac{1}{2\gamma_{\nu-1}} \|y - \varphi_\nu(x)\|^2 = \{\varphi_{n+1}(x)\} \\ &= \text{Arg max}_{y \in Y} F(x, y) \end{aligned}$$

for every $x \in X$, where the last equality holds thanks to Lemma 5.4.1. Consider a new procedure whose follower's initial point is the continuous function $\varphi_\nu = \gamma_0$ and with $(\varphi_{\nu+n})_n = (\gamma_n)_n$. Applying Theorem 5.5.3, we have $\varphi_n = \varphi_\nu$ for any $n > \nu$. Given the above and by the continuity of φ_ν , arguing as in the last part of Theorem 5.5.3, it follows that φ_ν is the strategy chosen by the follower according to Theorem 5.5.1. □

5.6 Comparison with the strong and weak Stackelberg equilibria

The Cost-to-Move procedure offers a significant advantage to the leader by eliminating the need to know the follower's best reply correspondence. In contrast, when computing weak and strong Stackelberg equilibria, knowledge of the best reply correspondence is necessary. It is important to note that the SPNEs obtained through the

Cost-to-Move procedure do not generally coincide with the strong and weak equilibria, as shown in the following example.

Example 5.6.1. Let Γ be the game defined in Example 7.0.2.

$$M(x) = \begin{cases} [-1, 1], & \text{if } x \in [\frac{1}{2}, 1] \\ \{-1\}, & \text{if } x \in (1, 2]. \end{cases}$$

So,

$$\max_{x \in M(x)} L(x, y) = -x + 1, \quad \min_{y \in M(x)} L(x, y) = \begin{cases} -x - 1, & \text{if } x \in [\frac{1}{2}, 1] \\ -x + 1, & \text{if } x \in (1, 2], \end{cases}$$

then

$$\text{Arg max}_{x \in [\frac{1}{2}, 2]} \max_{y \in M(x)} L(x, y) = \{\frac{1}{2}\}$$

and

$$\text{Arg max}_{x \in [\frac{1}{2}, 2]} \min_{y \in M(x)} L(x, y) = \emptyset.$$

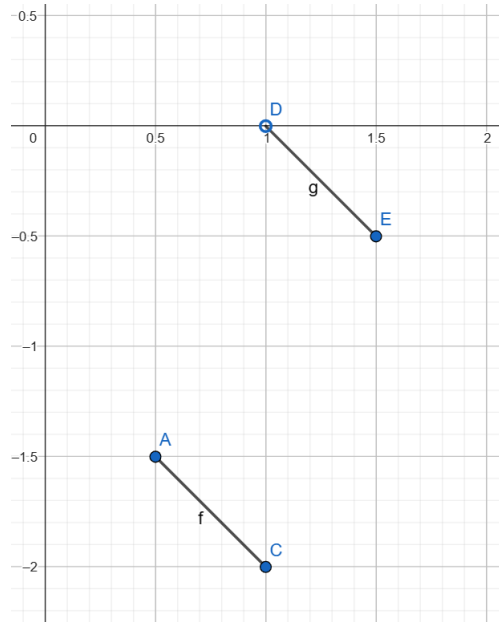


Figure 5.1: $\min_{y \in M(x)} L(x, y)$

Thus, the weak Stackelberg equilibrium does not exist, and the strong one is $(\frac{1}{2}, -1)$ since

$$\operatorname{Arg\,max}_{y \in M(\frac{1}{2})} L(\frac{1}{2}, y) = \{-1\}.$$

The sequence $(\bar{x}_n, \varphi_n(\bar{x}_n))_n$ generated by Procedure S_n converges to $(1, -1)$, which is neither a strong nor a weak Stackelberg equilibrium.

Chapter 6

Other approaches

The objective of this chapter is to explore alternative procedures for selecting a follower's strategy in cases where $M(x)$ is not a singleton.

Let us recall the Stackelberg problem: the leader faces the problem

$$\max_{x \in X} L(x, y(x)), \quad (\text{S})$$

where $y(x) \in M(x)$ is a solution of the problem

$$\max_{y \in Y} F(x, y). \quad (P_x)$$

In the case where $M(x)$ is not a singleton, the goal is to choose a single element from $M(x)$ to insert into (S).

6.1 General selection method

In this section, our goal is to generate an SPNE of Γ without making assumptions about the single-valuedness of M or the continuous convergence of $(\bar{\varphi}_n)_n$. Specifically, we will only assume the pointwise convergence of $(\bar{\varphi}_n)_n$. Let us begin by formulating a result where (L_n) and (F_n) are not derived from a specific selection method.

Theorem 6.1.1. *Let $(\bar{x}_n, \bar{\varphi}_n)$ be an SPNE of Γ_n for any $n \in \mathbb{N}$. Assume that:*

(H_1) the sequences $(L_n)_n$ and $(F_n)_n$ continuously converge to L and F , respectively,

(H_2) the sequence $(\bar{\varphi}_n)_n$ pointwise converges to $\bar{\varphi} : X \rightarrow Y$.

If the sequence $(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n))_n \subseteq X \times Y$ converges to $(\bar{x}, \bar{y})$ and

$$\hat{\varphi}(x) = \begin{cases} \bar{\varphi}(x) & \text{if } x \neq \bar{x} \\ \bar{y} & \text{if } x = \bar{x}, \end{cases}$$

then the strategy profile $(\bar{x}, \hat{\varphi})$ is an SPNE of $\Gamma = (L, F)$.

Proof. We start by proving that $\hat{\varphi}(x) \in M(x)$ for any $x \in X$. Let $x \in X \setminus \{\bar{x}\}$ and $y \in Y$. By definition of $\hat{\varphi}$, assumptions (H_1) and (H_2), it follows that

$$\begin{aligned} F(x, \hat{\varphi}(x)) &= F(x, \bar{\varphi}(x)) = \lim_{n \rightarrow \infty} F_n(x, \bar{\varphi}_n(x)) \\ &\geq \lim_{n \rightarrow \infty} F_n(x, y) = F(x, y), \end{aligned}$$

where the inequality follows because $(\bar{x}_n, \bar{\varphi}_n)$ is an SPNE of Γ_n .

Now, let $x = \bar{x}$ and $y \in Y$. By definition of $\hat{\varphi}$ and assumption (H_1) it follows that

$$\begin{aligned} F(\bar{x}, \hat{\varphi}(\bar{x})) &= F(\bar{x}, \bar{y}) = \lim_{n \rightarrow \infty} F_n(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n)) \\ &\geq \lim_{n \rightarrow \infty} F_n(\bar{x}_n, y) = F(\bar{x}, y), \end{aligned}$$

where the inequality follows because $(\bar{x}_n, \bar{\varphi}_n)$ is an SPNE of Γ_n . So $\hat{\varphi}(x) \in M(x)$ for any $x \in X$ and the first condition of SPNE in Definition 2.2.3 is satisfied.

Let us show that

$$\bar{x} \in \operatorname{Arg\,max}_{x \in X} L(x, \hat{\varphi}(x)).$$

As $\hat{\varphi}(\bar{x}) = \bar{y}$, this is equivalent to prove that

$$L(\bar{x}, \bar{y}) \geq L(x, \bar{\varphi}(x))$$

for any $x \in X \setminus \{\bar{x}\}$. By definition of $(\bar{x}, \bar{y})$, assumptions (H_1) and (H_2), we have that

$$L(\bar{x}, \bar{y}) = \lim_{n \rightarrow \infty} L_n(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n)) \geq \lim_{n \rightarrow \infty} L_n(x, \bar{\varphi}_n(x)) = L(x, \bar{\varphi}(x)),$$

for any $x \in X \setminus \{\bar{x}\}$, where the inequality holds since $(\bar{x}_n, \bar{\varphi}_n)$ is an SPNE of Γ_n .

Therefore, $(\bar{x}, \hat{\varphi})$ satisfies also the second condition of SPNE in Definition 2.2.3. $\square$

Let us show that the hypotheses of this theorem are satisfied if we generate (L_n) and (F_n) with the Cost-to-Move Procedure. To achieve such a goal, we consider the following assumptions on Γ :

$$\left\{ \begin{array}{l} X \text{ and } Y \text{ are non-empty compact subsets of } \mathbb{R}^n, \text{ and } Y \text{ is convex;} \\ L \text{ is continuous on } X \times Y; \\ F \text{ is continuous on } X \times Y \text{ and } F(x, \cdot) \text{ is concave on } Y \text{ for any } x \in X. \end{array} \right. \quad (\text{A})$$

Remark 6.1.2. If conditions (A) hold, then the assumptions of Definition 5.2.1 hold. It is enough to prove (F2) since the other proprieties trivially follow from (A). Suppose, by contradiction, that exists $(x, y) \in X \times Y$ and $x_k \rightarrow x$ such that for every $\tilde{y}_k \rightarrow y$

$$\lim_{k \rightarrow \infty} F(x_k, \tilde{y}_k) < F(x, y).$$

Consider the sequence $\tilde{y}_k \subseteq Y$, $\tilde{y}_k = y$. Then, for the continuity of F ,

$$\lim_{k \rightarrow \infty} F(x_k, \tilde{y}_k) = \lim_{k \rightarrow \infty} F(x_k, y) = F(x, y),$$

which leads us to an absurd.

Then, (H_2) holds thanks to Theorem 5.4.3, and (H_1) is trivial since L and F are continuous, $\lim_{n \rightarrow \infty} \gamma_n = \lim_{n \rightarrow \infty} \beta_n = +\infty$ and X and Y are compact sets.

Remark 6.1.3. In Chapter 5, we proved the same result under weaker assumptions.

6.2 Tikhonov regularization

The aim of this section is to investigate the use of Tikhonov regularization and introduce two distinct procedures that reframe the lower-level problem.

In the first approach, we will select the least norm solution of (P_x) .

In the second approach, we will apply the Tikhonov regularization to perturb L and F with respect to the player's variable only. Specifically, we will subtract from each player's function the Euclidean norm of their corresponding variable.

6.2.1 First approach: least norm

In convex optimization problems with an infinite number of solutions, a well-known approach is to select the solution with the *least norm*.

In the context of Stackelberg's problems, this idea has led to a new problem, denoted as Q_x :

$$\min_{y \in M(x)} \|y\|. \quad (Q_x)$$

Thus, assuming that (Q_x) has a unique solution $\bar{y}(x)$ for each $x \in X$, the problem (S) becomes:

$$\max_{x \in X} L(x, \bar{y}(x)). \quad (\bar{S})$$

Unfortunately, this new problem can be ill-posed, as shown in the following example:

Example 6.2.1. Let us choose $X = Y = [0, 1]$, and

$$L(x, y) = xy, \quad F(x, y) = y \left(\frac{1}{2} - x \right).$$

Then,

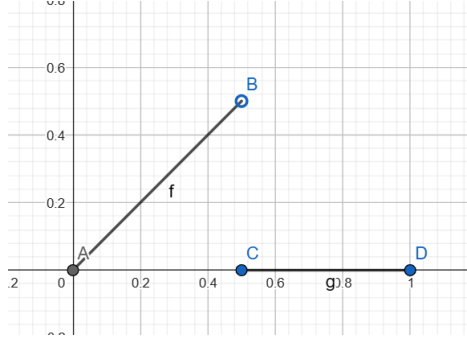
$$M(x) = \operatorname{Arg} \max_{y \in [0, 1]} y \left(\frac{1}{2} - x \right) = \begin{cases} \{1\} & \text{if } x \in [0, \frac{1}{2}) \\ [0, 1] & \text{if } x = \frac{1}{2} \\ \{0\} & \text{if } x \in (\frac{1}{2}, 1] \end{cases}$$

is not a singleton for $x = \frac{1}{2}$, and the solution with the least norm is

$$\bar{y}(x) = \begin{cases} \{1\} & \text{if } x \in [0, \frac{1}{2}) \\ \{0\} & \text{if } x \in [\frac{1}{2}, 1]. \end{cases}$$

So we obtain a new Stackelberg problem

$$\max_{x \in X} L(x, \bar{y}(x)) = \max_{x \in X} \begin{cases} x & \text{if } x \in [0, \frac{1}{2}) \\ 0 & \text{if } x \in [\frac{1}{2}, 1] \end{cases} = \emptyset.$$

Figure 6.1: $L(x, \bar{y}(x))$

This issue has been managed in [16], where the authors investigate the concept of regularized solutions.

6.2.2 Second approach: Tikhonov regularization

Let us consider $\Gamma = (L, F)$ in the family G (see Definition 5.2.1), and let $\hat{y}(x)$ be the minimum norm solution to (P_x) :

$$\hat{y}(x) = \operatorname{Arg} \min_{y \in M(x)} \|y\|.$$

The minimum norm solution exists because $M(x)$ is closed, convex, and non-empty. For any $x \in X$, we can consider the *Tikhonov regularized* lower level problem:

$$\max_{y \in Y} [F(x, y) - \alpha \|y\|^2] \quad (P_x^\alpha)$$

that has a unique solution denoted by $\bar{y}_\alpha(x)$. The quadratic function $\alpha \|y\|^2$ represents a penalty term that increases as the size of the player's variable increases.

Lemma 6.2.2. *Let $\Gamma \in G$, $\alpha_n \rightarrow 0^+$ and*

$$y_{\alpha_n}(x) = \operatorname{Arg} \max_{y \in Y} (F(x, y) - \alpha_n \|y\|^2). \quad (6.1)$$

Then,

$$\lim_{\alpha_n \rightarrow 0^+} y_{\alpha_n}(x) = \hat{y}(x).$$

Proof. Let us observe that, by utilizing (6.1) and considering that $\hat{y} \in M(x) \subseteq Y$,

$$F(x, y_{\alpha_n}(x)) - \alpha_n \|y_{\alpha_n}(x)\|^2 \geq F(x, \hat{y}) - \alpha_n \|\hat{y}\|^2,$$

so

$$\|y_{\alpha_n}(x)\|^2 \leq \frac{F(x, y_{\alpha_n}(x)) - F(x, \hat{y})}{\alpha_n} + \|\hat{y}\|^2.$$

Being $\hat{y} \in M(x)$, $F(x, y_{\alpha_n}(x)) - F(x, \hat{y}) \leq 0$, so

$$\|y_{\alpha_n}(x)\|^2 \leq \|\hat{y}\|^2.$$

Furthermore, since $y_{\alpha_n}(x) \subseteq Y$ and Y is a compact set, there exists a subsequence (which, for simplicity, we will still denote as α_n) such that

$$y_{\alpha_n}(x) \rightarrow y^* \in Y,$$

and it holds that

$$\|y^*\|^2 = \lim_{\alpha_n \rightarrow 0^+} \|y_{\alpha_n}(x)\|^2 \leq \|\hat{y}\|^2. \quad (6.2)$$

Let us prove that $y^* \in M(x)$: applying (6.1) we have that

$$\begin{aligned} F(x, y_{\alpha_n}(x)) - \alpha_n \|y_{\alpha_n}(x)\|^2 &\geq F(x, y) - \alpha_n \|y\| \quad \text{for every } y \in Y \\ \Rightarrow \limsup_{\alpha_n \rightarrow 0^+} (F(x, y_{\alpha_n}(x)) - \alpha_n \|y_{\alpha_n}(x)\|^2) &\geq F(x, y) \quad \text{for every } y \in Y. \end{aligned}$$

Because of the continuity of $y \rightarrow \alpha_n \|y\|^2$, if $\Gamma \in G$ then $(x, y) \rightarrow F(x, y) - \alpha_n \|y\|^2$ is an upper semicontinuous function, so

$$\limsup_{\alpha_n \rightarrow 0^+} (F(x, y_{\alpha_n}(x)) - \alpha_n \|y_{\alpha_n}(x)\|^2) \leq F(x, y^*),$$

and we can conclude that

$$F(x, y^*) \geq F(x, y)$$

for every $y \in Y$, so $y^* \in M(x)$. This, together with (6.2), imply that $y^* = \hat{y}$, so

$$\lim_{\alpha_n \rightarrow 0^+} y_{\alpha_n}(x) \rightarrow \hat{y}(x).$$

This confirms the thesis and completes the proof. $\square$

Now we can consider the regularized Stackelberg problem:

$$\max_{x \in X} L(x, \bar{y}_\alpha(x)), \quad (S_\alpha)$$

and the following theorem holds:

Theorem 6.2.3. *Let $\Gamma \in G$, $\alpha_n \rightarrow 0^+$ and*

$$\begin{cases} \bar{x}_n \text{ be a solution to } (S_{\alpha_n}) \\ \bar{y}_n \text{ be the solution to } (P_{\bar{x}_n}^{\alpha_n}) \end{cases}$$

If $(\bar{x}_n, \bar{y}_n) \rightarrow (\bar{x}, \bar{y}) \in X \times Y$, then $(\bar{x}, \bar{y})$ generates an SPNE $(\bar{x}, \tilde{y}(\cdot))$, where $\tilde{y}(\cdot)$ is defined as

$$\tilde{y}(x) = \begin{cases} \bar{y} & \text{if } x = \bar{x} \\ \hat{y}(x) & \text{if } x \neq \bar{x}. \end{cases}$$

Proof. First of all, note that because of the continuity of $y \rightarrow \alpha_n \|y\|^2$, if $\Gamma \in G$ then $F(x, y) - \alpha_n \|y\|^2$ is an upper semicontinuous function that satisfies (F2). So Theorem 2.1.4 holds and $\bar{y}_{\alpha_n}(x) = \text{Arg max}_{y \in Y} [F(x, y) - \alpha_n \|y\|^2]$ is a continuous function. Thus (S_{α_n}) has a solution thanks to Theorem 1.2.13. Assume that $(\bar{x}_n, \bar{y}_n) \rightarrow (\bar{x}, \bar{y})$, then we have to show that Definition 2.2.3 is satisfied. For the first condition, it is sufficient to note that $M(\bar{x})$ is closed (shown in the proof of Theorem 2.1.3). When $x \neq \bar{x}$, from the definition it is obvious that $\tilde{y}(x) = \hat{y}(x) \in M(x)$.

For the second condition, we have to prove that

$$L(\bar{x}, \bar{y}) \geq L(x, \tilde{y}(x))$$

for every $x \in X$. By definition, $(\bar{x}_n, \bar{y}_n)$ satisfies $L(\bar{x}_n, \bar{y}_n) \geq L(x, \bar{y}_{\alpha_n}(x))$ for every $x \in X$, and $\bar{y}_{\alpha_n}(x) \rightarrow \hat{y}(x)$ for $\alpha_n \rightarrow 0^+$ for Lemma 6.2.2. Hence,

$$L(\bar{x}, \bar{y}) \geq \limsup_{n \rightarrow \infty} L(\bar{x}_n, \bar{y}_n)$$

thanks to the upper semicontinuity of L . Moreover,

$$\limsup_{n \rightarrow \infty} L(\bar{x}_n, \bar{y}_n) \geq \liminf_{n \rightarrow \infty} L(x, \bar{y}_{\alpha_n}(x)) \geq L(x, \hat{y}(x))$$

for the lower semicontinuity of L on Y . □

Notice that this result provides a constructive way to approach an SPNE.

Remark 6.2.4. As in the case of Cost-to-Move Procedure, under the assumptions (A), the hypothesis of Theorem 6.1.1 are satisfied for (F_n) , (L_n) generated via Tikhonov regularization:

- (H_1) holds since L and F are continuous, $\lim_{n \rightarrow \infty} \alpha_n = 0$ and X and Y are compact sets,
- (H_2) has been shown in the proof of Theorem 6.2.3: $y_{\alpha_n}(x) \rightarrow \hat{y}(x)$ for any $\alpha_n \rightarrow 0$.

Nevertheless, the resulting limit is not necessarily a weak or strong Stackelberg solution, as shown in the following example:

Example 6.2.5. Let $X = [-2, 2]$, $Y = [-1, 1]$ and

$$L(x, y) = x + y, \quad F(x, y) = \begin{cases} -(x + \frac{7}{4})y & \text{if } x \in [-2, -\frac{7}{4}) \\ 0 & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ (-x + \frac{7}{4})y & \text{if } x \in (\frac{7}{4}, 2]. \end{cases}$$

Let us compute

$$M(x) = \operatorname{Arg} \max_{y \in [-1, 1]} F(x, y) = \begin{cases} \{1\} & \text{if } x \in [-2, -\frac{7}{4}) \\ [-1, 1] & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ \{-1\} & \text{if } x \in (\frac{7}{4}, 2] \end{cases}$$

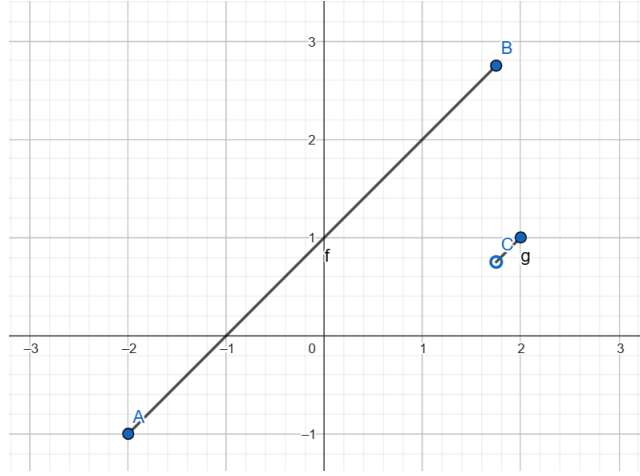
and observe that the minimum norm solution to (P_x) is

$$\hat{y}(x) = \begin{cases} \{1\} & \text{if } x \in [-2, -\frac{7}{4}) \\ \{0\} & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ \{-1\} & \text{if } x \in (\frac{7}{4}, 2]. \end{cases}$$

To get the strong Stackelberg solution, let us calculate

$$\operatorname{Arg} \max_{y \in M(x)} L(x, y) = \operatorname{Arg} \max_{y \in [-1, 1]} \begin{cases} x + 1 & \text{if } x \in [-2, -\frac{7}{4}) \\ x + y & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ x - 1 & \text{if } x \in (\frac{7}{4}, 2] \end{cases} = \begin{cases} x + 1 & \text{if } x \in [-2, -\frac{7}{4}] \\ x - 1 & \text{if } x \in (\frac{7}{4}, 2]. \end{cases}$$

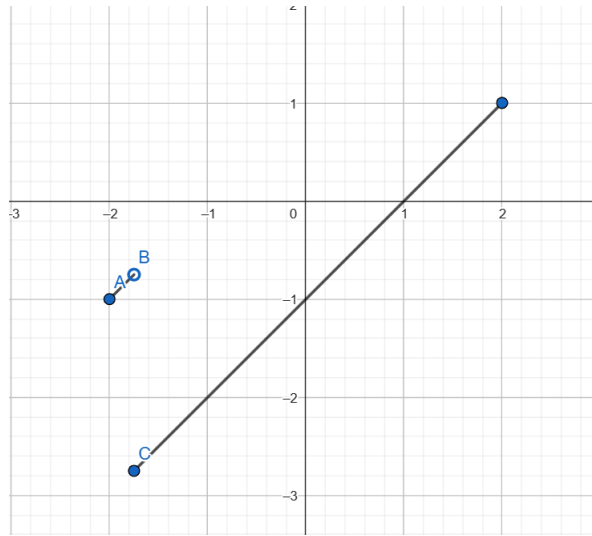
So, the strong Stackelberg equilibrium is $(\frac{7}{4}, 1)$:

Figure 6.2: $\text{Arg max}_{y \in M(x)} L(x, y)$

Conversely, to calculate the weak Stackelberg solution, we compute

$$\text{Arg min}_{y \in M(x)} L(x, y) = \text{Arg min}_{y \in [-1, 1]} \begin{cases} x + 1 & \text{if } x \in [-2, -\frac{7}{4}) \\ x + y & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ x - 1 & \text{if } x \in (\frac{7}{4}, 2] \end{cases} = \begin{cases} x + 1 & \text{if } x \in [-2, -\frac{7}{4}) \\ x - 1 & \text{if } x \in [-\frac{7}{4}, 2]. \end{cases}$$

And the weak Stackelberg equilibrium is (2, -1):

Figure 6.3: $\text{Arg max}_{y \in M(x)} L(x, y)$

On the other hand, with regard to the regularised solution, we calculate

$$y_n(x) = \operatorname{Arg max}_{y \in [-1, 1]} F(x, y) - \alpha_n y^2 = \operatorname{Arg max}_{y \in [-1, 1]} \begin{cases} -(x + \frac{7}{4})y - \alpha_n y^2 & \text{if } x \in [-2, -\frac{7}{4}) \\ -\alpha_n y^2 & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ (-x + \frac{7}{4})y - \alpha_n y^2 & \text{if } x \in (\frac{7}{4}, 2] \end{cases}$$

$$= \begin{cases} \{1\} & \text{if } x \in [-2, -\frac{7}{4} - 2\alpha_n) \\ \{\frac{-x - \frac{7}{4}}{2\alpha_n}\} & \text{if } x \in [-\frac{7}{4} - 2\alpha_n, -\frac{7}{4}) \\ \{0\} & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ \{\frac{x - \frac{7}{4}}{-2\alpha_n}\} & \text{if } x \in (\frac{7}{4}, \frac{7}{4} + 2\alpha_n] \\ \{-1\} & \text{if } x \in (\frac{7}{4} + 2\alpha_n, 2]. \end{cases}$$

Furthermore,

$$\operatorname{Arg max}_{x \in [-2, 2]} L(x, y_n) = \operatorname{Arg max}_{x \in [-2, 2]} \begin{cases} x + 1 & \text{if } x \in [-2, -\frac{7}{4} - 2\alpha_n) \\ x + \frac{-x - \frac{7}{4}}{2\alpha_n} & \text{if } x \in [-\frac{7}{4} - 2\alpha_n, -\frac{7}{4}) \\ x + 0 & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ x + \frac{x - \frac{7}{4}}{-2\alpha_n} & \text{if } x \in (\frac{7}{4}, \frac{7}{4} + 2\alpha_n] \\ x - 1 & \text{if } x \in (\frac{7}{4} + 2\alpha_n, 2] \end{cases}$$

$$= \begin{cases} \{-\frac{7}{4} - 2\alpha_n\} & \text{if } x \in [-2, -\frac{7}{4} - 2\alpha_n) \\ \{-\frac{7}{4} - 2\alpha_n\} & \text{if } x \in [-\frac{7}{4} - 2\alpha_n, -\frac{7}{4}) \\ \{\frac{7}{4}\} & \text{if } x \in [-\frac{7}{4}, \frac{7}{4}] \\ \{\frac{7}{4}\} & \text{if } x \in (\frac{7}{4}, \frac{7}{4} + 2\alpha_n] \\ \{2\} & \text{if } x \in (\frac{7}{4} + 2\alpha_n, 2] \end{cases}.$$

so, evaluating the functions, we obtain that the solution for the regularized problem converges to $(\frac{7}{4}, 0)$, which is different from the strong and the weak solutions.

6.2.3 Comparison with the Cost-to-Move procedure

We observe that when constructing an SPNE using the Tikhonov regularization, there is no step-by-step learning occurring. Neither the leader nor the follower updates

their strategies taking into account their previous choices. Therefore, the anchoring effect as well as other behavioral motivations are absent in this alternative approach.

Moreover, the two procedures do not generate the same SPNE:

Example 6.2.6. Let Γ be the game defined in Example 7.0.2. We want to construct the SPNE using the Tikhonov regularization.

$$\begin{aligned} \operatorname{Arg max}_{y \in [-1, 1]} F(x, y) &= \operatorname{Arg max}_{y \in [-1, 1]} \begin{cases} 0 & \text{if } x \in [\frac{1}{2}, 1] \\ (1-x)y & \text{if } x \in (1, 2] \end{cases} \\ &= \begin{cases} [-1, 1] & \text{if } x \in [\frac{1}{2}, 1] \\ \{-1\} & \text{if } x \in (1, 2], \end{cases} \end{aligned}$$

so the minimum norm solution is

$$\hat{y}(x) = \begin{cases} \{0\} & \text{if } x \in [\frac{1}{2}, 1] \\ \{-1\} & \text{if } x \in (1, 2]. \end{cases}$$

$$\begin{aligned} y_n(x) &= \operatorname{Arg max}_{y \in [-1, 1]} [F(x, y) - \alpha_n ||y||^2] = \operatorname{Arg max}_{y \in [-1, 1]} \begin{cases} -\alpha_n y^2 & \text{if } x \in [\frac{1}{2}, 1] \\ (1-x)y - \alpha_n y^2 & \text{if } x \in (1, 2] \end{cases} \\ &= \begin{cases} \{0\} & \text{if } x \in [\frac{1}{2}, 1] \\ \{\frac{1-x}{2\alpha_n}\} & \text{if } x \in (1, 1 + 2\alpha_n) \\ \{-1\} & \text{if } x \in [1 + 2\alpha_n, 2]. \end{cases} \end{aligned}$$

$$\begin{aligned} \operatorname{Arg max}_{x \in [\frac{1}{2}, 2]} -x - y_n(x) &= \operatorname{Arg max}_{y \in [-1, 1]} \begin{cases} -x & \text{if } x \in [\frac{1}{2}, 1] \\ -x - \frac{1-x}{2\alpha_n} & \text{if } x \in (1, 1 + 2\alpha_n) \\ -x + 1 & \text{if } [1 + 2\alpha_n, 2] \end{cases} \\ &= \begin{cases} \{\frac{1}{2}\} & \text{if } x \in [\frac{1}{2}, 1] \\ \{1 + 2\alpha_n\} & \text{if } x \in (1, 1 + 2\alpha_n) \\ \{1 + 2\alpha_n\} & \text{if } [1 + 2\alpha_n, 2]. \end{cases} \end{aligned}$$

So, the SPNE generated by Theorem 6.2.3 is $(1, \tilde{y})$, where

$$\tilde{y}(x) = \begin{cases} 0 & \text{if } x \in [\frac{1}{2}, 1) \\ -1 & \text{if } x \in [1, 2], \end{cases}$$

that does not coincide with the SPNE found out in Exercise 7.0.2.

6.3 Slightly altruistic behavior

Let us consider a perturbation based on an *altruistic behavior* of the players. For any $n \in \mathbb{N}$ we define

$$\Gamma_n = (L_{\alpha_n}, F_{\alpha_n})$$

where $\alpha_n \geq 0$, $\beta_n > 0$, $\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n = 0$ and

$$L_{\alpha_n}(x, y) = L(x, y) + \alpha_n F(x, y), \quad F_{\beta_n}(x, y) = F(x, y) + \beta_n L(x, y).$$

Namely, we modify the payoff functions by incorporating a “small” proportion of the other player payoff function. In this scenario, players also consider the other player’s benefit. When selecting their own actions, both the leader and the follower make a compromise between maximizing their original payoff functions and slightly improving the payoff functions of the other player.

Now we show that the SPNEs of Γ_n converge to an SPNE of Γ .

Proposition 6.3.1. *Assume that assumptions (A) hold and $F(x, \cdot)$ is strictly concave on Y for any $x \in X$. Then,*

- *the sequence of strategy profiles $(\bar{x}_n, \bar{\varphi}_n)_n$ where*

$$\{\bar{\varphi}_n(x)\} = \operatorname{Arg} \max_{y \in Y} F_{\beta_n}(x, y), \quad \bar{x}_n \in \operatorname{Arg} \max_{x \in X} L_{\alpha_n}(x, \bar{\varphi}_n(x))$$

is well-defined and $(\bar{x}_n, \bar{\varphi}_n)$ is an SPNE of Γ_n for any $n \in \mathbb{N}$,

- *the sequences $(L_{\alpha_n})_n$, $(F_{\beta_n})_n$, and $(\bar{\varphi}_n)_n$ satisfies the assumptions (H_1) and (H_2) in Theorem 6.1.1 with the pointwise limit of the sequence $(\bar{\varphi}_n)_n$ defined by*

$$\{\bar{\varphi}(x)\} = \operatorname{Arg} \max_{y \in M(x)} L(x, y), \tag{6.3}$$

where M denotes the follower’s best reply correspondence.

Proof. Note that $\bar{\varphi}_n$ is well-defined thanks to Theorem 1.2.13, and it is a continuous function. Indeed, it is upper semicontinuous thanks to the Berge Maximum Theorem 2.1.3; to show that it is lower semicontinuous, we use Remark 1.2.12 and it is sufficient to prove that $\text{lev}_{\leq \alpha} \varphi_n$ is closed for every α , so we have to show that the set

$$\{x \in X \mid \text{Arg max}_{y \in Y} F_{\beta_n}(x, y) \leq \alpha\}$$

is closed, and this follows from Berge Theorem, which proves that $\bar{\varphi}_n$ has nonempty compact values.

Hence, in light of assumptions (A), the function $L_{\alpha_n}(\cdot, \varphi_n(\cdot))$ is continuous over the compact set X , so $\bar{x}_n$ is well-defined and the first item is proved. By assumption (A), the sequences $(L_{\alpha_n})_n$ and $(F_{\beta_n})_n$ continuously converge to L and F , respectively, so assumption (H_1) in Theorem 6.1.1 is satisfied.

Now let us show that $(\bar{\varphi}_n)_n$ continuously converge to $\bar{\varphi}$ defined in (6.3). Firstly, note that $\bar{\varphi}$ is well defined since $L(x, \cdot)$ is continuous and strictly concave over Y for any $x \in X$, M has compact values thanks to Berge Theorem, and it can be proved that it has also convex values: let $t, v \in M(x)$, so it holds that

$$\begin{cases} F(x, t) \geq F(x, z) & \text{for every } z \in Y, \\ F(x, v) \geq F(x, z) & \text{for every } z \in Y. \end{cases}$$

We have to prove that $F(x, \lambda t + (1 - \lambda)v) \geq F(x, z)$, for every $z \in Y$; we use the concavity of F :

$$F(x, \lambda t + (1 - \lambda)v) \geq \lambda F(x, t) + (1 - \lambda)F(x, v) \geq F(x, z).$$

Fix $x \in X$ and consider a subsequence $(\bar{\varphi}_{n_k}(x))_k$ of $(\bar{\varphi}_n(x))_n$ converging to $\bar{y}$, whose existence is guaranteed by the compactness of Y . On the one hand, in light of the definition of $\bar{\varphi}_n$, we have

$$F(x, \bar{y}) = \lim_{k \rightarrow \infty} F_{\beta_{n_k}}(x, \bar{\varphi}_{n_k}(x)) \geq \lim_{k \rightarrow \infty} F_{\beta_{n_k}}(x, y) = F(x, y),$$

for any $y \in Y$. So $\bar{y} \in M(x)$, and the definition of $\bar{\varphi}$ implies that

$$L(x, \bar{y}) \leq L(x, \bar{\varphi}(x)). \quad (6.4)$$

On the other hand, we get

$$\begin{aligned} F(x, \bar{\varphi}_{n_k}(x)) + \beta_{n_k} L(x, \bar{\varphi}_{n_k}(x)) &= F_{\beta_{n_k}}(x, \bar{\varphi}_{n_k}(x)) \geq F_{\beta_{n_k}}(x, \bar{\varphi}(x)) \\ &= F(x, \bar{\varphi}(x)) + \beta_{n_k} L(x, \bar{\varphi}(x)) \geq F(x, \bar{\varphi}_{n_k}(x)) + \beta_{n_k} L(x, \bar{\varphi}(x)). \end{aligned}$$

Since $(\beta_n)_n \subseteq (0, +\infty)$, it follows that

$$L(x, \bar{\varphi}_{n_k}(x)) \geq L(x, \bar{\varphi}(x))$$

which, taking the limit as k goes to $+\infty$, implies

$$L(x, \bar{y}) \geq L(x, \bar{\varphi}(x)). \quad (6.5)$$

In light of the uniqueness of the maximizer of $L(x, \cdot)$ over $M(x)$, from the inequalities (6.4) and (6.5), it follows that $\bar{y} = \bar{\varphi}(x)$. Since this holds for any convergent subsequence of $(\bar{\varphi}_n(x))_n$ in the compact set Y , then $\lim_{n \rightarrow \infty} \bar{\varphi}_n(x) = \bar{\varphi}(x)$. Hence, the sequence $(\bar{\varphi}_n)_n$ pointwise converges to $\bar{\varphi}$, and the second item is proved. $\square$

Theorem 6.3.2. *Suppose that assumptions (A) holds and $F(x, \cdot)$ is strictly concave on Y for any $x \in X$. Let $(\bar{x}_n, \bar{\varphi}_n)_n$ be the sequence of SPNEs of Γ_n . If the sequence $(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n))_n \subseteq X \times Y$ converges to $(\bar{x}, \bar{y})$ and*

$$\tilde{y}(x) = \begin{cases} \text{Arg max}_{y \in M(x)} L(x, y) & \text{if } x \neq \bar{x} \\ \bar{y} & \text{if } x = \bar{x}, \end{cases}$$

then the strategy profile $(\bar{x}, \tilde{y})$ is an SPNE of Γ .

This theorem follows straightly from Proposition 6.3.1.

In order to remove the hypothesis of strict concavity of $L(x, \cdot)$, we combine the slightly altruistic perturbation with the Tikhonov regularization. Let $\alpha_n \geq 0$, $\beta_n > 0$, $\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n = 0$ and

$$L_{\alpha_n}(x, y) = L(x, y) + \alpha_n F(x, y), \quad F_{\beta_n}^T(x, y) = F(x, y) + \beta_n [L(x, y) - \|y\|^2].$$

Replacing F_{β_n} with $F_{\beta_n}^T$ in Proposition 6.3.1, we only need the concavity of $L(x, \cdot)$ in order to guarantee the assertion of the proposition, where the pointwise limit of the

sequence $\{\bar{\varphi}_n^T(x)\} = \text{Arg max}_{y \in Y} F_{\beta_n}^T(x, y)$ is

$$\{\bar{\varphi}^T(x)\} = \text{Arg max}_{y \in M(x)} [L(x, y) - ||y||^2].$$

Then, the following theorem holds:

Theorem 6.3.3. *Assume that assumptions (A) holds and $L(x, \cdot)$ is concave on Y for any $x \in X$. Let $(\bar{x}_n, \bar{\varphi}_n)$ be the sequence of SPNEs of $\Gamma_n = (L_{\alpha_n}, F_{\beta_n}^T)$. If the sequence $(\bar{x}_n, \bar{\varphi}_n(\bar{x}_n))_n \subseteq X \times Y$ converges to $(\bar{x}, \bar{y})$ and*

$$\{\hat{\varphi}(x)\} = \begin{cases} \text{Arg max}_{y \in M(x)} [L(x, y) - ||y||^2] & \text{if } x \neq \bar{x} \\ \bar{y} & \text{if } x = \bar{x}, \end{cases}$$

then the strategy profile $(\bar{x}, \hat{\varphi})$ is an SPNE of Γ .

Remark 6.3.4. Note that this modification allows us to weaken the strict concavity assumption on $L(x, \cdot)$, but makes the follower “less altruistic”.

Chapter 7

Appendix

In this chapter, we provide examples that illustrate notable properties of the SP-NEs obtained through Procedure S_n .

Example 7.0.1. Let $X = Y = [-1, 1]$ and $\Gamma = (L, F)$, where

$$L(x, y) = y, \quad F(x, y) = -xy.$$

The follower's best reply correspondence M is defined on $[-1, 1]$ by

$$M(x) = \begin{cases} \{1\} & \text{if } x \in [-1, 0) \\ [-1, 1] & \text{if } x = 0 \\ \{-1\} & \text{if } x \in (0, 1]. \end{cases}$$

Let $(\bar{x}_0, \bar{y}_0) = (1, 1)$ be the initial point of the procedure and let $\beta_n = \gamma_n = 2^n$ for any $n \in \mathbb{N} \cup \{0\}$. The Procedure S_n generates the following sequence of strategy profiles:

$$\bar{x}_n = 0, \quad \varphi_n(x) = \begin{cases} 1 & \text{if } x \in [-1, 0) \\ 1 - a_n x & \text{if } x \in [0, \frac{2}{a_n}) \\ -1 & \text{if } x \in (\frac{2}{a_n}, 1], \end{cases} \quad (7.1)$$

where the sequence $(a_n)_n$ is recursively defined by

$$\begin{cases} a_1 = 1 \\ a_{n+1} = a_n + 2^n \text{ for any } n \geq 1. \end{cases}$$

Firstly, note that $\Gamma \in G$. We prove (7.1) by induction on n . Let $n = 1$, then

$$\begin{aligned} \operatorname{Arg max}_{y \in [-1, 1]} F_1(x, y) &= \operatorname{Arg max}_{y \in [-1, 1]} -xy - \frac{(y-1)^2}{2} \\ &= \operatorname{Arg max}_{y \in [-1, 1]} \frac{-y^2}{2} + y(1-x) \\ &= \begin{cases} 1 & \text{if } x \in [-1, 0) \\ 1-x & \text{if } x \in [0, 1], \end{cases} \end{aligned}$$

since $y \rightarrow \frac{-y^2}{2} + y(1-x)$ is a parabola with vertex $V = (\frac{1-x}{1}, \cdot)$, and

$$\{\bar{x}_1\} = \operatorname{Arg max}_{x \in X} L_1(x, \varphi_1(x)), \quad \text{where } L_1(x, \varphi_1(x)) = \begin{cases} -\frac{x^2-2x-1}{2} & \text{if } x \in [-1, 0) \\ -\frac{x^2-1}{2} & \text{if } x \in [0, 1], \end{cases}$$

so

$$\{\bar{x}_1\} = \operatorname{Arg max}_{x \in [-1, 1]} \begin{cases} -\frac{x^2}{2} + x & \text{if } x \in [-1, 0) \\ -\frac{x^2}{2} & \text{if } x \in [0, 1] \end{cases} = 0,$$

since $x \rightarrow -\frac{x^2}{2} + x$ is a parabola with vertex $(2, \cdot)$ and $x \rightarrow -\frac{x^2}{2}$ is a parabola with vertex $V = (0, 0)$. The base case is fulfilled as $a_1 = 1$.

Assume that (7.1) holds for $n > 1$. So

$$F_{n+1}(x, y) = \begin{cases} P_1(x, y) = -\frac{y^2}{2^{n+1}} - \left(x - \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}} & \text{if } x \in [-1, 0) \\ P_2(x, y) = -\frac{y^2}{2^{n+1}} - \left(x + \frac{a_n x - 1}{2^n}\right)y - \frac{(1-a_n x)^2}{2^{n+1}} & \text{if } x \in \left[0, \frac{2}{a_n}\right] \\ P_3(x, y) = -\frac{y^2}{2^{n+1}} - \left(x + \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}} & \text{if } x \in \left(\frac{2}{a_n}, 1\right], \end{cases}$$

and

$$L_{n+1}(x, y) = -\frac{x^2}{2^{n+1}} + y. \quad (7.2)$$

If $x \in [-1, 0)$, then the unique maximizer of $P_1(x, \cdot)$ on $Y = [-1, 1]$ is 1 since the abscissa of the vertex of the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x - \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}$ is $V = (1 - 2^n, \cdot)$,

and $1 - 2^n > 0$ for every $x \in [-1, 0)$. If $x \in [0, \frac{2}{a_n}]$, we have the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x + \frac{a_n x - 1}{2^n}\right)y - \frac{(1 - a_n x)^2}{2^{n+1}}$ with vertex $V = (1 - x(2^n + a_n), \cdot)$.

$$\begin{aligned} 1 - x(2^n + a_n) < -1 &\iff x > \frac{2}{2^n + a_n} \\ &\iff 1 - x(2^n + a_n) < 1 \\ &\iff x(2^n + a_n) > 0 \\ &\iff x > 0. \end{aligned}$$

So, in the first case, if $x \in [\frac{2}{2^n + a_n}, \frac{2}{a_n}]$, the unique maximizer is -1 , while if $x \in [0, \frac{2}{2^n + a_n})$, the unique maximizer is $1 - x(2^n + a_n)$.

If $x \in (\frac{2}{a_n}, 1]$, then the unique maximizer of the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x + \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}$ is -1 since the vertex is $V = (-2^n x - 1, \cdot)$, and $-2^n x - 1 < -1$ for every $x \in (\frac{2}{a_n}, 1]$. Thus, since $2^n + a_n = a_{n+1}$,

$$\{\varphi_{n+1}(x)\} = \operatorname{Argmax}_{y \in Y} F_{n+1}(x, y) = \begin{cases} 1 & \text{if } x \in [-1, 0) \\ 1 - a_{n+1}x & \text{if } x \in [0, \frac{2}{a_{n+1}}] \\ -1 & \text{if } x \in (\frac{2}{a_{n+1}}, 1], \end{cases} \quad (7.3)$$

for any $x \in [-1, 1]$. Evaluating the function L_{n+1} given in (7.2) at $(x, \varphi_{n+1}(x))$, we get

$$\begin{aligned} L_{n+1}(x, \varphi_{n+1}(x)) &= \begin{cases} -\frac{x^2}{2^{n+1}} + 1 & \text{if } x \in [-1, 0) \\ -\frac{x^2}{2^{n+1}} - a_{n+1}x + 1 & \text{if } x \in [0, \frac{2}{a_{n+1}}] \\ -\frac{x^2}{2^{n+1}} - 1 & \text{if } x \in (\frac{2}{a_{n+1}}, 1] \end{cases} \\ &= \begin{cases} T_1(x) & \text{if } x \in [-1, 0) \\ T_2(x) & \text{if } x \in [0, \frac{2}{a_{n+1}}] \\ T_3(x) & \text{if } x \in (\frac{2}{a_{n+1}}, 1]. \end{cases} \end{aligned}$$

Since

- the abscissa of the vertices of the parabolas $x \rightarrow -\frac{x^2}{2^{n+1}} + 1$ and $x \rightarrow -\frac{x^2}{2^{n+1}} - 1$ is 0, and there the two functions take the values, respectively, 1 and -1 ,

- the abscissa of the vertex of the parabola $x \rightarrow -\frac{x^2}{2^{n+1}} - a_{n+1}x + 1$ is $-2^n a_{n+1} < 0$,
- $L_{n+1}(\cdot, \varphi_{n+1}(\cdot))$ is continuous on $[-1, 1]$,

it follows that T_2 can not take values higher than 1 for $x \in [0, \frac{2}{a_{n+1}}]$.

For example, let us consider $n = 1$:

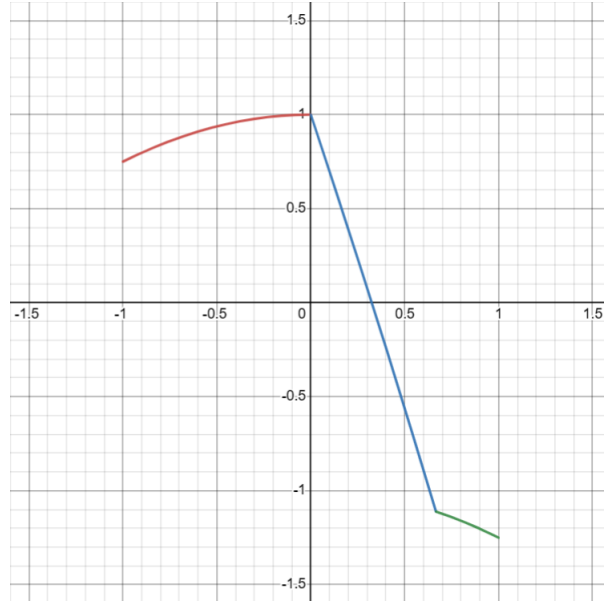


Figure 7.1: $L_2(x, \varphi_2(x))$

then

$$\{\bar{x}_{n+1}\} = \text{Arg max}_{x \in [-1, 1]} L_{n+1}(x, \varphi_{n+1}(x)) = \{0\}. \quad (7.4)$$

Equalities (7.2)-(7.4) demonstrate the inductive step, so (7.1) holds. From $\lim_{n \rightarrow +\infty} a_n = +\infty$, we get

$$\bar{x} = 0, \quad \bar{\varphi}(x) = \begin{cases} 1 & \text{if } x \in [-1, 0] \\ -1 & \text{if } x \in (0, 1]. \end{cases}$$

Since $\lim_{n \rightarrow +\infty} \varphi_n(\bar{x}_n) = 1$, then the SPNE constructed according to Theorem 5.5.1 is $(0, \bar{\varphi})$. In this case, $\bar{\varphi}$ coincides with the pointwise limit of $(\varphi_n)_n$:

$$\lim_{n \rightarrow \infty} \varphi_n(\bar{x}_n) = 1 = \lim_{n \rightarrow \infty} \varphi_n(\lim_{n \rightarrow \infty} \bar{x}_n)$$

since $\bar{x}_n = 0$ for every $n \in \mathbb{N}$.

Let us note that Γ has infinitely many SPNEs. In fact, if we denote by $\hat{\varphi}^\alpha$ the function defined on $[-1, 1]$ by

$$\hat{\varphi}^\alpha(x) = \begin{cases} 1 & \text{if } x \in [-1, 0) \\ \alpha & \text{if } x = 0 \\ -1 & \text{if } x \in (0, 1], \end{cases} \quad (7.5)$$

the two conditions of Definition 2.2.3 are satisfied for

$$\{(\hat{x}, \hat{\varphi}^\alpha) \mid \hat{x} \in [-1, 0), \alpha \in [-1, 1]\}, \quad \{(0, \hat{\varphi}^1)\}.$$

Indeed, $\hat{\varphi}^\alpha(x) = M(x)$ when $\alpha \in [-1, 1]$, and in order to maximize L the leader should choose $\hat{x} \in [-1, 0)$. If $\alpha = 1$, then

$$\hat{\varphi}^\alpha(x) = \begin{cases} 1 & \text{if } x \in [-1, 0] \\ -1 & \text{if } x \in (0, 1]. \end{cases}$$

Therefore the leader will choose $\hat{x} \in [-1, 0]$, with zero included. Only one of these SPNEs is obtained via our method, so the selection is effective. Moreover, the SPNE obtained via the Procedure S_n is the closest to the starting point among the SPNEs: $\bar{x} = 0$ is the nearest point to $\bar{x}_0 = 1$ among all the follower's strategies being part of the SPNE, and also $\bar{\varphi}(x)$ minimizes the distance from $\bar{y}_0 = 1$ for any $x \in [-1, 1]$.

Example 7.0.2. Let $X = [\frac{1}{2}, 2]$, $Y = [-1, 1]$ and $\Gamma = (L, F)$, where

$$L(x, y) = -x - y, \quad F(x, y) = \begin{cases} 0 & \text{if } x \in [\frac{1}{2}, 1] \\ (1 - x)y & \text{if } x \in (1, 2]. \end{cases}$$

The follower's best reply correspondence M is given by

$$M(x) = \begin{cases} [-1, 1] & \text{if } x \in [\frac{1}{2}, 1] \\ \{-1\} & \text{if } x \in (1, 2]. \end{cases}$$

Let $(\bar{x}_0, \bar{y}_0) = (1, 1)$ and $\beta_n = \gamma_n = n + 1$ for any $n \in \mathbb{N} \cup \{0\}$. Then, Procedure S_n results in the following sequence of strategies:

$$\begin{aligned} \bar{x}_n &= \begin{cases} \frac{1}{2} & \text{if } n = 1 \\ 1 + \frac{2}{a_n} & \text{if } n \geq 2 \end{cases} \\ \varphi_1(x) &= \begin{cases} 1 & \text{if } x \in [\frac{1}{2}, 1] \\ 2 - x & \text{if } x \in (1, 2], \end{cases} \quad \varphi_n(x) = \begin{cases} 1 & \text{if } x \in [\frac{1}{2}, 1] \\ a_n + 1 - a_n x & \text{if } x \in (1, 1 + \frac{2}{a_n}] \\ -1 & \text{if } x \in (1 + \frac{2}{a_n}, 2], \end{cases} \end{aligned} \quad (7.6)$$

where the sequence $(a_n)_n$ is recursively defined by

$$\begin{cases} a_1 = 1 \\ a_{n+1} = a_n + n + 1 \quad \forall n \in \mathbb{N}. \end{cases}$$

Firstly, note that $\Gamma \in G$. We prove (7.6) by induction on n . First, we analyze the case $n = 1$:

$$\{\varphi_1(x)\} = \operatorname{Arg} \max_{y \in Y} F_1(x, y),$$

where

$$\begin{aligned} F_1(x, y) &= \begin{cases} 0 - \frac{(y-1)^2}{2\gamma_0} & \text{if } x \in [\frac{1}{2}, 1] \\ (1-x)y - \frac{1}{2\gamma_0}(y-1)^2 & \text{if } x \in (1, 2] \end{cases} \\ &= \begin{cases} -\frac{(y-1)^2}{2} & \text{if } x \in [\frac{1}{2}, 1] \\ -\frac{y^2}{2} + (2-x)y - \frac{1}{2} & \text{if } x \in (1, 2]. \end{cases} \end{aligned}$$

Note that $y \rightarrow -\frac{(y-1)^2}{2}$ is a parabola with vertex $V = (1, \cdot)$ and $y \rightarrow -\frac{y^2}{2} + (2-x)y - \frac{1}{2}$ is a parabola with vertex $V = (2-x, \cdot)$. Also

$$\begin{cases} 2-x > -1 & \iff x < 3 \\ 2-x < 1 & \iff x > 1 \end{cases} \Rightarrow 2-x \in [-1, 1] \quad \forall x \in (1, 2].$$

So

$$\varphi_1(x) = \begin{cases} 1 & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ 2 - x & \text{if } x \in (1, 2]. \end{cases}$$

Moreover

$$\{\bar{x}_1\} = \operatorname{Arg\,max}_{x \in X} L_1(x, \varphi_1(x)),$$

where

$$\begin{aligned} L_1(x, \varphi_1(x)) &= -x - \varphi_1(x) - \frac{1}{2}(x-1)^2 \\ &= -\frac{1}{2}x^2 - \varphi_1(x) - \frac{1}{2} \\ &= \begin{cases} -\frac{1}{2}x^2 - 1 - \frac{1}{2} & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -\frac{1}{2}x^2 - (2-x) - \frac{1}{2} & \text{if } x \in (1, 2] \end{cases} \\ &= \begin{cases} -\frac{x^2+3}{2} & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -\frac{x^2-2x+5}{2} & \text{if } x \in (1, 2]. \end{cases} \end{aligned}$$

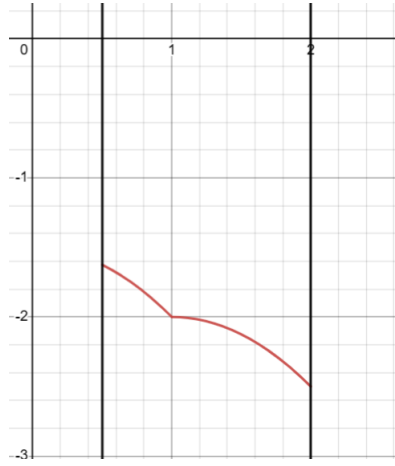


Figure 7.2: $L_1(x, \varphi_1(x))$

that is $\bar{x}_1 = \frac{1}{2}$, as shown in figure (7.2). So the case $n = 1$ is satisfied. Now suppose $n = 2$:

$$\{\varphi_2(x)\} = \operatorname{Arg\,max}_{y \in Y} F_2(x, y),$$

where

$$F_2(x, y) = \begin{cases} 0 - \frac{(y-1)^2}{2\gamma_1} & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ (1-x)y - \frac{1}{2\gamma_1}(y-2+x)^2 & \text{if } x \in (1, 2] \end{cases} = \begin{cases} -\frac{(y-1)^2}{4} & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -\frac{1}{4}y^2 + y(2 - \frac{3}{2}x) & \text{if } x \in (1, 2]. \end{cases}$$

Note that $y \rightarrow -\frac{(y-1)^2}{4}$ is a parabola with vertex $V = (1, \cdot)$ and $y \rightarrow -\frac{1}{4}y^2 + y(2 - \frac{3}{2}x)$ is a parabola with vertex $V = (4 - 3x, \cdot)$. Also

$$\begin{cases} 4 - 3x > -1 & \iff x < \frac{5}{3} \\ 4 - 3x < 1 & \iff x > 3. \end{cases}$$

So

$$\varphi_2(x) = \begin{cases} 1 & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ 4 - 3x & \text{if } x \in (1, \frac{5}{3}] \\ -1 & \text{if } x \in (\frac{5}{3}, 2]. \end{cases}$$

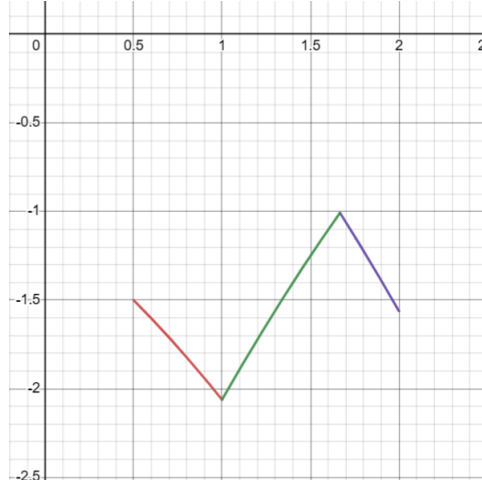
Moreover

$$\{\bar{x}_2\} = \operatorname{Arg} \max_{x \in X} L_2(x, \varphi_2(x)),$$

where

$$\begin{aligned} L_2(x, \varphi_2(x)) &= -x - \varphi_2(x) - \frac{1}{2}\left(x - \frac{1}{2}\right)^2 \\ &= \begin{cases} -x - 1 - \frac{1}{4}\left(-\frac{1}{2}\right)^2 & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -x - 4 + 3x - \frac{1}{4}\left(-\frac{1}{2}\right)^2 & \text{if } x \in (1, \frac{5}{3}] \\ -x + 1 - \frac{1}{4}\left(-\frac{1}{2}\right)^2 & \text{if } x \in (\frac{5}{3}, 2]. \end{cases} \end{aligned}$$

As shown in the next picture, the maximum is reached at $x = \frac{5}{3} = 1 + \frac{2}{a_2}$, as $a_2 = 3$.

Figure 7.3: $L_2(x, \varphi_2(x))$

So the base case is fulfilled. Assume that (7.6) holds for $n > 1$:

$$F_{n+1}(x, y) = F(x, y) - \frac{1}{2\lambda_n}(y - \varphi_n)^2$$

$$= \begin{cases} -\frac{(y-1)^2}{2(n+1)} & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -\frac{y^2}{2(n+1)} + \left(1 - x + \frac{a_{n+1}-a_n x}{n+1}\right)y - \frac{(a_{n+1}-a_n x)^2}{2(n+1)} & \text{if } x \in \left(1, 1 + \frac{2}{a_n}\right] \\ -\frac{y^2}{2(n+1)} + \left(1 - x - \frac{1}{n+1}\right)y - \frac{1}{2(n+1)} & \text{if } x \in \left(1 + \frac{2}{a_n}, 2\right] \end{cases}$$

and

$$L_{n+1}(x, y) = L(x, y) - \frac{1}{2(n+1)}(x - x_n)^2$$

$$= -\frac{x^2}{2(n+1)} - \left(1 - \frac{a_n + 2}{(n+1)a_n}\right)x - \frac{(a_n + 2)^2}{2(n+1)a_n^2} - y$$

If $x \in \left[\frac{1}{2}, 1\right]$, then the unique maximizer of $y \rightarrow -\frac{(y-1)^2}{2(n+1)}$ on $Y = [-1, 1]$ is 1 since the abscissa of the vertex of the parabola $y \rightarrow -\frac{(y-1)^2}{2(n+1)}$ is equal to the abscissa of the vertex of the parabola $y \rightarrow -y^2 + 2y - 1$, that is $V = (1, \cdot)$, so the maximizer is 1. If $x \in \left(1, 1 + \frac{2}{a_n + n + 1}\right]$, the vertex of the parabola $y \rightarrow -\frac{y^2}{2(n+1)} + \left(1 - x + \frac{a_{n+1}-a_n x}{n+1}\right)y - \frac{(a_{n+1}-a_n x)^2}{2(n+1)}$ is $V = (a_n + n + 2 - (n + 1 + a_n)x, \cdot)$. Note that

$$a_n + n + 2 - (n + 1 + a_n)x < 1 \iff x > 1,$$

and this holds for every $x \in (1, 1 + \frac{2}{a_n})$. Moreover

$$a_n + n + 2 - (n + 1 + a_n)x > -1 \iff x < \frac{3 + a_n + n}{1 + a_n + n} = 1 + \frac{2}{1 + a_n + n}.$$

Thus, for any $x \in (1, 1 + \frac{2}{a_n + n + 1}]$ the unique maximizer is $a_n + n + 2 - (n + 1 + a_n)x$, and for any $x \in (1 + \frac{2}{a_n + n + 1}, 1 + \frac{2}{a_n}]$ the unique maximizer is -1 . Finally, if $x \in (1 + \frac{2}{a_n}, 2]$, then the vertex of the parabola $y \rightarrow -\frac{y^2}{2(n+1)} + \left(1 - x - \frac{1}{n+1}\right)y - \frac{1}{2(n+1)}$ is $V = (n - (n + 1)x, \cdot)$; since

$$n - (n + 1)x < -1 \iff x > 1,$$

the unique minimizer in $(1 + \frac{2}{a_n}, 2]$ is -1 . Given the above, since $n + 1 + a_n = a_{n+1}$

$$\{\varphi_{n+1}(x)\} = \operatorname{Argmax}_{y \in Y} F_{n+1}(x, y) = \begin{cases} 1 & \text{if } x \in [\frac{1}{2}, 1] \\ a_{n+1} + 1 - (a_{n+1})x & \text{if } x \in (1, 1 + \frac{2}{a_{n+1}}] \\ -1 & \text{if } x \in (1 + \frac{2}{a_{n+1}}, 2], \end{cases} \quad (7.7)$$

for any $x \in [\frac{1}{2}, 2]$. From (7.0.2) we get

$$\begin{aligned} & L_{n+1}(x, \varphi_{n+1}(x)) \\ &= \begin{cases} -\frac{x^2}{2(n+1)} - \left(1 - \frac{a_n+2}{(n+1)a_n}\right)x - \frac{(a_n+2)^2}{2(n+1)a_n^2} - 1 & \text{if } x \in [\frac{1}{2}, 1] \\ -\frac{x^2}{2(n+1)} - \left(1 - a_{n+1} - \frac{a_n+2}{(n+1)a_n}\right)x - \frac{(a_n+2)^2}{2(n+1)a_n^2} - a_{n+1} - 1 & \text{if } x \in (1, 1 + \frac{2}{a_{n+1}}] \\ -\frac{x^2}{2(n+1)} - \left(1 - \frac{a_n+2}{(n+1)a_n}\right)x - \frac{(a_n+2)^2}{2(n+1)a_n^2} + 1 & \text{if } x \in (1 + \frac{2}{a_{n+1}}, 2] \end{cases} \\ &= \begin{cases} T_1(x) & \text{if } x \in [\frac{1}{2}, 1] \\ T_2(x) & \text{if } x \in (1, 1 + \frac{2}{a_{n+1}}] \\ T_3(x) & \text{if } x \in (1 + \frac{2}{a_{n+1}}, 2]. \end{cases} \end{aligned}$$

Since

- the abscissa of the vertexes of the parabolas $y = T_1(x)$ and $y = T_3(x)$ is $\frac{2}{a_n} - n < \frac{1}{2}$, so the first parabola assume his maximum in $x = \frac{1}{2}$, and the second one assume his minimum in $x = 2$,

- $T_1\left(\frac{1}{2}\right) < T_3(2)$,
- $L_{n+1}(\cdot, \varphi_{n+1}(\cdot))$ is continuous on $\left[\frac{1}{2}, 2\right]$,

then

$$\{\bar{x}_{n+1}\} = \operatorname{Arg\,max}_{x \in [-1, 1]} L_{n+1}(x, \varphi_{n+1}(x)) = \left\{1 + \frac{2}{a_{n+1}}\right\}. \quad (7.8)$$

For example, for $n = 1$ we have

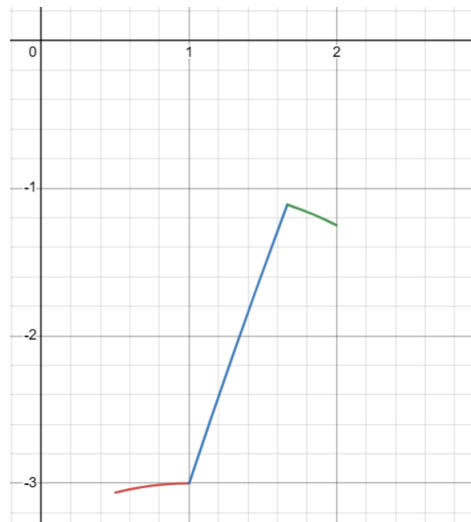


Figure 7.4: $L_2(x, \varphi_2(x))$

Equivalences (7.7)-(7.8) prove the inductive step, so (5.2) holds.

As $\lim_{n \rightarrow +\infty} a_n = +\infty$, we get

$$\bar{x} = \lim_{n \rightarrow +\infty} \bar{x}_n = 1, \quad \varphi(x) = \lim_{n \rightarrow +\infty} \varphi_n(x) = \begin{cases} 1 & \text{if } x \in \left[\frac{1}{2}, 1\right] \\ -1 & \text{if } x \in (1, 2]. \end{cases}$$

Since

$$\begin{aligned}
 \lim_{n \rightarrow +\infty} \varphi_n(\bar{x}_n) &= \lim_{n \rightarrow +\infty} \varphi_n\left(1 + \frac{2}{a_n}\right) \\
 &= \lim_{n \rightarrow +\infty} \begin{cases} 1 & x_n \in [\frac{1}{2}, 1] \\ a_n + 1 - a_n(1 + \frac{2}{a_n}) = -1 & x_n \in (1, 1 + \frac{2}{a_n}] \\ -1 & x_n \in (1 + \frac{2}{a_n}, 2] \end{cases} \\
 &= -1,
 \end{aligned}$$

then the SPNE constructed according to Theorem 5.5.1 is $(1, \bar{\varphi})$, where

$$\bar{\varphi}(x) = \begin{cases} 1 & \text{if } x \in [\frac{1}{2}, 1) \\ -1 & \text{if } x \in [1, 2]. \end{cases}$$

In this case

$$\lim_{n \rightarrow \infty} \varphi_n(\bar{x}_n) = -1 \neq 1 = \lim_{n \rightarrow \infty} \varphi_n(\lim_{n \rightarrow \infty} \bar{x}_n).$$

Furthermore, $(1, \varphi)$ is not even an SPNE of Γ , since

$$\begin{aligned}
 \text{Arg max}_{x \in [\frac{1}{2}, 2]} L(x, \varphi(x)) &= \text{Arg max}_{x \in [\frac{1}{2}, 2]} -x - \varphi(x) \\
 &= \text{Arg max}_{x \in [\frac{1}{2}, 2]} \begin{cases} -x - 1 & \text{if } x \in [\frac{1}{2}, 1] \\ -x + 1 & \text{if } x \in (1, 2] \end{cases} \\
 &= \emptyset,
 \end{aligned}$$

as shown in the following picture:

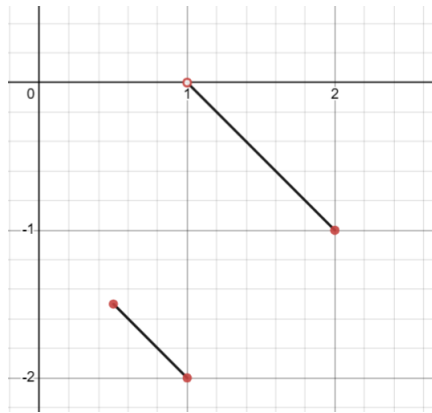


Figure 7.5: $L(x, \varphi(x))$

Example 7.0.3. Let $X = Y = [-1, 1]$ and $\Gamma = (L, F)$, where

$$L(x, y) = x, \quad F(x, y) = -xy.$$

The follower's best reply correspondence M is given by

$$M(x) = \begin{cases} \{1\} & \text{if } x \in [-1, 0) \\ [-1, 1] & \text{if } x = 0 \\ \{-1\} & \text{if } x \in (0, 1]. \end{cases}$$

Let $(\bar{x}_0, \bar{y}_0) \in [-1, 1] \times [-1, 1]$ be the initial point of the procedure and let $\beta_n = \gamma_n = 2^n$ for any $n \in \mathbb{N} \cup \{0\}$. Then, Procedure S_n generates the following sequence $(\bar{x}_n, \varphi_n)_n$ of strategy profiles:

$$\bar{x}_n = \begin{cases} \min \{1 + \bar{x}_0, 1\} & \text{if } n = 1 \\ 1 & \text{if } n \geq 2 \end{cases}, \quad \varphi_n(x) = \begin{cases} 1 & \text{if } x \in [-1, \frac{\bar{y}_0 - 1}{a_n}) \\ \bar{y}_0 - a_n x & \text{if } x \in [\frac{\bar{y}_0 - 1}{a_n}, \frac{\bar{y}_0 + 1}{a_n}] \\ -1 & \text{if } x \in (\frac{\bar{y}_0 + 1}{a_n}, 1] \end{cases} \quad (7.9)$$

where the sequence $(a_n)_n$ is recursively defined by

$$\begin{cases} a_1 = 1 \\ a_{n+1} = a_n + 2^n \end{cases} \quad \text{for any } n \geq 1.$$

Firstly, note that $\Gamma \in \mathcal{G}$. We prove (7.9) by induction on n . Let $n = 1$, then

$$\begin{aligned}
\{\varphi_1(x)\} &= \operatorname{Arg\,max}_{y \in Y} F_1(x, y) \\
&= \operatorname{Arg\,max}_{y \in [-1, 1]} \left(-xy - \frac{(y - \bar{y}_0)^2}{2} \right) \\
&= \operatorname{Arg\,max}_{y \in [-1, 1]} \left(-\frac{1}{2}y^2 + y(y_0 - 1) \right) \\
&= \begin{cases} 1 & \text{if } x \in [-1, \bar{y}_0 - 1) \\ \bar{y}_0 - x & \text{if } x \in [\bar{y}_0 - 1, \bar{y}_0 + 1] \\ -1 & \text{if } x \in (\bar{y}_0 + 1, 1] \end{cases}
\end{aligned}$$

and

$$\begin{aligned}
\{\bar{x}_1\} &= \operatorname{Arg\,max}_{x \in [-1, 1]} L_1(x, \varphi_1(x)) \\
&= \operatorname{Arg\,max}_{x \in [-1, 1]} x - \frac{(x - \bar{x}_0)^2}{2} \\
&= \operatorname{Arg\,max}_{x \in [-1, 1]} -\frac{x^2}{2} + x(x_0 + 1) \\
&= \begin{cases} 1 + \bar{x}_0 & \text{if } \bar{x}_0 \in [-1, 0] \\ 1 & \text{if } \bar{x}_0 \in (0, 1] \end{cases}
\end{aligned}$$

The base case is fulfilled as $a_1 = 1$. Assume that (7.9) holds for $n > 1$. So

$$\begin{aligned}
F_{n+1}(x, y) &= F(x, y) - \frac{1}{2\gamma_n}(y - \varphi_n)^2 \\
&= \begin{cases} P_1(x, y) = -\frac{y^2}{2^{n+1}} - \left(x - \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}, & \text{if } x \in \left[-1, \frac{\bar{y}_0 - 1}{a_n}\right) \\ P_2(x, y) = -\frac{y^2}{2^{n+1}} - \left(x + \frac{a_n x - \bar{y}_0}{2^n}\right)y - \frac{(\bar{y}_0 - a_n x)^2}{2^{n+1}}, & \text{if } x \in \left[\frac{\bar{y}_0 - 1}{a_n}, \frac{\bar{y}_0 + 1}{a_n}\right] \\ P_3(x, y) = -\frac{y^2}{2^{n+1}} - \left(x + \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}, & \text{if } x \in \left(\frac{\bar{y}_0 + 1}{a_n}, 1\right] \end{cases}
\end{aligned}$$

and

$$\begin{aligned}
L_{n+1}(x, y) &= L(x, y) - \frac{1}{2\beta_n}(y - x_n)^2 \\
&= \frac{x^2}{2^{n+1}} + \left(1 + \frac{1}{2^n}\right)x - \frac{1}{2^{n+1}}.
\end{aligned}$$

If $x \in [-1, \frac{\bar{y}_0-1}{a_n})$, the vertex of the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x - \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}$ is $V = (1 - 2^n x, \cdot)$, that is greater than 1 since $\frac{\bar{y}_0-1}{a_n} < 0$ for every $\bar{y}_0 \in [-1, 1]$; so the unique maximizer of $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x - \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}$ in $Y = [-1, 1]$ is 1.

If $x \in \left[\frac{\bar{y}_0-1}{a_n}, \frac{\bar{y}_0+1}{a_n}\right]$, then the vertex of the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x + \frac{a_n x - \bar{y}_0}{2^n}\right)y - \frac{(\bar{y}_0 - a_n x)^2}{2^{n+1}}$ is $V = (\bar{y}_0 - x(a_n + 2^n))$; we have that

$$\begin{aligned}\bar{y}_0 - x(a_n + 2^n) < 1 &\iff x > \frac{\bar{y}_0 - 1}{a_n + 2^n} \\ \bar{y}_0 - x(a_n + 2^n) > -1 &\iff x < \frac{\bar{y}_0 + 1}{a_n + 2^n}\end{aligned}$$

So, if $x \in [\frac{\bar{y}_0-1}{a_n}, \frac{\bar{y}_0+1}{2^n+a_n}]$, then the unique maximizer is 1. If $x \in [\frac{\bar{y}_0-1}{2^n+a_n}, \frac{\bar{y}_0+1}{2^n+a_n}]$ the unique maximizer is $\bar{y}_0 - (2^n + a_n)x$. If $x \in [\frac{\bar{y}_0+1}{2^n+a_n}, \frac{\bar{y}_0-1}{a_n}]$, the unique maximizer is -1 . Finally, if $x \in [\frac{\bar{y}_0+1}{a_n}, 1)$, the vertex of the parabola $y \rightarrow -\frac{y^2}{2^{n+1}} - \left(x + \frac{1}{2^n}\right)y - \frac{1}{2^{n+1}}$ is $V = (-1 - 2^n x, \cdot)$; Since $-1 - 2^n x < -1$, the unique maximizer is -1 .

Given the above, since $2^n + a_n = a_{n+1}$,

$$\{\varphi_{n+1}(x)\} = \operatorname{Arg\,max}_{y \in Y} F_{n+1}(x, y) = \begin{cases} 1 & \text{if } x \in [-1, \frac{\bar{y}_0-1}{a_{n+1}}) \\ \bar{y}_0 - a_{n+1}x & \text{if } x \in [\frac{\bar{y}_0-1}{a_{n+1}}, \frac{\bar{y}_0+1}{a_{n+1}}] \\ -1 & \text{if } x \in (\frac{\bar{y}_0+1}{a_{n+1}}, 1] \end{cases} \quad (7.10)$$

for any $x \in [-1, 1]$.

Since $L_{n+1}(x, \varphi_{n+1}(x)) = L_{n+1}(x, y)$ for any $(x, y) \in X \times Y$ and the abscissa of the vertex of the parabola $x \rightarrow L_{n+1}(x, \varphi_{n+1}(x))$ is $2^n + 1 > 1$, then

$$\{\bar{x}_{n+1}\} = \operatorname{Arg\,max}_{x \in [-1, 1]} L_{n+1}(x, \varphi_{n+1}(x)) = \{1\}. \quad (7.11)$$

Equalities (7.10)-(7.11) prove the inductive step, so (7.9) holds.

As $\lim_{n \rightarrow +\infty} a_n = +\infty$, we get

$$\bar{x} = \lim_{n \rightarrow +\infty} \bar{x}_n = 1, \quad \varphi(x) = \lim_{n \rightarrow +\infty} \varphi_n(x) = \begin{cases} 1 & \text{if } x \in [-1, 0) \\ \bar{y}_0 & \text{if } x = 0 \\ -1 & \text{if } x \in (0, 1]. \end{cases}$$

Since $\lim_{n \rightarrow +\infty} \varphi_n(\bar{x}_n) = \lim_{n \rightarrow +\infty} \varphi_n(\bar{x})$, then the SPNE constructed according to Theorem 5.5.1 is $(1, \bar{\varphi}) = (1, \varphi)$.

Note that we obtain the SPNEs of Γ by varying $y_0 \in [-1, 1]$, so $(1, \bar{\varphi})$ is the one that minimizes the distance from the follower's initial point $\bar{y}_0$.

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